DEGREE PROJECT

A Human-Centered Approach to Radar Maintenance

Applying usability principles to evaluate and improve service and repair

Albin Sämfors
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Industrial Design Engineering, master's level
2026

Luleå University of Technology
Department of Social Sciences, Technology and Arts

LULEÅ UNIVERSITY OF TECHNOLOGY

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CIVILINGENJÖR I TEKNISK DESIGN
Master of Science Thesis in Industrial Design Engineering

A Human-Centered Approach to Radar Maintenance
Applying usability principles to evaluate and improve maintenance

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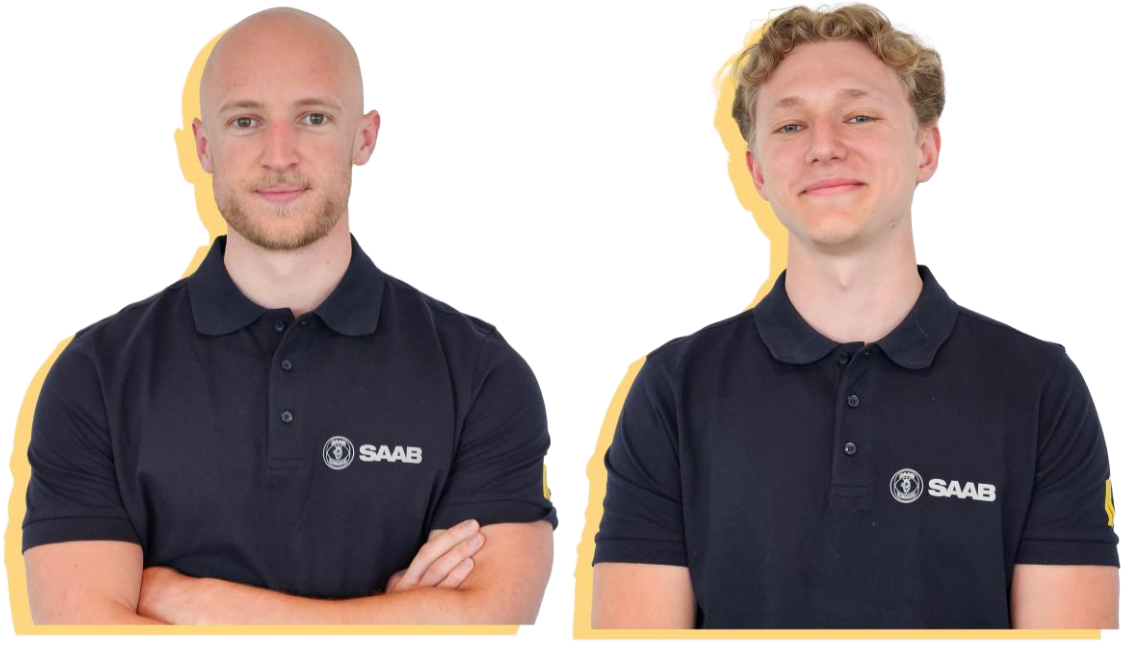
We would like to express our sincerest gratitude to all individuals who have contributed and supported us throughout this project. We are grateful to Luleå University of Technology for the opportunity to write this Industrial Design Engineering master's thesis, and to our university supervisor, Peter Törlind. Peter's knowledge and experience in product development and design have been invaluable to us, and we extend our sincerest thanks.

This research would also not have been possible without the support of the Saab personnel working with the Giraffe 1X radar. They generously participated in interviews, granted us access to observe their work, and shared insightful information that we would have otherwise missed. We would like to specially thank Maria Köhler, and our supervisors at Saab. Erik Lindälv, with his immense knowledge of engineering, functionality, and creativity, and Erik Halldin, for his insightful perspective on human interaction and processes.

We would also like to thank the external group who participated in the workshop.

Working with the people involved in the development of an innovative product like the Giraffe 1X at an evolving company like Saab has been an invaluable experience. Connecting with such talented individuals and learning about engineering has given us great insight into the future that lies ahead of us.

Gothenburg June 11, 2025
Simon Tollesson & Albin Sämfors



Abstract

This master thesis has been made with Luleå University of Technology in collaboration with Saab AB to research and improve the usability in the Giraffe 1X radar in service and reparation. In these worrying times, security and protection is in high priority, the threat of flying targets and their location is therefore of high interest. Giraffe 1X have the ability to fulfill this need continuously, although the need for service and repairs occurs. Increasing the usability of the product in service scenarios, would reduce the risk of failure by human factors, reduce operators training time, and imply faster service and repairs.

We have cooperated with Saab to map the existing usability challenges, as well as to identify areas with potential for further improvement. Based on these challenges, they were addressed and examined to determine what would lead to the most effective improvements, and subsequently, design proposals were developed to eliminate these challenges. These design proposals are intended to be easily integrated into today's product, but also include suggestions that could be implemented in future products. A more concise and focused formulation can be found in the following research questions:

1. What are the current usability issues, regarding repairability and assembly of the radar components?
2. How can we, with relatively small means, effectively increase the usability and make the product more intuitive to use?
3. How can design solutions be effectively integrated into the existing product?

The thesis follows a six-phase process designed to promote user-centered product development. The work began with a contextual investigation of the product's components and functions. In parallel, usability challenges were identified through interactions with users, from which primarily qualitative, but also some quantitative, data were collected. This information was then analyzed, and design proposals were developed. After careful consideration in consultation with experts, it was concluded that a communicative toolkit with associated icons for the radar interface was of particular interest. Several prototypes were developed using various materials, both for the functional design of the toolkit and for different icon systems. These tools and their design were evaluated and tested, along with conceptual suggestions for integration into both current and future radar systems.

KEYWORDS: Industrial Design Engineering, Human-centered Design, Co-Design, Usability

Sammanfattning

Detta examensarbete har utförts genom Luleå Tekniska Universitet i samarbete med Saab AB för att undersöka och förbättra användbarheten i radarn Giraffe 1X vid service och reparation. I dessa oroliga tider är säkerhet och skydd i hög prioritet, hotet av flygande föremål och vart de befinner sig är därför av högt intresse. Giraffe 1X är utformad för att kontinuerligt möta detta behov, men kräver emellanåt service eller reparation. Genom att förbättra produktens användbarhet i dessa situationer kan risken för fel orsakade av mänskliga faktorer minska, utbildningstiden förkortas, och serviceprocessen påskyndas.

Vi har samarbetat med Saab för att kartlägga vad som är de befintliga utmaningarna i användbarheten och även vad som har potential bli ännu bättre. Baserat på de utmaningarna, behandlades de och undersöktes för att se vad som skulle innebära mest effektiv förbättring, för att sedan utveckla förslag på designmässiga förändringar som skulle eliminera dessa utmaningar. Designmässiga förslag som på ett enkelt sätt kan integreras i dagens produkt, men även förslag som skulle kunna implementeras i framtida produkter. För att definiera omfattningen av arbetet så formulerades dessa tre forskningsfrågor:

1. Vad är de nuvarande användbarhetsproblemen, angående reperationsbarhet och montering av radarkomponenterna.
2. Hur kan vi, med relativt små medel, effektivt förbättra användbarheten och göra produkten mer intuitiv att använda.
3. Hur kan designlösningar bli effektivt integrerade i den existerande produkten.

Examensarbetet följer en process i sex faser, utformad för att främja användarcentrerad produktutveckling. Arbetet inleddes med en kontextuell undersökning av produktens komponenter och funktioner. Parallellt identifierades användbarhetsutmaningar genom interaktioner med användare, varvid främst kvalitativa men även vissa kvantitativa data samlades in. Därefter analyserades informationen och designförslag togs fram. Efter noggranna överväganden i samråd med experter konstaterades att ett kommunikativt verktygskit med tillhörande ikoner för radargränssnitt var av särskilt intresse. Flera prototyper utvecklades i olika material, både för kittets funktionella design och för olika ikonssystem. Dessa verktyg och deras utformning utvärderades och testades, tillsammans med konceptuella förslag för hur lösningen kan integreras i både nutida och framtida radarsystem.

NYCKELORD: Teknisk Design, Människo-centrerad design, Co-design, Användbarhet

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Thesis Glossary

This project utilizes several technical terms and descriptive words that may be unfamiliar. To enhance understanding of these terms within the project's context, please refer to the following glossary:

Radar – A radar system is used for detecting and tracking objects through transmission and reception of radio waves. In this context, the term refers to the complete product solution, encompassing all components and parts as a unified whole.

Giraffe 1X – refers to the complete system, encompassing both the antenna and other integrated components. When this solution is combined and integrated into various vehicles or structures, it retains the name Giraffe 1X.

Antenna – is the rotating component of the radar system. It's responsible for transmitting and receiving radio waves, which allows the radar to detect objects and determine their location, speed, and other characteristics.

Sensor Kit - encompasses not only the antenna but also essential external components required for the radar's operation. This includes elements like power distribution systems and computing units.

CRM – The CRM, or Compact Radar Module, is a compact kit that includes all of the external components required for the operation of the Giraffe 1X radar. The two are often combined and then go under the name Giraffe 1X CRM.

Operator – is a soldier trained to operate the radar and troubleshoot errors. Their domain is the radars' functionality; they utilize its features but are not necessarily responsible for repairs or maintenance.

Technician / Maintainer – is a user with further technical training. They interact with the radar primarily for repairs and maintenance.

ESD - stands for electrostatic discharge. This term is used to identify components that are sensitive to even small amounts of static electricity. The charge that can build up on humans can be enough to damage these sensitive components.

01 INTRODUCTION

1.1 Background
1.2 Stakeholders
1.3 Objective & Aims
1.4 Limitations
1.5 Thesis Outline



Photo: Saab

1 Introduction

During these worrying times in the world, protection and security is of high priority. Achieving this requires advanced systems that can identify when threats are inbound. It is important that those systems are in operation as much as possible. When the systems eventually need service or repairs, it is important that this can be performed as quickly as possible and with a low risk of damage.

The Giraffe 1X radar is one of Saab AB's radar systems and focuses primarily on short-range air defense. The antenna is incredibly versatile and has set new standards for the industry, which will impact the entire surveillance industry. Saab has designed the Giraffe 1X radar with maintainability in mind, minimizing downtime and ensuring maximum operational readiness, but strives to further improve the radar system. Therefore, improving usability and making the system easier to service intuitively will be the central focus of this master thesis. It will be carried out by two Industrial Design Engineer students during the spring of 2025 in collaboration with Luleå University of Technology and Saab AB.



Figure 1.1: Mounted Giraffe 1X CRM (Saab, u.d.)

1.1 BACKGROUND

Many are worried about the current geopolitical situation with the tension between major powers constantly rising. The situation is causing a sense of global uncertainty, which creates an increasing interest for countries to expand their defense. This is where Saab ABs wide range of radars comes into play. With solutions that range from utilizing larger aircrafts, such as the GlobalEye, to radars being stationed on the ground to spot mortars, Saab is always finding new solutions for the situation. One of their newest products, the Giraffe 1X, is an easily transportable radar which has set new standards for what a radar of that size is capable of. The antenna system is able to withstand almost any environment and can be used for a variety of missions, including naval operations, fixed land-based installations, and vehicle-mounted ground deployments.

The Giraffe 1X is able to operate all hours of the day, but every now and then the system is in need of maintenance or repairs. It is crucial to reduce the downtime of the radar as much as possible, which requires the maintenance to be quick and easy. Improving usability during service primarily reduces the risk of errors and lowers training demands, which are critical in high-stress environments. Faster time-on-task during repairs is also beneficial, but the time-on-task is a minimal part of the duration of the entire repair.

Operators working with the Giraffe 1X in high-stress situations face a variety of challenges, making a user-centered design approach crucial for safety and efficiency. This calls for a more uniform and intuitive design, which could prevent errors, save resources and maybe even lives by increasing availability of the systems.

1.2 STAKEHOLDERS

This project is part of a master's education as Industrial Design Engineers at Luleå University of Technology (LTU). Since the thesis is the final examination for the master's degree and is directly affecting the university's reputation, they are one of the closest stakeholders. We have direct contact and support from one of our teachers who is both making sure the project is carried out in a correct manner, but also providing help and solutions to our problems.

In the same manner as LTU, Saab has a high interest in the work being done. They're the ones who directly benefits of the work by being provided new propositions and solutions to their product. Therefore, they pose as a major

stakeholder with high interest and influence on the thesis. They are, to some degree, able to direct the work to specific areas, for example by recommending to focus on finding solutions for a specific challenge.

Since the work is aimed towards the user experience of service and maintenance, there will also be some external stakeholders. The technicians and operators of the systems will be affected by the changes made and while they may not have high influence on what changes are made, their opinion and thoughts are highly valued.

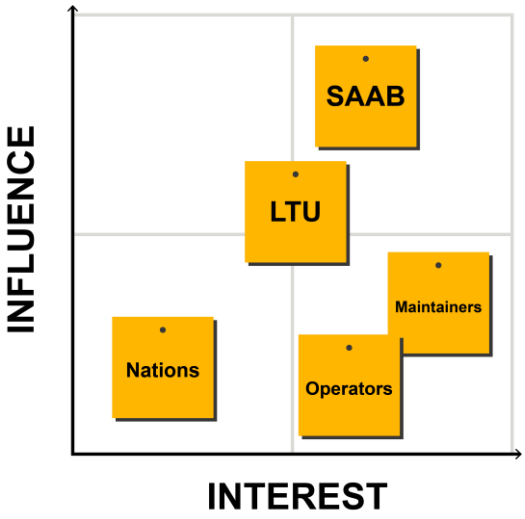


Figure 1.2: Project stakeholders Influence-Interest matrix. Illustration: A. Sämfors

The thesis will be classed as an academic research paper and uses researched methodologies and theories. Hopefully the results and findings from this thesis can contribute to further research being done within the field of Industrial Design Engineering.

1.3 OBJECTIVE & AIMS

The project aims to analyze today's design from a user perspective and develop design proposals to make the product more intuitive to maintain. It also aims to find improvements that can increase usability in the interaction when maintaining or repairing the product. These goals can reduce errors and downtime while repairing, reduce need for education, and increase availability.

To achieve these goals, we will be using a design process which involves analyzing and understanding the current issues, explore new solutions, and finally create a prototype to validate its functionality. This will be done in three separate project phases, which have two sub-phases each.

The aim is also to provide concepts and prototypes of the solutions found. These will then be evaluated and tested with consideration to the current system, which will generate results suitable for further analysis and discussion.

The following research questions were chosen to guide the thesis toward the sought-after goal:

1. What are the current usability issues, regarding repairability and assembly of the radar components?
2. How can we, with relatively small means, effectively increase the usability and make the product more intuitive to interact with?
3. How can design solutions be effectively integrated into the existing product?

1.4 PROJECT SCOPE

This thesis focuses on improving the physical aspects of user experience for the Giraffe 1X radar unit, with focus on usability and guessability during assembly and maintenance. The aim is to identify areas for improvement that could enhance intuitive use and reduce user errors, especially during infrequent but critical servicing operations.

To clarify the project's boundaries and manage expectations, we distinguish between external limitations and deliberate delimitations:

External Limitations

Limited Access to End Users:
Due to security protocols at Saab AB and the sensitive nature of the radar system, access to a broad and diverse user base is restricted. For this thesis only a small group of experienced users is available for user testing and feedback. This constraint means that each user interaction must be strategically planned and fully utilized.

Design Constraints of the Existing Radar Unit:
The Giraffe 1X radar unit has strict spatial limitations and uses standardized components. As a result, proposed design improvements must work within the existing physical framework and cannot introduce major structural changes. This may limit the range of implementable solutions but ensures compatibility with the current system.

Industry Standards and Regulations:
The defense industry imposes strict regulations on design elements such as color usage, iconography, and interaction patterns to prevent user errors in high-stress environments. These standards guide usability but also restrict creative freedom. Since the thesis aims to explore potential improvements rather than deliver a finalized, deployable solution, full compliance with these regulations is outside the current scope.

Deliberate Delimitations

Focus on Cognitive Rather than Physical Ergonomics: While ergonomics is considered in the design process, the limited time users spend interacting with the radar during maintenance reduces the risk of long-term physical strain. Therefore, the project does not focus on load ergonomics or related physical injury prevention. Instead, attention is directed toward cognitive ergonomics to support intuitive understanding and operation.

Scope of Design Recommendations: Given the regulatory and technical constraints, this thesis does not attempt to deliver a complete redesign. Instead, the focus is on generating well-founded design suggestions and usability improvements that could inform future development of the radar system, whether for the current or next-generation models.

1.5 THESIS OUTLINE

The thesis is organized into chapters with headings and subheadings to ensure clarity and structure. Each chapter is briefly summarized below, outlining its purpose and how it was accomplished.

Introduction

The first chapter (this chapter) is an introduction to the thesis and is meant to give an overview and concise understanding of what the thesis will be about. It contains information about the project's background, objective and aims, and the limitations of the thesis.

Context

Chapter two presents a comprehensive overview of the project's context; it gives a foundational knowledge of the report and information about the terms that will be used. It primarily focuses on describing Saab as a company and the context of the Giraffe 1X-radars.

Theoretical Framework

The third chapter presents the theories and literature relevant to the project It aims to give a basic understanding of the scientific theories used and how they affected the decisions that were made. It covers areas such as Industrial design engineering, Human-centered design and User experience. The sub-headings give more detailed information about theories, such as Usability, Ergonomics, Learnability and Guessability.

Method & Implementation

Chapter four explains the methods used during the project and how they affected the work. It also gives an overall understanding of the process and timeline, including the different phases and what were done during them. The chapter ends with a discussion of what affected the results and if there was anything that could have been done different.

Results

The fifth chapter is structured identical to the method chapter but presents the results and conclusions from each phase. The content of this chapter is also what the coming discussion and conclusion is based and built upon.

Final Design

In this chapter the final results of the thesis will be described and explained.

Discussion

The discussion, which is the seventh chapter, explains the projects relevance, the relevance of the results, and final reflections and conclusions.

References

The eighth chapter contains references to all of



02 CONTEXT

2.1 Saab

2.2 Giraffe 1X

2.2.1 Work Environment

2.2.2 Antenna Components

Photo: Saab

2 Context

This chapter provides the context for the master's thesis. It includes a description of the client company, an overview of the Giraffe 1X radar and the environments that influence this project.

2.1 SAAB

Saab is an established and versatile company specializing in defense systems. Founded in the 1930s, during the lead-up to World War II, Saab initially focused on developing and manufacturing airplanes. Since then, they have expanded significantly and now play a major role in the Swedish defense and security industry.

Figure 2.1: Saab AB logo. Illustration: Saab AB

Beyond air systems, Saab operates in naval systems, including submarines and ships, and ground systems such as missiles, camouflage technology, and radars. In 2006, Saab acquired Ericsson Microwave Systems, known for its radar technology. This acquisition led to Saab being involved in further development of already existing world-leading products like the Giraffe series of ground-based radars, but also the creation of new products such as the airborne early warning system known as GlobalEye. Saab continues to innovate, with the Giraffe 1X being one of their latest radar systems.

2.2 GIRAFFE 1X

Giraffe 1X is a ground-based multifunctional 3D radar. It can detect threats such as drones, airplanes, artillery and rockets. It can also be used for surface surveillance in naval applications. Its compact and lightweight design allows for integration into various mobile platforms or fixed structurers, even with low power requirements. Weighing less than 150 kg, the Giraffe 1X is significantly lighter than any other radar system with comparable performance (Saab, u.d.).

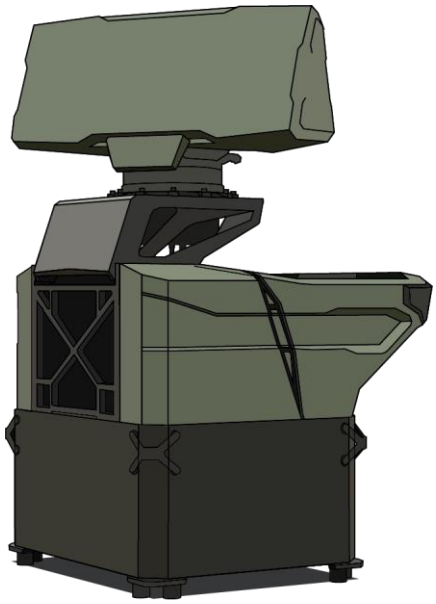


Figure 2.2: Giraffe 1X on CRM. Illustration: A. Sämfors

The Compact Radar Module (CRM), shown in Figure 2.2, is the setup which has been mainly worked with during this thesis. It's a complete radar system where the antenna is mounted on a module, with folding capabilities, and allows for integration into almost any vehicle the size of a pickup truck or larger.

2.2.1 Work Environment

The design's versatility and lightweight construction enables integration into a wide range of environments. This means the radar encounters a diverse set of weather and climate conditions, as well as physical surroundings that influence user ergonomics during repair and service. Examples include military off-road vehicles, offshore wind farms, buildings, and ships.

These settings often present significant challenges for maintainers. They may need to work in confined spaces, at heights, or in harsh weather. Furthermore, users frequently face demanding conditions such as sleep deprivation, hunger and mental fatigue, all of which contribute to elevated stress levels and increase the risk of errors. Therefore, an efficient, intuitive design is critical for reducing time-on-task, preventing accidents, and minimizing potential risk of errors.

2.2.2 Antenna Components

This project analyzes the multiple components of the radar antenna unit. While the components themselves have functionalities that are beyond the scope of this project, understanding their basic functions is crucial for grasping their importance and relevance in making a significant impact. Therefore, it's essential to grasp which components were mainly considered during the project. In addition, most of the components which were taken into account are all located under, or near, the back cover of the antenna and are all shown in Figure 2.3.

The Transmit and Receive Module (TRM) houses both transmitters and receivers responsible for generating outgoing radar pulses and capturing the returning echo signals. Once received, the echo is amplified and then digitized, enabling further signal processing and interpretation by the radar system.

Located in the center of the antenna, the Central Magazine (CM) contains multiple cassettes that manage signal distribution and power regulation. It serves as a central hub for distributing electrical and communication signals to various parts of the antenna.

The Turntable consists of the turntable unit and the rotary joint. The turntable unit itself contains a turntable control unit (TCU), an angle and rotation detector and a motor. The TCU is the central hub for information and electrical connections. The angle and rotation detector monitors the antenna's rotation speed, while the motor converts electrical energy into rotational force.

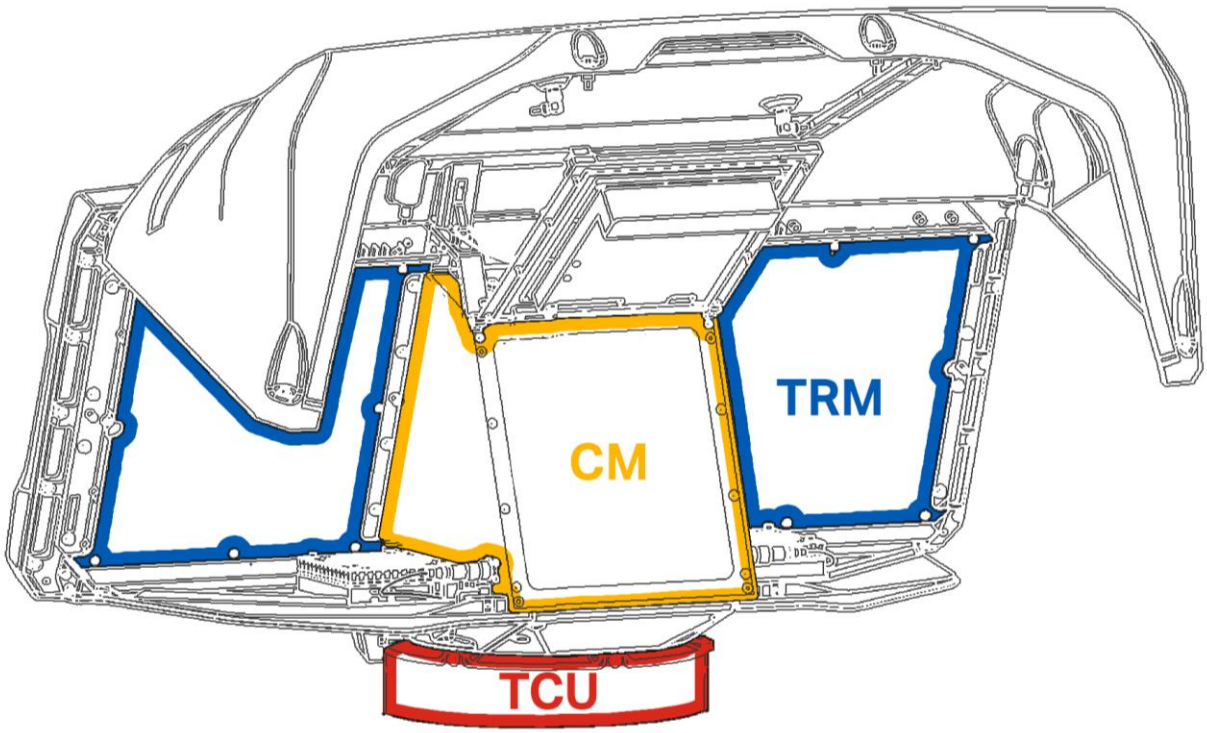
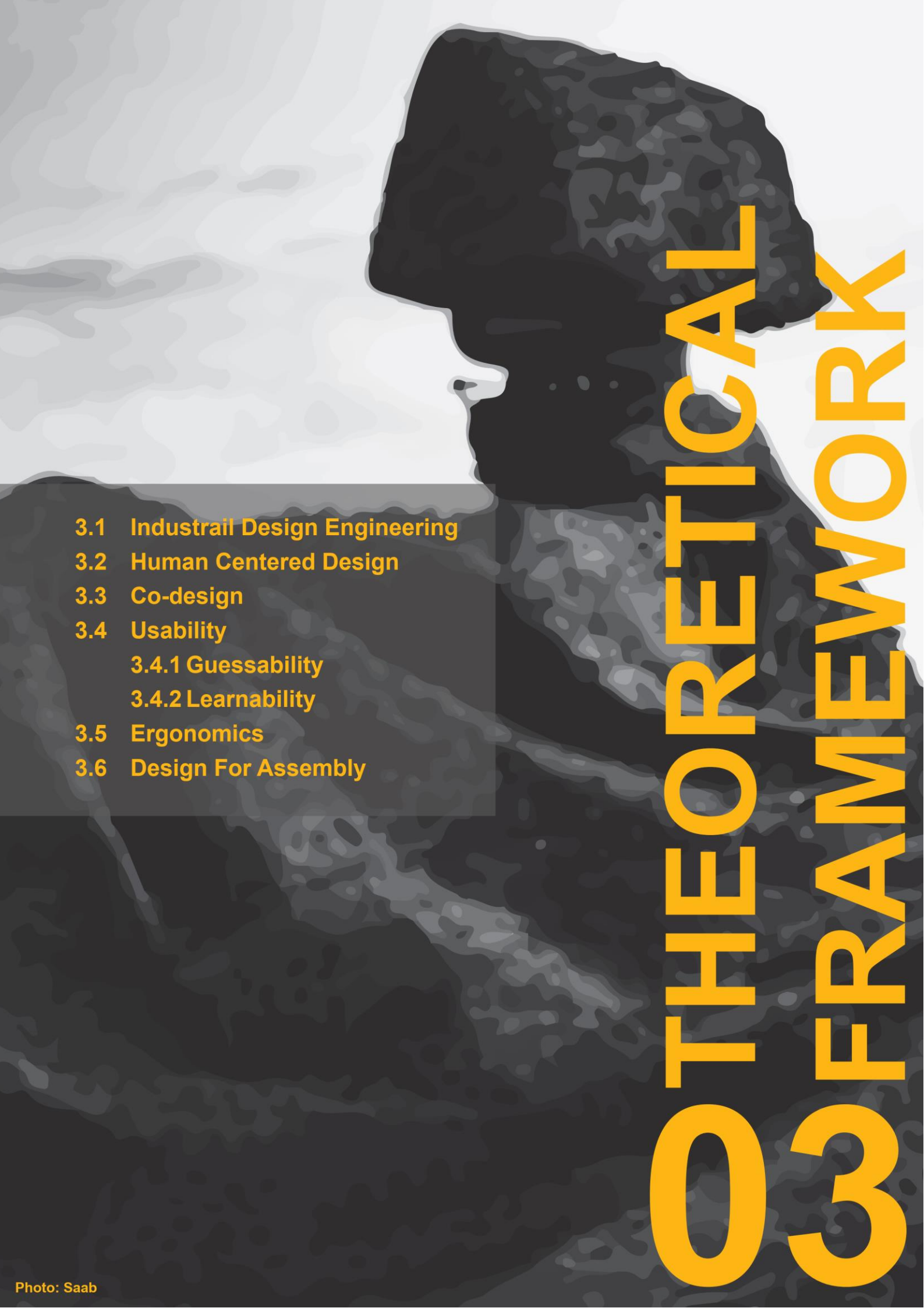


Figure 2.3: Relevant antenna components. Illustration: S. Tollesson



03 THEORETICAL FRAMEWORK

3.1 Industrail Design Engineering
3.2 Human Centered Design
3.3 Co-design
3.4 Usability
3.4.1 Guessability
3.4.2 Learnability
3.5 Ergonomics
3.6 Design For Assembly

Photo: Saab

3 Theoretical framework

This chapter establishes the theoretical framework behind this master thesis. It describes key concepts and principles that have guided the development of the project and provides foundational knowledge to understand the choices made throughout the project.

The chapter contains an overview of Industrial Design Engineering and Human-Centered Design and the subcategories within the theory.

3.1 INDUSTRIAL DESIGN ENGINEERING

Industrial Design Engineering (IDE), as IDSA (u.d.) describes, is a professional practice of designing products, devices, objects, and services used by millions of people around the world every day. Practitioners in this field are not only concerned with the aesthetics of a product, but also how it functions, is manufactured and the overall experience it provides the user (Ulrich & Eppinger, 2019).

Industrial Design Engineering distinguishes itself from other engineering disciplines through its unwavering focus on the user experience. IDE prioritizes understanding user needs, behaviors, and preferences throughout the design process. This ensures that products are not only functional but also intuitive, enjoyable, and ultimately meet the specific requirements of the intended audience. IDE practitioners achieve this user-centricity through a blend of creative and empathetic skills, coupled with robust methodologies. These include user research, ergonomic principles, and usability testing, which ensure that designs are not only aesthetically pleasing but also functional and user friendly (Chalmers, 2025).

> "Industrial Design is a trans-disciplinary profession that harnesses creativity to resolve problems and co-create solutions with the intent of making a product, systems, service, experience, or a business, better."
>
> (World Design Organization, u.d.)

This master's thesis draws directly on the principles of Industrial Design Engineering and uses the taught academic foundation to bridge theories and practices within areas such as ergonomics, usability and human-centered design.

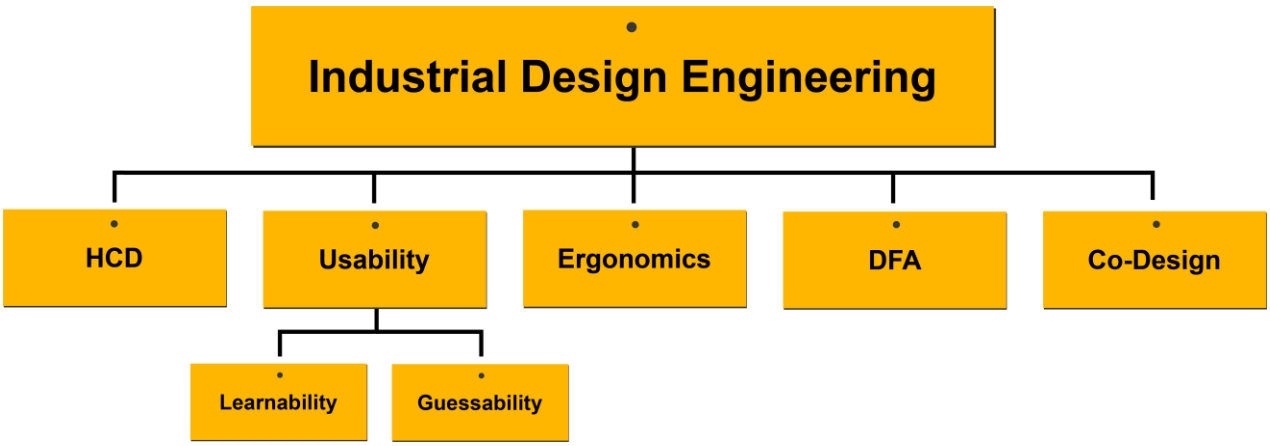


Figure 3.1: Structure of the theoretical framework. Illustration: A. Sämfors

3.2 HUMAN CENTERED DESIGN

Human-centered design (HCD) provided the overarching framework for this thesis, ensuring that every phase, from initial problem framing to prototype evaluation, remained focused on human abilities, limitations and needs. By positioning users at the core of the process, the project maintained alignment between design decisions and the real contexts in which the solutions would be applied.

Harte (2017) describes HCD as a philosophy of Industrial Design Engineering that places the human actor at the center of product development, enhancing rather than constraining user capabilities. Wikberg (2015) further argues that effective design must accommodate human diversity, covering factors such as age, size, cultural background and prior experience. The ISO 9241-210 standard (2019) formalizes HCD as an iterative approach that integrates human factors, ergonomics and usability techniques to produce systems that are both usable and useful.

ISO 9241-210 outlines four interrelated principles of HCD:



Figure 3.2: Depiction of HCD principles. Illustration: S. Tollesson

Table 3.1: ISO 9241-210 HCD Principles

1.	Explicit understanding of users, tasks, and environments
2.	Active user involvement throughout the process
3.	User-driven evaluation and refinement
4.	Iterative design cycles

In this thesis, these principles heavily influenced project planning, stakeholder engagement, prototype development and evaluation strategies, ensuring that each activity reinforced a consistent human-centered perspective.

Human-centered design served as the umbrella under which related methods were applied: co-design workshops, usability testing and ergonomic analysis. Co-design workshops introduced fresh viewpoints from individuals outside the immediate user group, broadening the solution space. Usability testing validated design assumptions against real-world scenarios, highlighting areas for improvement. Ergonomic analysis ensured that physical, cognitive and environmental factors were accounted for in every concept iteration.

By contrast, technology- or feature-driven approaches often begin with system capabilities and involve users only to confirm a nearly finished design. Such processes risk creating products that, while technically sophisticated, may be hard to learn, prone to misuse or misaligned with actual work practices. Adopting HCD from the outset reduced these risks, fostering designs that better matched user expectations and minimized costly late-stage revisions.

The following chapters will explore several key methodologies closely related to the broader framework of HCD, including usability, ergonomics, guessability, and learnability.

3.3 USABILITY

To improve usability through research and redesign, it is first necessary to define what usability means and how it can be evaluated. A clear understanding of the concept and its components forms the basis for selecting appropriate methods and criteria for assessing different design solutions. Since the aim of this thesis is to enhance usability, particularly in the early stages of product use, establishing a strong theoretical foundation is essential.

Usability is described by ISO 9241-11 (2018) as; "the extent to which a system, product or service can be used by specified users to achieve specified goals with effectiveness, efficiency and

satisfaction in a specified context of use". The measurements are described by Jordan (1998) and ISO 9241-11 (2018) as; Effectiveness is the extent to which a goal, or task is achieved; Efficiency is the amount of effort required to accomplish a goal; Satisfaction is the level of comfort that the user feels when using a product and how acceptable the product is to users as vehicle for achieving their goals. These factors can be measured and evaluated by using methods, such as, observations, interviews, and prototypes.

Usability can greatly vary in a single product depending on the goals, types of users and use environment. Bevan et al. (1991) adds to this by stating, usability should be seen as a function of the ease of use and the acceptability of the product, which in turn determines whether a product can be used, will be used, and how it can be used. But often it is not the product in its entirety that decides this, but rather the attributes and characteristics in a particular environment. Bevan et al. (1991) further explains that the characteristics doesn't have to be ergonomics related, but can be any kind of design element or detail that can impact the users ease of use.

Arguably, usability is frequently underestimated in its potential to drive commercial success, particularly for technologically advanced products where a seamless and intuitive user experience can be a significant competitive advantage. Jordan (1998) reinforces this point, arguing that in a competitive market with lots of similar products, a highly user-friendly solution can be the deciding factor for customers. Small changes with the users' experience can make incredible differences in safety, annoyance and time spent on tasks. ISO 9241-11 (2018) adds that it can also greatly reduce the risk of undesirable consequences of use errors and maintenance. Which is connected to Jordans (1998) "Components of Usability", where he describes Guessability, Learnability, and Experienced user performance (EUP), as stages of users' knowledge of a system or product. Two additional components, System potential and Re-usability, are also mentioned by Jordan (1998) and are connected to achieving the optimal obtainable performance when a system or product is used frequently. The usability of an item can be optimized around one or more of these components depending on whether it should be easy to use first-try, easy to learn or increase efficient use for experienced users.

The usability components which will be taken into consideration during this thesis are the ones regarding the early use of the product, and will be further described in the following headings.

3.3.1 Guessability

Jordan (1998) defines guessability as "the effectiveness, efficiency and satisfaction with which specified users can complete specified tasks with a particular product for the first time". This can be measured by for example comparing time-on-task and errors before and after changes have been made. Enhancing guessability leads to a smoother first-time use. The more effectively a product communicates its intended use, the higher its guessability will be (Jordan, 1998).

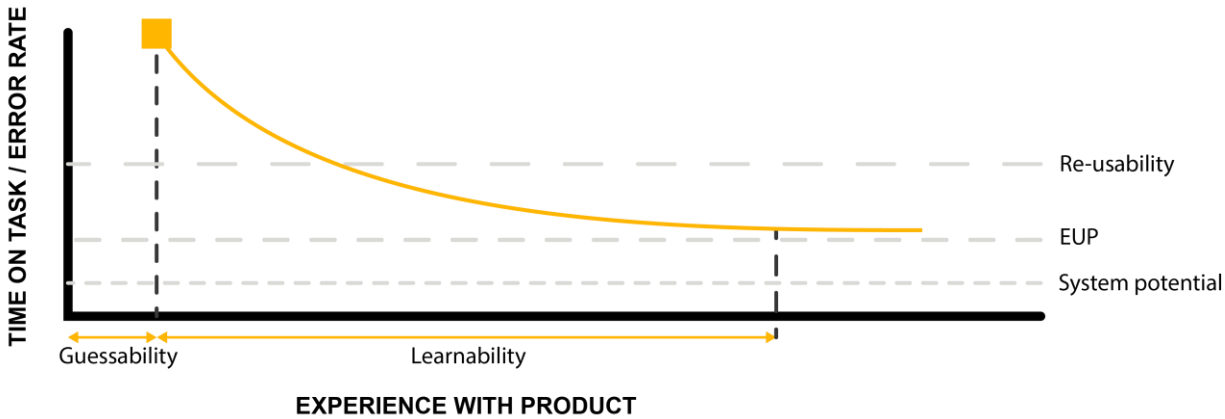


Figure 3.3: Usability in a Time-on-task to experience-with-product matrix. Illustration: A. Sämfors

This communication is closely related to the term intuition. Daniel Kahneman (2011) describes intuition, in his book Thinking, Fast and Slow, as the ability to spot patterns and to integrate insights from past experiences. Intuition is automatic, effortless, and subconscious. In other words, an intuitive design should make a product easier to use and helps the customer understand the design without needing a tutorial or other help, which is closely related to guessability. Therefore, by creating an intuitive design, the products guessability should also increase.

Insufficient guessability can be particularly problematic for tasks performed infrequently by diverse users. Jordan (1998) illustrates this point with the example of fire extinguishers, where immediate, intuitive use is crucial. Conversely, military equipment often prioritizes other usability factors, as it is typically operated by highly trained personnel. However, Jordan (1998) acknowledges that even in such contexts, there may be rare scenarios where guessability becomes advantageous, ensuring that personnel can handle unfamiliar tasks effectively.

Even though military systems are typically designed for highly trained users, real-world conditions often limit the effectiveness of manuals and procedural training. In time-sensitive or stressful environments, users may not recall instructions or maybe perform maintenance tasks they haven't encountered for several years. Therefore, improving guessability in such systems is not only relevant, but it can also be critical.

To support first-time or infrequent use, product design should employ clear affordances and signifiers. As Norman (1999) and Gaver (1991) explain, affordances are the actual or perceived properties of an object that suggest how it can be used, while signifiers are the visible cues that indicate how to interact with those affordances. For instance, a visibly marked screw or a shaped handle provides immediate visual guidance on where and how an action should be performed. Integrating such cues improves guessability by reducing cognitive load and ensuring that even infrequent users can interact with the system efficiently and safely without having to rely on outdated memory or written instructions.

3.3.2 Learnability

Some systems, products and services require short training time or rely the user to self-learn. Learnability, as defined by Jordan (1998), measures how quickly users can become competent with using a system. This concept is closely tied to guessability.

> **"The effectiveness, efficiency and satisfaction with which specified users can achieve a competent level of performance on specified tasks with a product, having already completed those tasks once previously."**
> (Jordan, 1998; p. 12).

In work environments where employees often move between job tasks, it is important to take Learnability into account. In this setting, the employees are often exposed to new tasks and applications where they are expected to learn quickly (Jordan, 1998). This can be achieved by implementing recognition-based cues which can remind the user of previously used functionalities and interactions.

3.4 CO-DESIGN

Co-design, as the Interaction Design Foundation (2023) describes it, is a collaborative process where a diverse group, experienced and inexperienced, work together to create solutions. It aims to collect all thoughts and insights from all perspectives in order to innovate and solve problems effectively. The method is specifically used to combine these diverse perspectives which most often leads to more inclusive, creative and user-centered outcomes. They also add that Co-design can take many forms, such as workshops, brainstorming or as an ongoing collaboration.

Incorporating co-design into the design process can play a crucial role and has been adopted by numerous companies to ensure the development of profitable and successful products and services (Shaukat, 2022). By involving customers in design and development, particularly during the idea generation phase, companies can significantly increase the number of designs and solutions considered for production. This is achieved by making customers active members of the design team. As Shaukat (2022) notes, co-design has a high

probability of replacing market research methods, such as focus groups, demonstrating its effectiveness in gathering similar data.



Figure 3.4: Co-design. Illustration: A. Sämfors

In contrast to the conventional design process, where the marketing team primarily interacts and seeks opportunities, introducing industrial design departments at later stages, co-design fosters seamless interaction between customers and designers (Shaukat, 2022). By facilitating and encouraging contact between these groups, designers can become comfortable working with individuals who may not have formal design training, ultimately benefiting the final product.

Co-design is most effective when the issue involves multiple stakeholders and complex problems. As the stakeholders are often directly affected by the outcomes of the solutions of the proposed solutions, they have strong interest in making their voice heard. This fosters innovation and encourages collaboration with the end-users, which results in creative and resilient solutions.

In this thesis, co-design was used to introduce new perspectives and ideas from individuals outside the immediate user group. This collaborative effort provided fresh insights that challenged assumptions and expanded the range of possible solutions. The input gathered during this phase was used to explore alternative approaches and contributed to the development of concepts that could be evaluated and iterated upon in later stages of the design process.

3.5 ERGONOMICS

Understanding ergonomics is essential for optimizing the usability. In this thesis, ergonomics is used for analyzing how design decisions influence performance, reduce errors and enhance efficiency in radar operation.

As Wikberg-Nilsson (2015) explain, ergonomics prioritizes adapting tasks to users' needs and abilities rather than expecting users to adapt to poorly designed tasks. The International Ergonomics Association (n.d.) defines ergonomics as "the scientific discipline concerned with the understanding of interactions among humans and other elements of a system, and the profession that applies theory, principles, data, and methods to design in order to optimize human well-being and overall system performance." Korhan (2023) further divides ergonomics into three interconnected domains: physical, cognitive and organizational. Physical ergonomics addresses the body's capabilities and limitations; cognitive ergonomics explores mental processes such as perception, memory and decision-making; and organizational ergonomics examines how workflows, team structures and social factors influence performance.

Although ergonomics spans all three domains, this thesis focuses on cognitive ergonomics. Decision-making under pressure occurs when users face stressors such as time constraints, information overload, complexity or uncertainty, which tend to impair performance, increase reliance on heuristics and elevate risk-taking behaviors (Phillips-Wren & Adya, 2020). These effects often lead to poorer outcomes compared to low-stress situations, making it essential to reduce any cognitive demands placed on users.

Managing cognitive workload is one of the most effective strategies for mitigating the negative impact of stress on decision quality. By limiting the number of alternatives, extra information and presenting options in clear, digestible chunks, designers can reduce users' mental workload and support faster, more accurate choices. This approach aligns with recommended practices from the Nielsen Norman Group (Whitenton, 2013), which recommend avoiding visual clutter, leveraging

existing mental models and guiding users through progressive disclosure of information.

3.6 DESIGN FOR ASSEMBLY

Since this project focuses on processes related to service and repair, the assembly and disassembly of products is central. Applying theory on how to design products or components for easier assembly and servicing is therefore essential.

Design for assembly (DFA) is a subcategory of Design for manufacturing and focuses on reducing the cost of assembly (Ulrich & Eppinger, 2019). Although assembly costs typically represent a fractional overall product and process cost, implementing DFA can yield significant benefits. By incorporating DFA principles during the design phase, engineers can reduce the number of parts required, manufacturing complexity, and improve ease of assembly (Ulrich & Eppinger, 2019).

Products that have the same number of parts can differ in assembly time. The three affecting factors are; the time to grasp, orient and insert a part. These challenges can be addressed through thoughtful part geometry and mechanical design solutions. Ulrich & Eppinger (2019) describes a few guiding design principles that improves assembly efficiency. For example, parts requiring specific orientation, such as screws, can be improved by adding chamfers to both screws and corresponding holes. This modification eliminates the required fine motor skill during insertion. Similarly, parts requiring securing operations, such as fastening to other components, can be made easier to assemble if they are automatically secured upon insertion (Ulrich & Eppinger, 2019).

04 METHOD & IMPLEMENTATION

4.1 Design Thinking
4.2 Project Planning
4.3 Phase 1 – Empathize
4.3.1 Literature Study
4.3.2 Stakeholder Analysis
4.3.3 User observation
4.3.4 Interview
4.4 Phase 2 – Define
4.4.1 Criticality Analysis
4.4.2 Customer Journey
4.4.3 Root Cause Analysis
4.4.4 Expert Interview
4.5 Phase 3 - Ideate
4.5.1 Sketching
4.5.2 Brainstorming
4.5.3 Benchmarking
4.5.4 Workshop
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4.6.1 Mock-up
4.6.2 Tool Evaluation
4.6.3 Functional Prototypes
4.7 Phase 5 - Test
4.7.1 Think Aloud
4.7.2 Pugh Matrix
4.7.3 Usability Test
4.8 Phase 6 – Deliver
4.9 Method Discussion

Photo: Saab

4 Method & Implementation

This chapter covers the process and scientific methodology used during the project. Each heading provides a basic scientific understanding of the topic, explains the motivation behind its use, and outlines its implementation within the project. The chapter concludes with a discussion regarding the methods used.

4.1 DESIGN THINKING

The aim of this project is to improve an already existing product by implementing smaller usability-focused solutions. These improvements primarily target service and maintenance tasks of the system. To support this goal, the chosen design process was Design Thinking, which offers a structured yet flexible framework for user-centered innovation. It specializes and focuses on empathy with users, iterative development, and early prototyping, which are all crucial for improving usability in complex systems.

Design thinking uses a user-centric approach, which implies a focus on understanding and developing solutions to the users' needs around the product (Interaction Design Foundation, u.d.). The process follows 3 stages; Understanding, Exploring and Materializing. Within these stages there are 6 sub-phases; Empathize, Define, Ideate, Prototype, Test and Implement. The process is often not linear, it is meant to be iterative and cyclical, where you can jump in between the different stages, as seen in Figure 4.1. This allows for an increased flexibility which, in turn, increases the probability of success and innovational breakthroughs (Gibbons, 2016).

Empathize: This phase is all about understanding the users' needs and what can be improved with the product. The goal here is to reach out to a wide range of users and develop a deep knowledge of their needs.

Define: Use the gathered data from the Empathize-phase and highlight where there might be opportunity for innovation. Gather all opportunities and evaluate what challenge might be most beneficial to move forward with.

Ideate: Building on the improvement areas identified in earlier phases, this stage focused on generating creative solutions to enhance usability. Guided by co-design principles, the team explored ideas together with users to ensure relevance and practicality.

Prototype: Construct the concept found during the previous phase. The goal is to understand what components of the new idea work and which do not. Often this phase is very iterative, when something doesn't work, it is recommended to go back to the ideate-phase and change the idea based on the feedback you received.

Test: Throughout the process, multiple prototypes were tested with users to evaluate functionality, clarity, and overall usability. Early low-fidelity concepts were iteratively refined based on user feedback, which informed the continued development of more detailed solutions. Final concepts were also validated through testing to ensure alignment with user needs and expectations.

Deliver: While the traditional Design Thinking process concludes with "Implementation," our project's objectives differed. Instead of directly integrating the new design into the product, research, analysis, and the communication of our findings was prioritized. Therefore, the final

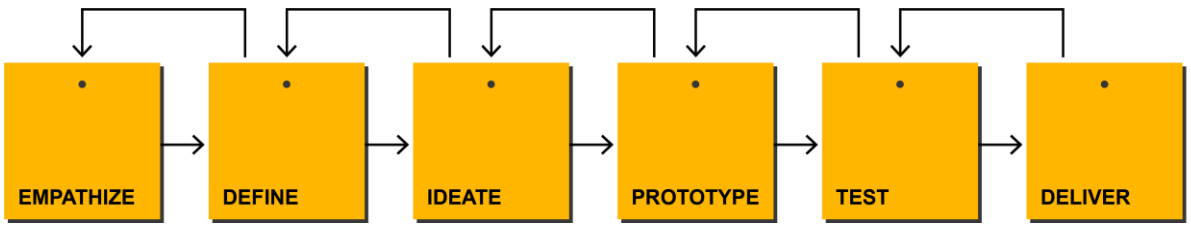


Figure 4.1: Design thinking process. Illustration: A. Sämfors

phase was renamed to "Deliver," reflecting our focus on creating a comprehensive presentation, a compelling showcase, and a detailed report. These deliverables aim to provide Saab with valuable insights and recommendations for future product improvement.

4.2 PROJECT PLANNING

Project planning is the first and one of the most important steps of a project. The purpose of this phase is to evaluate and plan the different stages of the project before initiating. This includes considering when certain milestones need to be met, which is done by exploring what methods can be used and approximating how long they will take to execute.

Thorough project planning ensures there's enough time for each phase, it eliminates uncertainty about deadlines and responsibility, and most importantly, establishes a clear goal (Coursera, 2024).

> "Failing to plan is planning to fail."
> (Rosala, 2024)

During this project it was decided to create a clear roadmap with each milestone and phase visualized using a Gantt chart, which can be seen in Figure 4.2. A Gantt chart is a horizontal bar chart used to visually represent a project over time (Teamgantt, u.d.). It was used as a tool to schedule the phases and when certain deadlines needed to be met. Each phase was represented by a horizontal bar and the deadlines by a vertical column.

The Gantt chart was deliberately structured into six main phases, each detailing the methods used and the scheduled completion deadlines for the respective subphases. These phases were chosen based on the Design Thinking methodology, and each is described in Chapter 4.1, Design Thinking.

Beneath the horizontal bars representing the primary project phases, deadlines for the written report were incorporated. This inclusion was essential due to the significance of the written report within the overall project, making it crucial to stick to a strict schedule. We then carefully reviewed the Gantt chart with the project supervisors to ensure it aligned with their goals of the project.

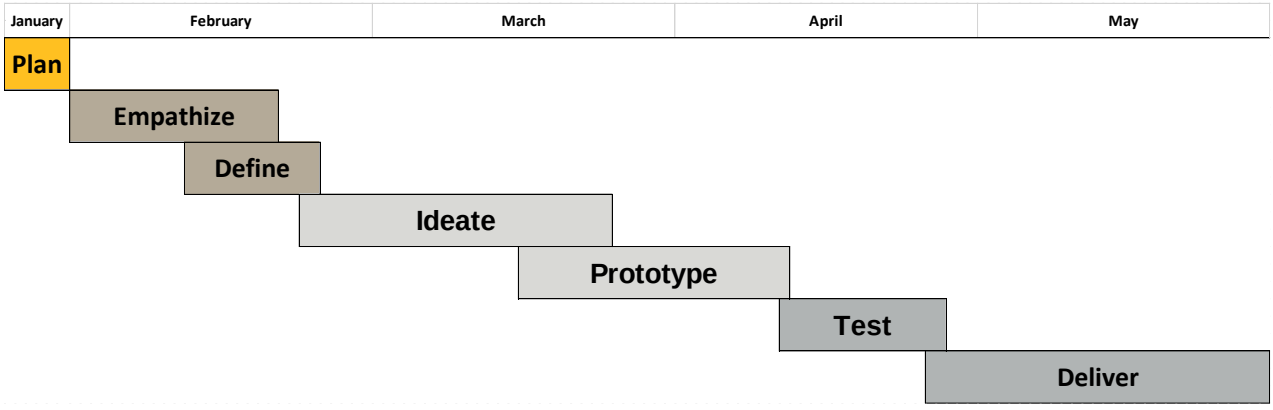


Figure 4.2: Down-scaled Gantt-scheme. Illustration: S. Tollesson

4.3 PHASE 1 – EMPATHIZE

The first phase focuses on learning about our users' needs and identifying areas for product improvement. We aim to gather insights from a diverse group of users to fully understand their challenges and pain points.

4.3.1 Literature Study

The method Literature study is, according to Snyder (2019), one of the best tools to provide answers. It can be useful when evaluating theories, collecting evidence or information, or examining the validity of certain concepts. This made it an appropriate method to use in the early stages of the project to help establish a strong theoretical foundation.

One of the more common and efficient ways to gather information is to search online, but this comes with certain challenges. The most common one is finding high-quality papers that are relevant to the project scope. This requires both a thorough evaluation of each source's validity and a careful selection of keywords. The terms used must cover the full scope of the topic while also narrowing down the results to a manageable number (Ulrich & Eppinger, 2019).

The information presented in the Theoretical Framework was gathered mainly through this method. Sources were accessed using the LTU digital library, the Directory of Open Access Books (DOAB), and Google Scholar.

To structure the literature study, we followed the five-step model advised by Karolinska Institutet (2022). This model was chosen due to its clarity and systematic structure, which made it easier to ensure methodological rigor throughout the process. The first step involves formulating research questions, which helps define the relevant theoretical areas to explore. The second step is to identify appropriate search terms to locate articles with relevant context. The third step includes performing a structured search, strategically adapting terms and applying them to multiple databases to ensure coverage and reliability. The fourth and fifth steps involve narrowing the search results and conducting a critical review of the selected material. This review focuses on evaluating quality and identifying possible limitations such as bias or outdated data (Karolinska Institutet, 2025).

The literature study was supported by three key sources. Two of these, Snyder (2019) and Ulrich & Eppinger (2019), helped clarify the nature and value of literature reviews, lending theoretical and academic legitimacy to the method. The third source, Karolinska Institutet (2022), was selected for its practical step-by-step structure, which guided the actual implementation of the review.

This combination of conceptual and procedural sources contributed to a robust and reliable literature study. The method was carried out with care, and the steps were followed according to guidance from established academic institutions. This provided confidence in the quality and relevance of the theoretical insights used in the project.

Looking forward, one potential area for improvement would be to further refine the keyword strategy and search filtering techniques. Although effective, additional testing with alternate terms or filters could enhance precision and breadth, especially in highly specific technical areas. This could be particularly valuable in collaborative projects involving sensitive or specialized domains, such as our work with Saab, where relevant academic literature may not always use the same terminology as industry practice.

4.3.2 Stakeholder Analysis

Effectively understanding who the user and key actors are of a product is crucial to identifying their interests and knowledge. This is incredibly important when developing or improving a product with the user in focus. Without knowing the user, it is not possible to make valid assumptions on what the customer desires.

Conducting a stakeholder analysis, as described by both Kennon et al. (2009) and Martins (2024), is done with four steps: Identify the stakeholders, Prioritize the parties, Understand their needs, and Develop a plan and communication.

During this project two stakeholder analyses were done, one to identify the stakeholders for the thesis outcome and one specific for the Giraffe 1X service stakeholders. The thesis stakeholders were found by performing a

brainstorm and the found parties were then placed in an Interest-Influence matrix. Their needs were identified and communication plans were established to ensure no information would be left out. The same process was followed when discovering the Giraffe 1X service stakeholders, but instead a brainstorm, expert interviews with relevant personnel within the field were conducted to identify its stakeholders.



Figure 4.3: Stakeholder Depiction. Illustration: A. Sämfors

The presented execution steps are quite straightforward. The stakeholder analysis was conducted according to Kennon, Howden, & Hartley (2009). The source's reliability is strengthened by its publication in an academic journal and its recency (2015). This method's execution, as described, makes its usage and purpose academically justifiable. However, to further bolster its legitimacy, incorporating an additional source would be beneficial.

4.3.3 User Observations

Understanding and empathizing with users is crucial when designing a system. User observation is a valuable method for achieving this. The focus is observing the user and often involves listening to, questioning, and experiencing user situations to identify their needs and areas of interest for development. Through observation, important aspects and relationships become apparent, providing valuable insights into the system (Wikberg-Nilsson, Ericson, & Törlind, 2015).

User observations can be conducted in various ways, and IxDF (2021) highlights two primary types of user observation: controlled and naturalistic.

Controlled observation focuses on specific tasks and is highly reproducible, allowing for easy data comparison. However, it is susceptible to the Hawthorne Effect, where users alter their behavior due to being observed (IxDF, 2016). Naturalistic observation, in contrast, takes place in a less controlled environment, allowing researchers to observe users during their typical daily activities. This approach yields richer qualitative data but can be more challenging to replicate and compare (IxDF, 2021).

Overall, user observation is a valuable method for understanding how a product is actually used and complements interviews effectively. If something is noteworthy during the observation, it should be written down and questioned through a structured interview after the observation is completed. This allows for deeper insights into user motivations and actions without disturbing the observation (IxDF, 2021).



Figure 4.4: Observed maintaining positions. Illustration: A. Sämfors

The observations conducted during this part of the thesis was aimed to give an understanding on how new users experienced the system. The information about experienced users would later be gathered with the help of interviews and were therefore not the primary focus with this method. Furthermore, the procedure of each task had already been researched before the observation to ensure full focus was held on the

participant instead of understanding the task. To find our targeted users, a trainer of the Giraffe 1X system was contacted and helped arrange visits during some lectures.

The observation-opportunity offered had a focus on the Giraffe 1X but was partly for a separate system outside of the project team's prior experience. Although only one opportunity was granted, the sessions included all necessary operations to evaluate the complete radar unit. The observation lasted two days and included three observable tasks that would be relevant to the thesis and two tasks that were outside of the scope. The tasks relevant to this project were the primary focus and described below:

Table 4.1: Steps of changing turntable gasket.

Changing the gasket on the neck of the Giraffe 1X	
1.	Unscrew gasketcover screws
2.	Remove gasketcovers
3.	Remove gasket
4.	Cut a new gasket
5.	Glue new gasket back on
6.	Put gasketcover back
7.	Fasten cover screws

Table 4.2: Steps of removing and replace TRMs.

Remove and replace the TRMs	
1.	Turn off antenna electronics
2.	Tie antenna using straps
3.	Unscrew screws to back cover
4.	Open back cover
5.	Unscrew TRM screws
6.	Remove TRM
7.	Replace with new TRM
8.	Fasten TRM screws again
9.	Close back cover
10.	Fasten back cover screws

Table 4.3: Steps of Rotary Joint removal.

Remove the Rotary joint	
1.	Turn off antenna electronics
2.	Tie antenna using straps
3.	Unscrew screws to back cover
4.	Open back cover
5.	Unscrew the CH screws
6.	Open CH
7.	Unscrew Radio Frequency Generator (RFG)
8.	Remove RFG
9.	Unscrew TCU screws and washers
10.	Remove TCU cover
11.	Disconnect cables from TCU
12.	Remove TCU
13.	Unscrew Angle and Rotation detector
14.	Remove Angle and Rotation detector
15.	Disconnect cables from the top of the Rotary joint
16.	Unscrew cables and screws from the bottom of the Rotary joint
17.	Remove Rotary Joint

The participants were split into two separate groups by the trainers, which proved advantageous. This strategy not only enabled the completion of more tasks than initially anticipated, within the limited timeframe, but also allowed the observers to split and focus on a single group each. This also allowed for a direct comparison of the groups' results, revealing their distinct approaches.

The observations were conducted as a combination of both Naturalistic and Controlled observations. The participants were allowed to follow the manual to achieve their tasked goals, but if they needed guidance the teacher would step in and help.

Information was gathered and noted using pen and paper, and the if any clarification questions were needed, they were asked to the teacher without disturbing the participants. Follow-up questions to the participants were saved for the end of the day to fully understand their thoughts and perspectives on specific segments of the tasks.

4.3.4 Interviews

After conducting the user observations, we identified several areas of improvement in how the Giraffe 1X system is used during training and maintenance. However, since these observations were made while the operators were still learning the system, we realized that the results might not fully represent how the system is used in real situations. Because of this, we saw the need to gather more input from other stakeholders involved in the project, such as operators, users, product experts, and trainers.

The goal of the interviews was to gain deeper understanding, confirm what we saw during the observations, and collect new perspectives. This allowed us to build a stronger foundation for the project by combining different types of insights.

Interview is a method of gathering information (Wikberg-Nilsson, 2015). The method is made up of a series of questions asked to users or stakeholders to figure out issues, opinions and behaviors in a product (Wikberg-Nilsson, 2015). It can be used both as a method to gather information, as well as to evaluate a concept of a future product. Jordan (2000), as well as Wikberg-Nilsson (2015), states that there are three different types of interview formats; structured, semi-structured and unstructured.

A structured interview uses predetermined questions and gives the participant a couple of options to choose as their answer, for example a numeric scale between one and five. This can be useful when quantitative data is required, but it can be difficult to predetermine the responses. This requires the interviewer to have a fairly clear idea of what the issues are to be able to ask precise questions and get the wanted information (Jordan, 1998).

Semi-structured interviews are similar to structured ones, but lets the participant answer freely and leaves more room for the interviewer to ask questions about relevant issues they find interesting. This format can ensure the respondent to answer a central set of questions while leaving room for evaluation if needed (Jordan, 1998).

Lastly, the unstructured interviews are an open format where the interviewer will ask open-ended questions which lets the participant to steer toward the challenges which they regard as important (Jordan, 1998). The unstructured format can be useful to collect data about topics where little information is known since before.

The interviews realized during this thesis were planned by following Wikberg-Nilsson's (2015) 10 step guide. In summary the guide consisted of creating a set of questions relevant to the topic, completing a pilot interview, finding appropriate participants and carry out the interviews with certain steps. If the guide and interview steps were followed correctly, the participant would be comfortable with answering the questions and appropriate information would be presented to the interviewer.

This project utilized multiple interview formats, as a single approach wasn't suitable for all situations.

Initial interviews:

The initial interviews were performed with the goal of gathering a fundamental understanding of the Giraffe 1X system. This was done very early in the empathize-phase to be able to better understand the needs and challenges that would later be investigated during the research. A small number of qualitative interviews with product experts were conducted to ensure the quality and validity of the data. To reach full explanatory potential, a physical Giraffe 1X antenna unit was available for the participants to use, and show where the parts mentioned were located.

The interviews were held in an unstructured format to ensure any question could be asked and answered without disturbing the flow of the interview. This also allowed the participant to decide how much information was necessary for each component.

The interviews covered topics such as, how the radar was assembled, why the parts were built and placed a certain way, and the names and importance of each one. There were three of these kinds of interviews performed which was deemed enough, as additional information was gathered alongside the interviews with the help of the instruction manual.

User interviews:
The main portion of the interviews were aimed to gather information about needs and challenges when maintaining and repairing the radar. From the newfound knowledge about the radar found from the initial interviews, we created a script adjusted for semi-structured interviews. The semi-structured type was chosen to allow for further explanation when needed, but the backbone of the interviews was kept the same to make the comparison between participants easier.

The structure of the interview consisted of an introduction of what the interview will be used for and what information is aimed to gather, followed by if the participant wished to be anonymous or not. The interview could then begin and with the questions regarding the following topics, the complete document can be found in Appendix B:

Table 4.4: Main topics of interviews

1.	Understanding the participant, their task and their experience with the system.
2.	What areas of improvement have they noticed? Do they know the reason why it exists?
3.	Do components often break by misuse? If so, why and how?
4.	Are there any parts that are more system-critical than others?
5.	What tools are currently used for maintenance?
6.	What change would you prefer to see in the antenna?

As Wikberg-Nilsson recommended in her 10-step guide, a pilot interview was performed to ensure the questions were correctly formulated and gave expected answers. Participants were then gathered with the help of the results from the Stakeholder analysis from chapter 4.3.2. The interviews were aimed to use experienced users of the system as participants, this could range from product expert to integration and verification personnel. This group was primarily chosen because they had the experience and knowledge of the system we needed, but also because they were the most easily accessible, which is of huge importance when the available users are already scarce. In total seven interviews were done with the targeted group.

With all the input gathered during the Empathize phase, we now felt ready to move on to the next step of the process.

4.4 PHASE 2 – DEFINE

The mission of this phase was to identify what areas of improvement should be researched and focused on. This was done by structuring and prioritizing the potential improvements found during the empathize-phase which were then evaluated with the help of our supervisors and stakeholders.

4.4.1 Criticality Analysis

The list of identified improvements resulted in several independent challenges. To best support Saab, researching and categorizing the most important challenge is crucial. Additionally, sorting the events to reveal the most common types would be valuable, as it might allow for a unified solution approach. Therefore, a criticality analysis was conducted.

Oguz (2018) argues that integrating human factors into the design process is crucial for creating resilient systems. Oguz's method expands upon standard Failure Mode, Effects and Criticality Analysis (FMECA) by incorporating human factors related to accessibility, repairability, and the probability and degree of human contact. He notes that the basic procedure for conducting a Criticality Analysis can be found in the Military Standard (MIL-STD-1629A).

Oguz's (2019) method consists of six steps. The first step involves selecting the system and defining the analysis depth. The second step requires gathering information about the system, including functionality, failure modes, and assembly logic. The third step focuses on collecting data about environmental conditions such as vibration and temperature. The fourth step involves assessing potential failures, their consequences, and their probabilities. Step five entails evaluating the impact of changes and modifications on other system functionalities. Finally, step six determines the probability of system failure considering human factors.

The chosen system to analyze is the Giraffe 1X and the analysis is concerned with usability challenges during maintenance. The data intended for step two and three was collected through the interviews and observations. Step four, which involved mapping potential failures and their consequences, also drew from those. The system of categorization can be found in Figure 4.5. Probability assessments were not possible to do the recommended way. During this stage, data revealing most frequently failing parts was not obtainable. Consequently, a category was created, tracking the number of times each event was mentioned. This was used as a substitute to mirror reality. Due to the lack of developed changes and concepts, steps five and six were excluded from the applied method.

Table 4.5 displays the consequences or severity of an event on the radar system's functionality. The Military Standard defines severity, while Oguz leaves it undefined. Therefore, a new definition focused on usability was created for this project.

Table 4.5: Severity levels

Description	Category
Critical	1
Marginal	2
Negligible	3
None	0

Table 4.6 indicates how often a failing event occurs, based on the times mentioned. "A" represents frequent occurrence, while "E" represents extremely unlikely occurrence.

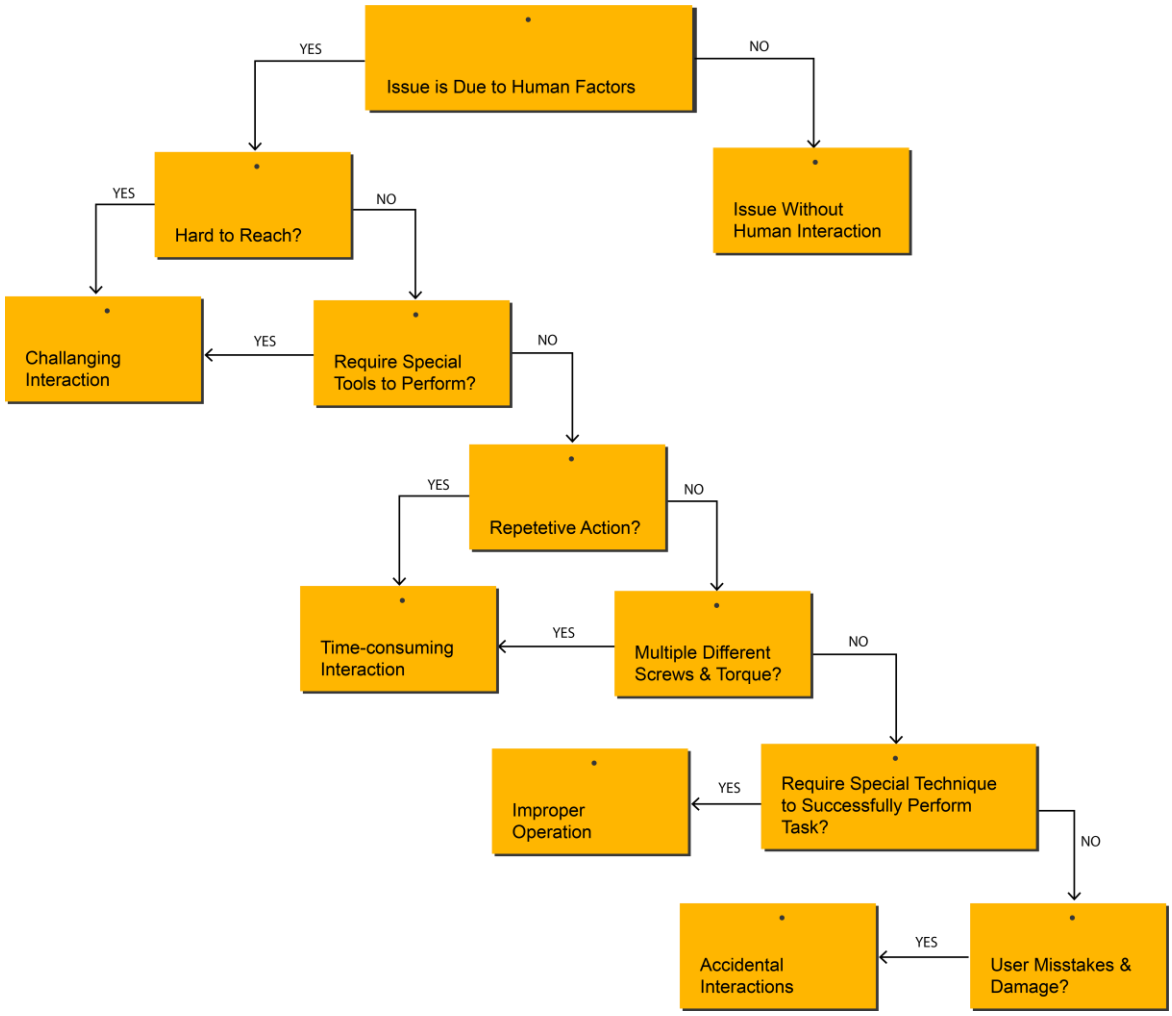


Figure 4.5: Event sorting process of criticality analysis. Illustration: A. Sämfors

Table 4.6: Probability of Failure Occurrence Levels

Level	Frequancy	Times mentioned
A	Frequent	5
B	Reasonable probable	4
C	Occasional	3
D	Remote	2
E	Extremely Unlikely	1

The classification system represented in table 4.7 describes how exposed a failing part is during an event. The more accessible a component, the fewer tasks are required to reach it.

Table 4.7: Accessibility Degree Classification

Accessibility Degree	Accessbility
1	Completely accessible
2	Accessible with minor difficulties
3	Accessible with major difficulties
4	Completely inaccessible

Combining these attributes resulted in two distinct tables. The first table combines FMECA and HDA, focusing on the failure mode. The second table focuses on the human factor and the type of damage caused by the event.

The FMECA method, as described by Oguz (2017), was developed with the U.S. Army and implemented by NASA during the Apollo missions. This history suggests its effectiveness in creating resilient designs. The fact that it remains part of the military standard further supports its legitimacy.

However, the definitions of catastrophic, critical, marginal, and negligible severity levels in the military standard method differ from those in this project. This discrepancy could lead to misapplication and potentially diminish the method's effectiveness. While FMECA is still used in engineering and human factors analysis, adapting the definitions is crucial for relevance and precision in this project. Although the differing definitions are acknowledged, they are not considered significantly detrimental to the overall results of the method.

4.4.2 Customer Journey

During observations, the user expressed emotions while performing certain tasks. To visualize and record these emotions and their level of satisfaction, a customer journey map was created. Satisfaction is crucial because it is a pillar of usability according to 3.4 Usability.

The goal of a customer journey is to map the experienced stages of use when interacting with a product, service or system. The result of the process is a visualized map with distinct key points where there is opportunity for improvement (Wikberg-Nilsson et al, 2015). According to Rosenbaum et al. 2017, the customer journey mapping is praised for its usefulness in understanding an organization's customer experience. The applied use of the method aligns with the six-step process of Wikberg-Nilsson (2015):

According to Wikberg-Nilsson (2015), the first step involves defining the target users for analysis. The second step requires gathering information, either through personal product or service use, observation, or user interviews. Steps three and four focus on constructing a matrix with a horizontal time axis and a vertical axis defined by experience-related questions. The specific questions depend on the project type and evaluation focus. Step five implies mapping the user's touchpoints, including background processes and attributes used to complete tasks. Finally, step six involves describing the journey chronologically through the time axis, using symbols to illustrate the user's experiences.

The Customer journey was mapped by selecting a component and analyzing the associated service process. Based on prior observations and interviews, the necessary steps to successfully complete the process is already know and documented. The required data implementation, such as emotions and experience, has also been collected mainly during the observations. In this case, the order of components interacted with will reflect the analyzed steps or key points. The evaluation categories include what the user interacts with, the actions they need to take, what they say, what they see or hear, the emotions they experience, the advantages and disadvantages of

the process, and potential areas for improvement. This method allows for mapping and illustrating user emotions and experience using a Satisfaction-Time matrix. As described by Wikberg-Nilsson (2015), a graph can then be created to visually represent the key points within the customer journey.

Both Wikberg-Nilsson (2015) and Rosenbaum (2017) highlight the customer journey as a valuable and effective method for understanding users, although they offer slightly different approaches and applications. While Wikberg-Nilsson's method excels at understanding users and informing concept and design development, Rosenbaum presents it more as a strategic management and marketing tool applicable on a broader scale.

Despite these differences, both authors recommend creating a vertical axis with columns representing aspects like emotions, thoughts, and beliefs. However, Wikberg-Nilsson (2015) emphasizes a purely time-based horizontal axis, whereas Rosenbaum describes it as "before, during, and after," which also relates to time but with a slightly different focus.

This application follows Wikberg-Nilsson's (2015) method, excluding touchpoints before and after, as Rosenbaum (2017) suggests. This discrepancy is not considered a significant flaw, as the core principles of the method remain consistent. Both authors emphasize the importance of time on the horizontal axis.

4.4.3 Root Cause Analysis

The purpose of Root Cause Analysis (RCA) is to identify the fundamental causes underlying observed problems, rather than merely addressing their visible symptoms. This ensures that any proposed solutions target the root cause and not just surface-level effects. RCA can be carried out using various tools and techniques, depending on the nature and complexity of the problem (Tableau, n.d.).

During the early stages of the project, it was considered essential to analyze the initially identified improvement areas in order to verify that no critical underlying causes had been overlooked. To support this, the 5 Whys technique was selected as the primary RCA method. This method was chosen for its simplicity, effectiveness in uncovering causal chains, and the team's existing familiarity with its application.

The 5 Whys technique is an iterative, qualitative method in which a problem is examined by repeatedly asking the question "Why?" at each level of causality until a root cause is identified. While five iterations are common, the actual number may vary depending on the complexity of the challenge. The process begins with a clearly defined problem statement and proceeds through sequential questioning to uncover deeper layers of causation.

Successful use of this method requires a solid understanding of the system under investigation to ensure that the reasoning is grounded in factual knowledge rather than assumptions. To support this, the method was conducted in structured brainstorming sessions involving team members familiar with the system's architecture and operational context.

The considerations selected for analysis were those identified during the Empathize phase, through stakeholder interviews, user observations, and internal assessments. These were organized by subsystem or component, and each was analyzed individually using the 5 Whys approach. The method followed the process outlined by Tonevås (n.d.), ensuring a structured and consistent approach across all problem areas.

Each RCA session was manually documented to ensure traceability of the reasoning process. To further support the method's validity and reduce the risk of inaccurate conclusions, the identified root causes were later reviewed in expert interviews (see Section 4.4.4). This additional step was included to confirm the plausibility and relevance of the findings before proceeding to the ideation phase.

4.4.4 Expert Interview

Expert interviews were conducted to evaluate and validate the outcomes of the Root Cause Analysis (RCA) and to ensure alignment between theoretical problem identification and practical feasibility. In this context, an expert is defined as an individual who holds specialized

knowledge of the system or product, and who has been directly involved in its development, implementation, or maintenance (Döringer, 2020). Such individuals are capable of providing insights that go beyond the information accessible through quantitative methods or publicly available documentation.

Expert interviews are recognized for their value in qualitative research, particularly when detailed, context-specific insights are needed (Windmill, 2022). They are often used to complement earlier research phases by providing validation, clarification, or deeper understanding of findings. Windmill (2022) recommends conducting expert interviews either at the beginning of a research project to inform its direction, or at the end to evaluate and contextualize its findings. In this study, the method was applied toward the latter purpose.

The interview format followed a semi-structured approach, which balances consistency across interviews with the flexibility to explore relevant details through follow-up questions. This approach is supported by Döringer (2020) and von Soest (2023), who emphasize that semi-structured expert interviews are particularly suitable for exploring complex, system-dependent topics. The interview guide was developed using the principles of the problem-centered interview outlined by Witzel (2000), incorporating open-ended questions to elicit in-depth responses.

The interviews were conducted with two product experts who possessed extensive knowledge of the radar system and its components. The interview questions were primarily based on the root causes identified during the RCA. These were used as a foundation for discussion, and additional questions were asked to clarify uncertainties and explore alternative perspectives. The experts were selected based on their involvement in the product's lifecycle and their ability to provide both technical and contextual insights.

All interviews were conducted in a professional setting and documented for traceability. Their purpose was to evaluate the plausibility, relevance, and practicality of the theoretical root causes identified earlier. This step helped ensure that the subsequent stages of development would be based on validated, actionable improvement areas.

4.5 PHASE 3 - IDEATE

This phase outlines the methodology employed in the creative process of this thesis. This methodology aimed to generate a wide range of ideas addressing the usability of multiple radar components. It encompassed analyzing current solutions on the market, identifying their strengths and weaknesses, and collaborating with external designers to leverage diverse perspectives.

4.5.1 Sketching

Sketching is a distinctive form of drawing which designers use to propose, explore, refine, and communicate ideas (Rojas, 2024). A sketch is often a good place to start when ideating as it has no requirements other than having a pen and paper available. In comparison to prototyping, sketches can be produced in abundance, costs very little and can be made very quickly without any specials tool or materials needed. With that said, sketches and prototypes complement each other and neither should be left out when coming up with the design as they are equally important but in different stages of the ideation (Rojas, 2024).

When drawing it is decided by the illustrator how detailed the sketch should be, it can be a simple sketch of the general shape or it can show every screw and every detail, which gives the designer freedom to explore any aspect of the object. The details can show functions or certain shapes or forms, and Rojas (2024) recommends it should be communicated using elements such as annotations, arrows or notes. These elements should complement the illustration without disturbing the design. They can be used to provide detailed explanation of the action required, how things move, or explain why certain design choices were made.

The methodology of sketching doesn't follow any certain steps and doesn't require any previous experience, but there are techniques that can be learned to more clearly visualize what you want to communicate. For example, Rojas (2024) mentions scribble sketching as a technique to capture the essence of an object while leaving out most details and decorations

to produce fast and easy illustrations. Other techniques include sketching in two or three dimensions or using different line weights, these can be used to give perspective to an object and to highlight the focused area of the object. These techniques are not scientifically researched but rather a suggestion to how it is possible to improve the visual communication of sketches.

The use of sketching and these techniques were mainly implemented during the brainstorming and workshop methodologies, but also whenever an idea needed further visual explanation. It was also used to illustrate the final concept which could then, with the help of prototyping, be crafted into its actual shape and size to then be further evaluated. As sketching was used in multiple other methodologies and chapter, the resulting illustrations will be shown in each individual result chapters.

4.5.2 Brainstorming

Following the Define phase, where key areas of improvement were identified and prioritized based on expert input and project goals, the next step involved generating potential solutions to address these challenges. The brainstorming method was selected as a suitable approach for ideation due to its ability to encourage creative thinking and produce a wide range of ideas, providing a foundation for subsequent concept development.

According to Besant (2016), the origins of brainstorming can be traced back to Alex Osborn in 1953, making it one of the most popular ideation processes. The process was initially designed to counter negative thinking. However, the term "brainstorming" has become widely used in the context of creative idea generation, diverging from Osborn's original definition. According to Osborn (1954), four mental powers are crucial: absorption, retention, reasoning, and creative thinking. While many approaches to problem-solving focus on learning, memorization, and the absorption of knowledge, Osborn emphasized the importance of creative thinking and reasoning. His process aimed not merely to support learning, but to foster innovative problem-solving through active ideation:

> **"We need to think up plenty of tentative ideas; because, in ideation, quantity helps breed quality"**
> (Osborn, 1954)

Both Wikberg-Nilsson (2015) and Osborn (1954) outline similar step-by-step methods for creative problem-solving. The first step is Orientation, which involves discovering and highlighting the problem. The second step is Preparation, focusing on gathering background data related to the problem. The third step is Analysis, where relevant and valuable information is broken down and sorted for the participants. The fourth step is Ideation, during which new ideas are generated and recorded. The fifth step is Incubation, a period of relaxation aimed at inviting new insights. The next step involves Synthesizing the ideas, putting the pieces together to make sense of them. Finally, the last step is Verifying the solutions. These stages, originally described by Osborn, align closely with the nine-step guide presented by Wikberg-Nilsson.

The brainstorming sessions followed the methods of Wikberg-Nilsson (2015) and Osborn (1954). First, we collected data on recorded challenges and recommendations from expert supervisors regarding areas of interest for further development. The experts specifically recommended components that would benefit from this brainstorming method. The data consisted of qualitative information from interviews and observations, as well as quantitative data extracted from those sources. The method was then applied through dedicated sessions for each component, aiming to find solutions for all areas of improvement, ideally identifying applicable solutions.

Sessions were divided by component and its related challenges; each was given fifteen minutes of brainstorming time. Ten-minute breaks followed each session. After the final session, all generated ideas were read aloud. The results were then considered based on factors such as cost, ease of implementation, and relevance for future systems. The sketches created during brainstorming were used to quickly visualize written ideas and to communicate concepts when presented to product experts. These sketches focused on the

core idea without detailing materials or unnecessary parts.

While the method was based on established and reliable sources, and the implementation followed these carefully, several limitations should be noted. The brainstorming was performed by only myself and a coworker, resulting in a very small group. This limited the diversity of perspectives and the potential for in-depth discussion that larger groups often provide. This constraint was largely due to the confidentiality requirements of Saab and the difficulty in involving personnel outside the immediate project team, which also reflects challenges inherent in applying co-design principles in restricted environments.

Despite the small group, the sessions benefited from a relaxed and trusting atmosphere, allowing us to share ideas openly without hesitation or fear of judgment. This comfort level fostered creativity, even if the breadth of ideas was somewhat limited.

Additionally, the brainstorming process was influenced by the supervisors' early input, who shared their thoughts on areas needing improvement. While this guidance helped focus the sessions and align ideas with project goals, it may have constrained the exploration of more diverse or unconventional solutions.

Although the final concept selected for further development centered on the toolkit, many other ideas generated during brainstorming were valued by the product experts. These ideas provide valuable inspiration for potential future improvements beyond the immediate scope of this thesis.

4.5.3 Benchmarking

The brainstorming session and supervisor evaluations revealed an interest in further developing an action toolkit. Naturally, examining how successful competitors construct their kits, their functions, and detailed designs would be a great source of inspiration, leading to the creation of a benchmark.

Benchmarking is the process of identifying the highest standards of products or processes, establishing the ground for innovative breakthroughs (Bhutta & Huq, 1999). By analyzing and researching competing systems, the development of solving a problem can take inspiration of already existing systems, and therefore focus on further improvement of the solution. There are different kinds of benchmarking: Performance, competitive, and generic to mention a few. The method most relevant to this project would be the functional benchmarking, this focuses on comparing technologies and solutions on an area with the purpose of developing the best possible solution. The procedure of benchmarking is, according to Bhutta & Huq (1999), the following steps:

Firstly, plan the study. This involves deciding what should be benchmarked, ensuring alignment with company interests. The second step is to form the benchmarking team. This entails deciding on team members, defining roles and responsibilities, and developing the execution plan. The third step is identifying potential competitors or role models within the relevant field. The fourth step is collecting and analyzing information. This involves not only gathering data but also analyzing and translating it into a comparable format. Step five focuses on adapting and improving. This means recording and drawing inspiration from best practices. Bhutta & Huq (1999) recommend implementing a method to evaluate the best practices, depending on the specific benchmarking objectives.

The benchmark was conducted following the method of Bhutta & Huq (1999), with some slight modifications. Firstly, the decision was made to include as many toolkit functions as possible, encompassing everything from storage and pockets to handles, material structure, and closing mechanisms. Secondly, the research team consisted of the authors of this thesis. Thirdly, due to the limited availability of known competitors specializing exclusively in military toolkits, the scope was expanded to include interesting, well-constructed kits found online, rather than focusing solely on competitors. The gathered information consisted of images of toolkits in various sizes, shapes, and materials, presented visually rather than in the list or table format insinuated by Bhutta & Huq (1999). Finally, the benchmark served as a platform for discussing pros, cons and as an inspirational foundation for further concept development.

The benchmark used in this project differed slightly from the way Bhutta and Huq (1999) described its visualization. In their framework, benchmarking is intended as a structured method for evaluating and ranking differences between solutions in order to determine which options offer the best functionality.

Although the visualization of the benchmark in this thesis deviated from their original approach, its application remained consistent in a more abstract form. Instead of using a table to display which product includes which functionality, the benchmark here served primarily as a source of inspiration.

Since its purpose was limited to informing the ideation process rather than producing comparative data, the deviation from the academic standard does not compromise the legitimacy of the results.

4.5.4 Workshop

A workshop is used to utilize the creative ability of the group in order to explore problems, solutions or scenarios (Wikberg-Nilsson, Ericson, & Törlind, 2015). Cirulis & Chang (2022) agrees with Wikberg-Nilsson by saying, "When we want to discuss problems, imagine new ideas, and brainstorm solutions, it's best to bring people together, whether in person or virtually." The method effectively enables participatory design, or in other words co-design, which is used to harness the insights and collective wisdom of anyone and everyone involved.

Westerlund (2007) argues for a workshop to fully utilize the participants' insight; he encourages participants to say, do and make. The reason for this is a theory that there's a difference in what people say and what they do. With the help of methods to make the participant express their thoughts in text, verbally, and by creating, the resulting suggestions can be expected to have much higher standards and understandability. By using these participatory design methods, diverse opinions can be united and developed into even more refined and relevant concepts (Westerlund, 2007).

This workshop followed a benchmarking phase, where we explored existing solutions and products relevant for the tool-kit. The benchmark served as a foundation for understanding the current market and usability trends. However, in order to go beyond what currently exists and gain fresh, unbiased perspectives, we initiated this workshop. The aim was to gather innovative and unconventional ideas that could challenge current assumptions and provide inspiration for the next phase: prototyping.

The workshop was planned using Cirulis & Chang's (2022) guide to creating a successful product design workshop. The guide included steps to clarify the goals of the workshop, identify participants, define the tangible outcomes, and decide how the workshop would be conducted. A document was created to note preparative thoughts and ideas, which can be found in Appendix F, "Workshop". The goals of the workshop were to gather new ideas by implementing co-design and to find new concept ideas for usability regarding:

Table 4.8: Main topics of workshop.

1.	How the tool bag can be attached
2.	How the tools can be arranged in the bag
3.	How the tools can be attached inside the bag.

To achieve the sought-after co-design, we invited five peers studying Industrial Design Engineering to participate. They had no previous background or knowledge of the work being done, which was intended to prevent overthinking and encourage fresh, unbiased ideas. In retrospect, this decision effectively supported creativity, although we later recognized that a brief introduction to typical operator tasks, tools, and procedures could have enhanced the relevance of some ideas without compromising the open-mindedness of the session.

Because of Saab's secrecy requirements, the scenario used during testing was adapted to resemble the intended context without disclosing sensitive information. During interviews with operators and trainers, it was explained that the radar system can be mounted in a variety of ways, including on masts or towers that are difficult to access. This would

sometimes require operators to climb and work in constrained positions to reach the radar unit.

Based on this insight, the adapted scenario involved an operator climbing a radio tower to repair electrical circuits, which was considered analogous in terms of required movements and challenges. This setup was particularly useful for validating requirements related to carrying the service kit hands-free and accessing tools with one hand without the risk of dropping them. However, it was later observed that this scenario influenced the participants to focus more on how the toolkit should be carried to the antenna, rather than the internal layout and usability of the tools themselves. For future iterations, choosing a more typical or routine scenario, such as working from the ground with the box placed on an elevated surface could better balance the workshop's focus and encourage discussion on internal arrangement as well as ergonomics.

The workshop was conducted online, using the meeting application Zoom and the online workspace Miro. During the sketching part of the workshop, participants used pen and paper, which they photographed and sent after the workshop. The schedule of the workshop was inspired by the guidelines of Cirulis & Chang (2022) and was conducted as follows:

The final part of the method involved collecting and organizing all the outcomes from the workshop. Keywords from the discussion were summarized, and all submitted sketches were gathered. These outputs were then reviewed to identify elements that could be implemented into previously developed concepts. The summary was presented to supervisors and product experts at Saab for further input and evaluation.

Table 4.9: Workshop script

Introduction	Welcomed the participants and showed the agenda.
Warm-up Exercises	Three quick brainstorms with easy topics, each lasting two minutes, to get the participants in the right mindset.
Inspiration	Showed benchmarked solutions to provide context about current market offerings and to stimulate participants' imagination.
Brainstorm	Two sessions were held, each with five minutes of brainstorming followed by five minutes of discussion, totaling twenty minutes. A break was held before sketching to prevent mental fatigue.
Sketching	Two sessions followed the same five-minute sketching and five-minute discussion format.
Conclusion	Participants were thanked, and questions around the newfound concepts were discussed.

The outcome of the workshop was a wide range of new inputs concerning the design of the toolkit. These insights helped us proceed to the next phase of the project prototyping where selected ideas from the workshop were further explored and tested. Despite the adapted scenario and limited context given to participants, the workshop successfully generated novel perspectives that extended beyond the benchmarked solutions and contributed directly to the concept development process.

4.6 PHASE 4 – PROTOTYPE

To develop different prototypes early in the design process is beneficial because the communicating and illustrative properties (Wikberg-Nilsson, 2015). According to Wikberg-Nilsson (2015), using prototypes can support product development in fields such as ideation, but also in understanding. Prototypes helps concretize what important changes to the design or how easy it is to use. According to Wikberg-Nilsson (2015), early prototypes include working with Lego, clay or cardboard.

Developing different prototypes early in the design process is beneficial due to their communicative and illustrative properties (Wikberg-Nilsson, 2015). Wikberg-Nilsson (2015) mention using prototypes can support product development in various fields, they also help concretize important design changes and demonstrate the ease of use. According to Wikberg-Nilsson (2015), materials commonly used for early prototypes include Lego, clay, cardboard and computer models. The prototypes developed in this chapter were made of paper or fabric.

4.6.1 Mock-up

The need to simulate fabric was crucial for understanding how the material would behave when rolled. At the same time, evaluating the size and layout of the tools, the space required to fit them, and how much space the kit would occupy when attached to a person, motivated the development and use of mock-ups.

According to Wikberg-Nilsson (2015), mocku-ups are fully scaled functional prototypes designed to evaluate the functional and ergonomically properties. These prototypes are beneficial because they allow subjects to test a concept's usability, which is central to this project.

According to Wikberg-Nilsson (2015), the process of developing and using a mockup involves the following steps: First, define the reason or functionality that is beneficial to evaluate. Second, develop ideas and sketch a concept. Third, build the mock-up using materials suited to the properties being tested. Fourth, test the functionalities. Finally, observe and evaluate the usage and functionality.

Idea development and concept sketching were carried out during the brainstorming session in the ideation phase. However, problem-solving related to specific details such as spacing, sizing, and compartmentalization was addressed in a separate brainstorming session. This session generated multiple potential solutions for each challenge. Through team discussions, design decisions were made for each detail, forming the basis for the next development step.

The construction of the mock-ups typically began with an ideation session focused on refining these detailed design elements. The mock-ups were constructed using paper and tape, and were briefly tested by attaching them to a user and observing their interaction with the prototype while lying down, sitting, and standing. In addition, 3D paper replicas of tools were created to evaluate how much space they would occupy next to each other, as well as to explore how they could be arranged during the folding and rolling of the kit.

The creation of the mockup aligned with the purpose of the method itself. However, the method is only supported by one source, Wikberg-Nilsson (2015). While the methods described in her book would be considered reliable due to her being a professor in the Industrial Design Engineering program, relying on a single source can be limiting.

The described execution steps are quite straightforward and lack detail. This could lead to varying interpretations and potentially unintended execution. However, given the method's strong dependence on chosen materials and concepts, a degree of flexibility in its description is necessary. Since the method is used solely for testing size and ergonomics, and doesn't constitute a major portion of the project's results, its usage in this project can be considered legitimate.

4.6.2 Tool Evaluation

The previous chapter had revealed a lacking knowledge of what tools were needed to perform the necessary repairs or actions by an operator. Therefore, it was decided to continue the research of what the most common tasks were for the users, what screws were being

handled, and the tools required to complete these actions. This was necessary to be able to evaluate different layouts of the tools and what the most user-friendly option would be.

To understand the tools needed for Giraffe 1X repairs and maintenance, it was decided to compile a list. This led to an initial literature study of the antenna manual, with the goal of finding the procedures performed by the operators and maintainers. These procedures gave information about the screws and their required torques, which were imported and sorted into an Excel-document.

From the Stakeholder Analysis performed in Phase 1 we had identified the most common tasks by each level of user. With this information and the fact that the kit was primarily aimed to satisfy the needs of the Level 1 Operator it was possible to identify what tools were most commonly used with the help of the Excel-document. To further evaluate the conclusions and assumptions drawn, it was decided to perform a couple of expert interviews which would provide valuable information regarding additional items needed to perform the procedures, other than the tools. The interviews led to a conceptual stage where multiple different options of tool combinations were created and then evaluated using our supervisors and additional interviews with the product experts. The concepts weighed aspects of user-friendliness and cognitive ergonomics in the sense of what the correct number of tools were needed, depending on if the torque wrenches should be fixed or not. The results ended up being a combination of fixed and adjustable torque screwdrivers, with some additional tools and items deemed necessary for uncommon, but possible scenarios. This allowed for continuation of the prototyping with further details as to what the layout could look like and how the tools should be divided in the kit.

4.6.3 Functional Prototypes

After identifying the necessary tools to include in the tool-kit, we proceeded to the prototyping phase. The purpose of creating real, tangible prototypes was to explore and evaluate several practical aspects of the design, such as the optimal layout for rolling up the kit, the appropriate stiffness of the fabric, the attachment mechanisms for tools, and general usability.

The decision to construct physical prototypes from fabric, rather than relying solely on computer models, required more time than initially expected. This was partly due to the initial assumption that computer-aided design software would be used, leading to a shorter planned prototyping phase. However, computer-aided design software is not inherently optimized for fabric materials, making it challenging to accurately represent their properties and dimensions. Additionally, crucial dimensional details and usability considerations often emerge only through physical construction, which justified the hands-on approach.

Constructing physical prototypes also enabled users and Saab experts to interact with tangible models, facilitating more detailed and insightful feedback. Experiencing the prototype's dimensions, weight, and handling firsthand allowed them to provide more nuanced evaluations, which might not have been possible through digital models alone.

Prototyping Process

The prototyping followed an iterative approach. The overall flow is illustrated in Figure 4.6 and began with planning the tool-kit, followed by rough sketches on paper. These sketches were reviewed to identify any obvious design flaws that could prevent functionality. If a concept seemed promising, we proceeded to create a fabric prototype.

Each prototype was evaluated and analyzed to identify design strengths, areas for improvement, and production challenges. Insights from each iteration informed the next, and the process was repeated until a satisfactory prototype was achieved. This approach enabled quick experimentation and refinement while focusing on one design question at a time.

Prototype Iterations

Several prototypes were developed, each exploring specific aspects of the design:

Table 4.10: Prototypes made in iteration process.

Black prototype	A first pilot prototype used to test the basic roll-up functionality and material stiffness.
Gray #1	An improved version with updated layout and new attachment methods. Introduced a separate pocket for uncommonly used tools.
Attachment test	A prototype focused specifically on tool attachments and the inclusion of a bit table.
Gray #2	A full-concept prototype integrating previous findings. Included revised attachment solutions, handle design, and closure mechanisms.
Final prototype	The final iteration featured updated spacing between tools to facilitate easier folding and rolling. A Velcro closure was added for the separate pocket to improve usability.

Each prototype contributed insights into both functional and structural aspects of the design, progressively moving us closer to the intended final result.

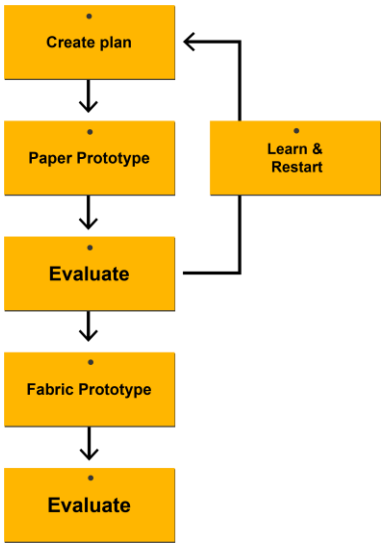


Figure 4.6: Iterative prototyping process. Illustration. A. Sämfors

The final prototype was considered to be of sufficient quality and completeness to proceed with user testing.

4.7 PHASE 5 - TEST

This chapter details two user tests conducted to evaluate the prototypes' functionality and usability. The first test focused on the effectiveness of kit's implemented symbols in conveying information. The second test assessed the final solution's usability and overall performance.

4.7.1 Think Aloud

According to Nielsen (2012), the Think Aloud methodology is a simple usability test which is robust, cheap, flexible, and easy to learn. Both Nielsen and Jordan (1998) describe Think Aloud as a method used to understand what a participant is thinking when using a system by encouraging the user to explain what they're doing while performing the task. The task being performed can either be planned beforehand or completely unstructured, both have their own advantages and can be useful in different cases. As explained by Jordan (1998), the unstructured format, also called free exploration format, involves giving the user a product and then asking them to do whatever they like with it, while the planned task uses predetermined steps to complete the set task. The planned format is helpful to uncover specific usability faults whilst the free exploration format can provide information about the product's intuitive use.

While performing the method with a user of the system, Nielsen (2012) insists it is important to let the participant use the system without interrupting. He further explains it is sometimes problematic since most participants forget to verbalize their thoughts, in which case it is allowed for the facilitator to prompt the users to keep them talking. Jordan (1998) agrees with this and encourages the facilitator to ask the participant questions, for example "What are you thinking now?" or "Why did you press that button?".

The Think Aloud method was used in this thesis to evaluate the use of symbols and how they were combined with corresponding tools and screws. The goal was to determine which of the concepts would perform best with regard to the usability measurements, such as: guessability, learnability, fail rate, and time-on-task.

The test, from the participant's perspective, consisted of the following: Get shown a row of screws with a tag consisting of a symbol, number or text next to it, then pick the corresponding tool that could be used for the screw, and finally state the bit-size and torque chosen for the screws. It was performed using the free exploration format, which meant the user was given a task and was then allowed to figure out how to solve it on their own. The only interaction with the participant was to either answer their questions or encourage them to explain their thoughts during the tests. The test order was varied between the participants to not introduce any biases, and only two of the concepts were tested on each one. All of the concepts were tested an equal number of times, varied equally in order, and the screws were tested equally from bottom-up or top-down.

The tests were performed on users that had previous experience within the antenna unit, most of them being considered product experts. In total, six tests were performed, excluding the first pilot test. We measured how many errors occurred, how long the participant spent on each screw, and what difficulties they had. The Think Aloud also ended with a semi-structured interview which asked questions such as what they found the most difficult, if there was anything they would like to see improved, and what concept they think could be implemented into the radar. This data was the most sought-after and therefore closely documented using a standardized Word document. When the participant had finished the tests, all of the gathered information was verified, summarized, and grouped to help ease the comparison between each participant.

While the Think Aloud method provided valuable insights into user reasoning, interpretation, and preferences, several methodological considerations should be acknowledged. The free exploration format was intentionally chosen to observe natural behavior and assess intuitive use, but the lack of task-specific guidance may have introduced inconsistencies in how participants approached each concept. Some users asked for clarification or relied on past experience, while others hesitated due to uncertainty. A more structured or hybrid format might have made it easier to compare results across participants.

The limited number of participants, while reasonable given the project's scope and timeframe, makes it difficult to draw broad conclusions. Although all participants were experienced within the antenna domain and offered highly relevant feedback, this also introduces a bias toward system-familiar perspectives. Including a wider range of participants in future studies would help evaluate how intuitive and accessible each concept is for users with less background knowledge.

Additionally, the controlled testing environment may have affected participant behavior compared to a more dynamic, real-world scenario. Printed visualizations and the absence of actual tools or tactile feedback made it harder to simulate the true assembly context. While this approach ensured consistent conditions, future tests in more realistic environments could provide deeper insights into usability.

Lastly, it is important to consider the influence of the facilitator, even if minimal. The presence of an observer and occasional prompts may have affected participants' responses. Some participants might have adjusted their behavior based on what they believed was expected, a phenomenon known as the Hawthorne effect. Although this is difficult to fully eliminate, being aware of it helps in interpreting the data.

Having gathered detailed insights from the Think Aloud tests regarding the strengths and weaknesses of each icon concept, the team now needed a structured way to compare them more systematically. While the qualitative data from user feedback was valuable, translating this into a clear decision required a more objective framework. To support the next design decisions and ensure alignment with project goals and stakeholder expectations, including those of Saab, a comparative evaluation of the concepts was the logical next step.

4.7.2 Pugh Matrix

The next step in the process was to evaluate which of the three icon concepts was the most promising. The evaluation considered multiple parameters, such as ease of understanding, cost, and aesthetics. To fairly and objectively compare the concepts, a Pugh matrix was implemented.

Guler & Petrisor (2021) propose an enhanced evaluation process using a Pugh matrix. They address a weakness of the traditional Pugh matrix by adding a step that enables logical weighting of criteria. This enhanced process begins by defining a list of design criteria and variables that could significantly influence the design process and the final product. This is typically done through research methods. Next, the strength of the relationships between each criterion is established, clarifying how design decisions in one criterion will affect others.

The criteria are then assigned a weight. This can be done through team discussions, expert consultations, or by using the method proposed by Guler & Petrisor (2021), which considers the total impact each criterion has on others. Relative weights are assessed on a scale of 1 to 3 (3 being highly essential, 2 being moderately essential, and 1 being less essential). Finally, existing products in the market are compiled, and reference design solutions that best meet each criterion are identified.

While this method followed the steps outlined by Guler & Petrisor (2021), there were some modifications. The design team generated a list of criteria proposals, which were then reviewed and refined with input from experts. The criteria were weighted based on team discussions, prioritizing aspects like usability (e.g., guessability, which received a weight of 3) over aesthetics (which received a weight of 1). Finally, one concept was selected as a reference, and the other concepts were evaluated in comparison to it. In this case, the concept using numbers served as the reference.

The method was largely implemented as intended. However, the objective rating system, as described by Guler & Petrisor (2021), could not be applied due to time constraints. Consequently, a standard weighted rating system was used instead. While this deviated from the original intention, the resulting evaluation can still be considered a valid Pugh matrix, as defined by Guler & Petrisor (2021).

It is important to note that the results may be influenced by the limited number of criteria used. Including additional criteria, such as "foolproofness," which supports the strengths of other systems, could lead to a more impartial evaluation.

4.7.3 Focus Group

With the toolkit concept well underway in its development, it became necessary to determine whether the entire solution is more usable than the current system for servicing the radar. Ultimately aiming to answer the research questions.

According to Morgan (2011), a focus group can be described as a form of group interview. However, unlike traditional interviews that center on the interaction between a researcher's question and a subject's response (Morgan, 2001), focus groups emphasize interaction and discussion among participants. In this setting, the researcher serves primarily as a moderator who guides the discussion around specific research questions. Morgan (2011) and Onwuegbuzie et al. (2009) highlight that such interaction generates unique qualitative data that other methods may not capture. Wikberg-Nilsson (2015) reinforces the method's value by noting that its strength lies in participants building on each other's contributions, making it especially effective for providing feedback on developed concepts.

According to Wikberg-Nilsson (2015), conducting a focus group begins with drafting a list of research questions, which should then be pilot-tested to identify potential communication issues. The next step involves inviting relevant participants, introducing the moderator and the purpose of the study, and allowing participants to introduce themselves. During the session, the moderator facilitates the discussion, ensuring everyone has a chance to contribute. All discussions should be thoroughly documented.

To evaluate this, a focus group consisting of experts in construction, sales, and training was conducted. One session for each department, with groups of 3-6 people. The focus group sessions examined the kit concept, discussing everything from pocket functionality and iconography to the potential integration into the Giraffe 1X system. The goal was to address the thesis' research questions, as well as to explore directions for future development. The questions included:

Table 4.11: Focus groups main areas of discussion.

1.	Is the kit intuitive? Do you think the additional pocket reduces the risk of selecting the wrong tools?
2.	Do the tools represent what you typically use in real-life scenarios?
3.	Are there any improvements you would suggest for the visual indicators?
4.	Are there other areas for improvement?
5.	What aspects do you believe require further exploration?
6.	Do you believe similar functionality could be implemented in the current Giraffe 1X / Giraffe 1X CRM system?
7.	Would the Giraffe 1X be easier to use if this kit were implemented?

The materials used included the final prototype design (as shown in Chapter 7), as well as an A3 paper representation of the Giraffe 1X antenna displaying how the icon system might be integrated. Since the participants were already familiar with each other and their respective fields, formal introductions were deemed unnecessary. All discussions were carefully documented.

Morgan (2011) argues that the reliability of this method is evidenced by its frequent use in academic research, cited in over 100 peer-reviewed articles annually. The consistency in how the method is described by all three sources suggests strong validity. However, none of these sources specifically address the use of a prototype or interface during the session. In this case, such visual materials may have deviated slightly from the traditional method but likely enhanced participant understanding. Ultimately, presenting both the kit and the interface likely improved the quality of the feedback compared to providing only a verbal explanation.

4.8 PHASE 6 – DELIVER

The final phase of the Design Thinking process, Deliver, involved concluding the project and presenting the outcomes in clear and engaging formats.

Report

The report compiled and summarized all aspects of the work completed during the project. It was developed continuously and finalized during the Deliver phase. It included various images, tables, and design choices, all intended to clarify the content and enhance the reader's understanding and experience.

Poster

To showcase our work at the exhibition at Luleå University of Technology, we created a poster. Its main purpose was to present the final results and briefly describe how they were achieved. The poster served as a visual summary and can be found in Appendix K, "Exhibiotion poster".

Delivering Solutions

As part of our collaboration with Saab, we delivered our key insights, results, and recommendations directly to the company. This included a summary of our findings, proposed solutions, and suggestions for potential next steps. Our intention was to ensure that the outcomes of the project could support Saab's ongoing development in a meaningful and constructive way.

These deliverables marked the conclusion of the thesis project and reflected the goals of the Deliver phase by effectively communicating our solution and its potential impact.

4.9 METHOD DISCUSSION

This section critically examines the validity and reliability of the methods used throughout the thesis. It analyzes the appropriateness of the selected approaches, acknowledges any deviations from standard procedures, and outlines potential refinements to enhance the defensibility of the results. The discussion is structured according to the six phases of the Design Thinking process: Empathize, Define, Ideate, Prototype, Test, and Deliver. Each phase is reviewed in light of the methods applied and the contextual limitations of the project.



Figure 4.7: Discussion. Illustration: S. Tollesson

Phase 1: Empathize

The Empathize phase employed literature studies, stakeholder analyses, user observations, and interviews. These methods were well-suited for establishing an initial understanding of user needs, system context, and stakeholder priorities. Interviews and observations, in particular, provided qualitative insights that enriched the problem framing.

However, more immersive methods such as shadowing users in their work environment could have yielded deeper contextual knowledge. Yet, given that the radar system is deployed in active military zones, such ethnographic approaches posed ethical and safety risks that rendered them infeasible. Within these boundaries, the chosen methods were both practical and informative.

Phase 2: Define

The Define phase utilized Criticality Analysis, Customer Journey Mapping, Root Cause Analysis, and Expert Interviews to frame the problem space and identify the most critical areas for development. These tools provided both structured and exploratory perspectives on system weaknesses and user pain points.

An alternative approach could have been to involve a broader cross-section of users or stakeholders through participatory co-creation workshops. This might have added further depth and diversity to the problem definition. However, the classified nature of the project and confidentiality policies at Saab limited external participation. Still, the use of iterative expert interviews helped validate the chosen direction.

Phase 3: Ideate

This phase featured a combination of sketching, brainstorming, benchmarking, and workshops. These methods encouraged creative thinking and grounded ideation in practical comparisons. Sketching and benchmarking, in particular, supported rapid idea generation and helped evaluate existing solutions.

The brainstorming sessions, however, were conducted solely by the thesis authors, which limited the range of input and discussion. Although the small group size was a constraint, it also created a psychologically safe environment that encouraged open and unfiltered idea generation. Saab's confidentiality requirements made wider involvement impractical. Furthermore, some degree of bias may have been introduced by prior supervisor input, which subtly guided the direction of certain discussions. Future efforts could benefit from additional structured creativity methods such as SCAMPER to stimulate diverse thinking within constrained groups.

Phase 4: Prototype

In the Prototype phase, methods such as mock-ups, functional prototypes, and tool evaluations enabled iterative testing of early concepts. These hands-on methods proved effective for exploring key usability aspects and gathering tangible feedback despite limited access to production resources.

Advanced digital prototyping, such as AR or VR simulations, was deemed unnecessary and impractical for this project due to its technical complexity and limited timeframe. However, such techniques could be reconsidered for future iterations where longer development cycles or additional resources are available.

Phase 5: Test

Evaluation was carried out using think-aloud sessions, a Pugh matrix, and focus groups. These methods provided a well-rounded mix of user-driven feedback and structured comparison. The think-aloud protocol captured user reactions in real time, while the Pugh matrix added objectivity to the evaluation of competing concepts.

While testing in actual operational settings was not viable, future research could explore testing in simulated military training environments. This would offer realistic conditions for assessing product usability and gathering contextual feedback. Biometric data or eye-tracking tools were considered unnecessary for the nature of this product and would likely offer limited additional insight.

Overall, the methods used were appropriate for the scope and limitations of the project. While there were inevitable trade-offs due to security constraints, access limitations, and time restrictions, the methodological framework was robust and well-aligned with the principles of the Design Thinking process. The sequence of methods ensured that each phase informed the next, enabling a structured yet flexible exploration of the problem space.

In future studies, increasing user involvement and integrating more iterative testing cycles could further strengthen the reliability and relevance of the results. Nonetheless, within the context of a highly specialized and sensitive domain, the methods applied in this thesis can be considered both valid and effective.

05 RESULTS

5.1 Results of Empathize
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5.1.2 Stakeholder Analysis
5.1.3 User Observation
5.1.4 Interview
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Photo:Saab

5 Results

This chapter showcases the project's outcomes, reporting findings from each method used. The results are presented in a phase-by-phase structure, with each phase acting as a header and the methods as subheadings. The chapter concludes with a comprehensive overview of the project's results and details the concepts functionalities.

5.1 RESULTS OF EMPATHIZE

The first phase was conducted as contextual research for understanding the radar and its potential usability improvements. The chapter will show the results from each of the methods used.

5.1.1 Literature Study

The literature study was done throughout the entire project with different focuses. The main area was the Theoretical framework closely followed by the research done for each method. The study gave insights to what methods would fit our work and was also used to strengthen the foundational theories the thesis was based upon.

5.1.2 Stakeholder Analysis

Two separate stakeholder analyses were done during the thesis. The first to identify the stakeholders of the thesis and the second one defining levels of maintainers.

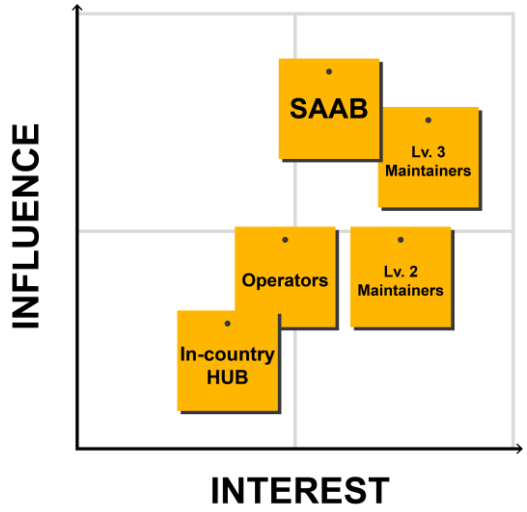


Figure 5.1: Influence-Interest of maintaining stakeholders. Illustration: A. Sämfors

The thesis stakeholders consisted of Saab, LTU, the supervisors, and the users and maintainers of the radar. Their levels of interest were identified and placed into an Interest-Influence matrix, which is shown in Figure 1.2. This analysis helped to clarify who the work would affect the most and who had high interest in the results. The biggest impact from the thesis would be on the engineers and maintainers, which meant they were very inclined on helping in any way they could. This was a great insight for gathering information about what participants would be appropriate for interviews and observations.

The maintainer analysis identified different levels of maintenance personnel based on the complexity of service tasks. These stakeholders can be found in Figure 5.1. It was found that, depending on the nature of the event, the appropriate level of maintainer needed to be assigned. In this thesis, four distinct maintainer levels were defined, categorized by the type of service they are authorized to perform. These levels represent a general depiction of service roles and may not reflect the exact structure used by all customers. The classification was based on the required level of training as well as the average time needed to replace a faulty part.

Level 1 was the operator who only controlled the performance of the radar and had no further training on how to repair the radar.



Figure 5.2: Operator stakeholder. Illustration: A. Sämfors

Level 2 was the base-level maintainer whose main responsibility is to read error codes and analyze what might be the problem and what part needed to be changed. This level of maintainer was only allowed to repair the radar if it would take less than an hour. If the error was identified to take longer than an hour or had a certain complexity, it was instead passed onto the level 3 maintainer.



Figure 5.3: Maintainer stakeholder. Illustration: A. Sämfors

Level 3 are the ones tasked to complete the larger repairs and services. These maintainers are the most educated ones and have to perform the repairs in workshops because of the complexity of the repairs.



Figure 5.4: Hubs stakeholder. Illustration: A. Sämfors

Level 4 is the highest level of maintainers and often have knowledge of the entire construction of the radar. Most customers do not have the training required to reach this level and if this level of repair is needed, Saab often sends one of their experts in relevant field to support with the task.



Figure 5.4: Saab stakeholder. Illustration: A. Sämfors

The final result from the stakeholder analysis was the decision to analyze simpler tasks on the first and second level, being operators and maintainers. But also, more complex scenarios of the level 3 maintainers, being larger repairs that sometimes require indoor environment or good weather. As seen in Figure 5.1 the level-2 maintainers are second to level 3 in most affected. Detailed information about the stakeholders can be found in appendix A.

5.1.3 User Observation

The user observations were done in two different groups which meant each task were performed twice by multiple users at once. The summarized notes from this method are presented below using each of the tasks as a header, with both of the groups combined into a summarized result for each. In the notes, the learning maintainers will be referenced as participants.

Table 5.1: Challenges of gasket change of turntable

Changing the gasket on the neck of the Giraffe 1X	
1.	Participants experienced difficulty accessing certain components due to limited space, requiring them to search for specific tools.
2.	The instructions lacked clarity regarding the length of a replacement part.
3.	A participant sought clarification on a lubrication procedure, which involved estimating the correct amount of lubricant without precise measuring tools.
4.	The intuitiveness for a control switch could be improved.

Table 5.2: Challenges of TRM change

Remove and replace the TRMs	
1.	Opening a cover on the equipment required a specific technique that was not immediately apparent to participants.
2.	Accessing certain screws proved challenging due to their location and the need for specialized tools.
3.	Some screws were obscured, causing difficulty in removing a module.

Table 5.3: Challenges of rotary joint removal

Remove the Rotary joint	
1.	Participants struggled to disconnect connectors due to limited space and positioning.
2.	The fragility of certain cables presented a risk of damage during disconnection.

Table 5.4: General Gathered challenges

General challenges	
1.	Even with ample lighting, visibility was a challenge when working in certain areas of the equipment.
2.	Task duration was significantly impacted by the time spent searching for the appropriate tools.
3.	The compact design of the equipment resulted in ergonomic challenges.

The User Observations were done to broaden the knowledge of what the usability challenges were. Being able to observe new users of the system gave insights to where the most time was spent, if tasks were intuitive and what questions they had. The observations revealed a number of areas for improvement and many closely correlated with the results found during the interviews. The most frequently cited and time consuming of which was related to screw placement and correctly identifying torque requirements of the screws.

5.1.4 Interview

To further develop an understanding of existing areas of improvement, it was decided to continue the phase by conducting interviews.

Initial Interviews

These initial interviews gave the foundational knowledge needed for the coming user interviews. The three different participants provided very similar answers and explanations to each of the components, how they worked and were assembled. At the end of each interview, they were asked if they had any suggestions for potential improvements of the antenna. Recurring themes emerged from the interviews, highlighting several key areas for potential enhancement. Key suggestions from the participants included:

Enhancing the mechanism securing a specific component in the open position and addressing alignment misplacement caused by screws on another component. They also highlighted the need for better tool accessibility for a particular sub-component and proposed a more standardized and efficient lubrication procedure. Additionally, they emphasized the importance of visual indicators for critical steps and components and suggested a standardized toolkit containing only essential tools. The feedback also said that the intuitiveness for a control switch could be improved and addressed difficulties in disconnecting certain cable connectors due to their location and tightness.

User interviews

The participants during the user interviews had similar knowledge of the Giraffe 1X, but some were specialized within certain areas. Because of their similar experience, many of the answers from the participants were very similar, which is why the results presented below are categorized by the discussed topics instead of what each participant said.

The results from the interviews were compiled into a excel sheet to ease the evaluation. This excel sheet was further used during the Define-phase to evaluate what areas of improvements was in Saabs biggest interest of being looked into.

Table 5.5: Areas of improvement, interviews.

Accessibility Optimization	Participants identified opportunities to enhance the accessibility of certain components, particularly in relation to their placement and the surrounding structural design. Increased ease of access could support more efficient maintenance procedures.
Component Identification and Visibility	Participants observed that identifying and locating some components could be further streamlined, especially for smaller elements or those with minimal visual contrast or labeling. They also suggested that visibility of system indicators, such as status and component health signals, could be enhanced to support quicker interpretation.
Component Handling	Some components were described as requiring careful handling, particularly in scenarios where instructions were not closely followed. This highlights the value of continued emphasis on intuitive design and user guidance to support correct interaction.
System Feedback Expansion	Participants expressed interest in an expanded feedback system that could offer more detailed insights into component status and early indications of potential improvements, supporting proactive maintenance.
Design for Maintainability	Suggestions were made to explore design refinements that could improve access to critical components, potentially reducing the extent of disassembly required during routine tasks.
Streamlined Tooling Approach	The idea of a standardized toolkit limited to essential tools was proposed to enhance efficiency and reduce the risk of tool-related mismatch or error during maintenance.
Lubrication Workflow Optimization	Participants suggested exploring a more standardized and efficient approach to lubrication procedures, aimed at supporting consistent and reliable execution.



Figure 5.5: Icons, areas of improvements. Illustration: S. Tollesson.

5.1.5 Phase Conclusion

The Empathize phase has provided a rich understanding of the Giraffe 1X radar's usability challenges and opportunities for improvement. The collected data highlights specific areas for refinement and offered valuable insights into user needs and experiences. Moving forward, the focus would be on analyzing and prioritizing these findings to guide the next stages of development.

5.2 RESULTS OF DEFINE

This chapter presents the results of the data filtration and analysis conducted during the first empathize phase. These results are used to narrow down and prioritize the most critical usability challenges from various perspectives.

5.2.1 Criticality Analysis

The criticality analysis structured the registered events from interviews and observations. It categorized the events according to severity, probability, and accessibility levels. This process concretized and highlighted the most critical challenges.

Figure 5.6 illustrates a key finding: some events are unrelated to human interaction and are simply registered occurrences. These events cannot be addressed through usability improvements because there were not a user triggering the event. A significant portion of the findings involve interactions that are time-consuming. While users find these interactions slow, they are not inherently difficult to perform.

The remaining events stem from two primary causes. First, users may not operate the components correctly, leading to unsuccessful task completion. Second, accidental

interactions, while allowing tasks to be technically performed, can result in damage due to a lack of caution.

This analysis suggests that redesigning components to mechanically guide users and prevent improper actions, or incorporating warnings and instructions on proper use. This could significantly reduce errors caused by human interaction.

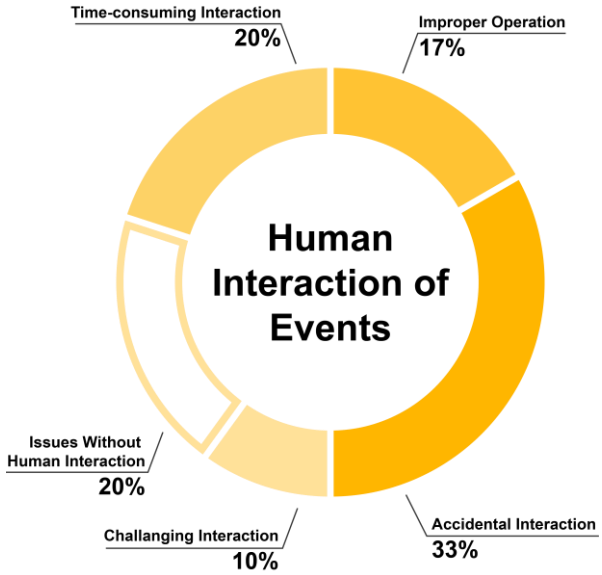


Figure 5.6: Pie chart, shares of human interaction. Illustration: A. Sämfors

5.2.2 Customer Journey

The first customer journey is about exchanging an old to a new TRM, for more written details such as emotions, the tasks and the users experience can be found in Appendix C, "Customer journey details". However, this mission is divided into four steps: two disassembly steps and two assembly steps, represented by each smiley in the Figure 5.7. The user's initial experience starts with excitement, which gradually decreases. In step two, the user might become impatient and overly excited, potentially missing some screws, leading to a few "errors". Step three can cause slight annoyance due to the need for a specific torque when tightening screws, requiring more precision. The experience improves in step four as the user becomes familiar with the process, which is a repetition of step one in reverse. However, the need for torque in this step can still cause some annoyance.

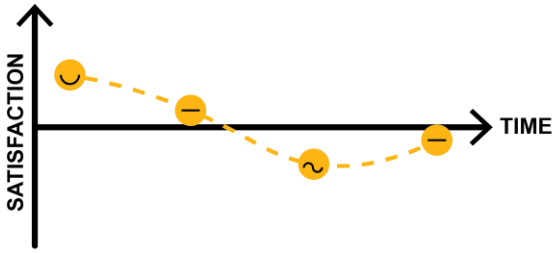


Figure 5.7: TRM change process. Illustration: A. Sämfors

The second customer journey focuses solely on the disassembly of the Rotary Joint, as illustrated in the Figure 5.8. Like previous process, it can be found in Appendix C. The smileys represent each component interacted with. Similar to the previous journey, the first step begins with an exciting experience. The second step remains positive but may evoke some nervousness in the user. The third step demands focus from the user, without eliciting strong positive or negative emotions. Steps four, five, and six require the user to work in a slightly uncomfortable position while screwing and lifting components.

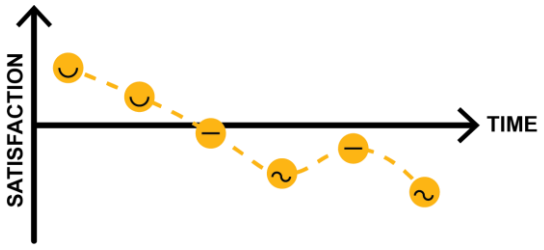


Figure 5.8: Rotary Joint removal process. Illustration: A. Sämfors

When analyzing the customer journeys from a usability perspective, screwing emerges as the most frequent task. Therefore, if the aim is to reduce repair time, regardless of the specific task, streamlining the screwing process presents an opportunity for improvement and hopefully increase satisfaction. This could involve evaluating the possibility of reducing the number of screws or optimizing the screwing task itself.

This leads to a deep dive into the customer's journey of choosing the right tool and adjusting the correct torque. This process is described in the steps outlined in Table 5.6 and illustrated in Figure 5.9. The user is in a situation where they need to reattach a component using screws with a specific torque requirement. Here's how they might proceed:

Table 5.6: Current tool choosing process

1	**Observe**: The user observes the component they need to replace.
2	**Consult Manual**: The user retrieves the manual and searches for the correct procedure for changing the component. They note the required tool size and torque.
3	**Gather Tools**: The user searches for the appropriate bit and torque tool in their backpack.
4	**Adjust Torque**: The user adjusts the torque tool according to the manual's specifications.
5	**Attach Bit**: The user attaches the bit to the screwdriver.
6	**Apply Torque**: Using the assembled tool, the user applies the correct torque to the screw



Figure 5.9: Current tool choosing process. Illustration: A. Sämfors

5.2.3 Root Cause Analysis

The result of the 5-Why-method was varied depending on the analyzed challenge. In some cases, identifying the root cause proved challenging due to gaps in available knowledge. In other instances, the root causes were more apparent and closely linked to other identified improvement areas. Notably, several top-level symptoms were found to share a common root cause, indicating that addressing a single challenge could potentially resolve multiple related ones.

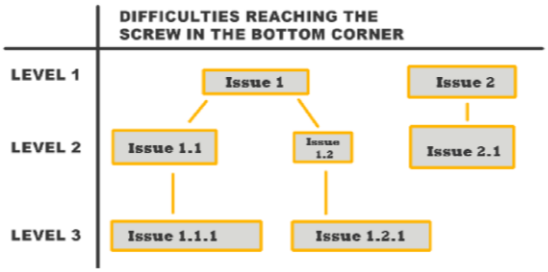


Figure 5.10: Root cause & 5-why. Illustration: S. Tollesson

Figure 5.10 represents how the method was noted, with each level diving deeper into what the root-cause might be. This was done for each of the evaluated components and the summarized results were the following:

The analysis revealed that several usability challenges stemmed from design considerations related to component protection, preventing part loss, or addressing other functional requirements. For example, friction between fasteners and a component was traced back to a design choice aimed at safeguarding internal elements. While some root causes were clearly identified, others remained inconclusive due to limited information or the complexity of the system.

This thorough analysis provided valuable insights of design choices, manufacturing processes, and user experience, and the next step was to evaluate them with the help of the product experts available.

5.2.4 Expert Interview

The product experts provided valuable feedback on our assumptions and questions regarding the identified root causes. One of the most important takeaways from these interviews was their input on what could realistically be addressed within the timeframe of our thesis. Many of the components shared a common root cause related to the need for protection against rain and harsh environmental conditions.

However, due to our limited time and limited knowledge of the system, addressing these events was deemed unfeasible. As a result, some root-causes were excluded from further consideration, and the preceding step in the 'Why-ladder' was instead treated as the actionable root cause.

In addition, the experts helped clarify areas where our understanding was lacking and confirmed that certain components would require more in-depth knowledge or time than we could allocate. Based on this, we decided to allocate less focus on the Turntable, Rotary Joint, Backcover, and Cassettes from further analysis and development.

In summary, the expert interviews validated the identified root causes and provided clarification on which challenges should be pursued further. This input helped us refine our focus and ensured that our efforts remained aligned with what was realistically achievable within the scope of the thesis.

5.2.5 Phase Conclusion

The Define phase has resulted in a focused set of target areas for improvement within the Giraffe 1X radar system. Based on a thorough analysis of user needs and system requirements, the TRM, CM, backcover, and toolkit have been identified as key components for further development.

User feedback has highlighted a need for streamlined tool access and clearer torque specifications to enhance efficiency and reduce user frustration.

The next phase will be concentrated on generating and evaluating innovative solutions for these specific components, ultimately prioritizing the most promising concept for implementation.

5.3 RESULTS OF IDEATE

This chapter presents the results of the ideation process, which focused on generating concrete ideas and solutions to address the areas of improvements identified in previous phases. The ideas center around component interaction, communication, and a toolkit.

5.3.1 Brainstorm

The brainstorming session generated a comprehensive list of conceptual solutions, presented both in writing and through illustrative sketches.

Several mechanical solutions were proposed with the intention of simplifying assembly and maintenance by applying Design for Assembly (DFA) principles. For instance, modifying the back cover hinges to include a guiding path could facilitate opening, thereby reducing the skill level required during this process.

To simplify the TRM attachment, the use of four guiding pins or rectangular pins with beveled edges was suggested. The chamfered areas on these pins help guide parts into position more easily, reducing the complexity of orientation and insertion during assembly (Ulrich & Eppinger, 2019). This design enhancement aims to support users by minimizing the fine motor skills needed for installation.

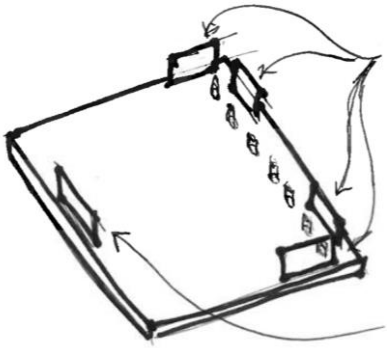


Figure 5.11: Rectangular guiding pins to TRM. Illustration: S. Tollesson

In response to user observations and interviews (Chapters 4.3.3 and 4.3.4), there is an opportunity to optimize the quantity and placement of screws across components. Conceptual solutions focus on improving fastener visibility and organization to promote more intuitive assembly and maintenance, while also exploring possibilities to reduce the number of fasteners where feasible.

Regarding fiber optic cable protection, reinforcing contacts was considered, but emphasizing cable importance through visual cues was prioritized. Implementing warning colors or specialized symbols on the cables could improve user awareness during handling.

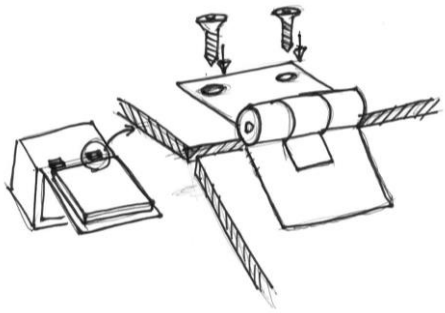


Figure 5.12: Alternative hinge. Illustration: S. Tollesson

For the CM component, incorporating data from the root cause analysis led to suggestions shown in Figure 5.13, by relocating diodes requiring observation to the exterior, enabling troubleshooting without opening the unit. Additionally, modifying CM hinges, as seen in Figure 5.12, was proposed to support fastening, complemented by adding warning symbols on screws that require special attention.

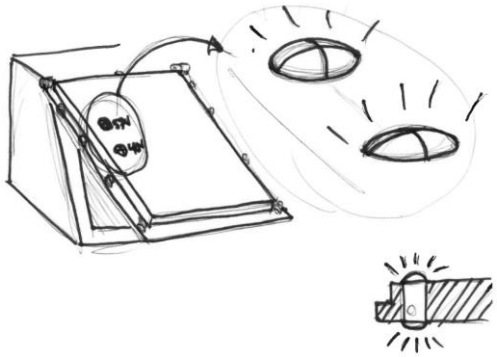


Figure 5.13: Exterior diodes. Illustration: S. Tollesson

Finally, the concept of a compact toolkit containing essential tools was explored. This toolkit, designed to be stored in the CRM and easily accessible to users, would include torque specifications and bit sizes, indicated through symbols on corresponding screws to further facilitate correct assembly.

The brainstorm was concluded by consulting Saab supervisors, who provided valuable feedback on which ideas were feasible within the scope of the thesis. They found many of the concepts promising for future consideration but recommended focusing on the toolkit as the most beneficial solution to pursue, given the relevance to our research questions.

5.3.2 Benchmarking

Building on the brainstorm outcomes, a benchmark was performed to explore existing solutions and gain inspiration for enhancing the newly identified focus: the toolkit. This research highlighted key functionalities and design implementations relevant to our project.

For example, various solutions are used to maintain order within tool sets. One common method involves foam padding with tool silhouettes cut out, designating a specific place for each tool and encouraging users to return them to their designated spots. While this solution requires considerable space, it is primarily found in toolboxes. In kit-style bags, the layout is often less sophisticated, utilizing pockets and compartments sized for the tools. This approach saves space but provides less visual guidance.

Regarding the surrounding design or bag used to store the tools, various types exist. These range from small to large hard cases, stiff foldable kits, backpacks, and rollable kits. When analyzing these different kit types, hard cases generally accommodate a large number of tools, often using foam for organization. Backpacks feature various sized pockets, allowing for storage of diverse tools. Foldable and rollable kits typically hold fewer tools, usually handheld ones. Foldable kits often include compartments for bits, while rollable kits generally lack such structure, resulting in a looser internal arrangement.

To access pockets or open the kits, mechanics like buckles, Velcro, and buttons are used, but zippers are the most prevalent. However, rollable kits utilize excess fabric that is rolled over the tools to secure them in place.

The benchmark's key takeaways suggest further development of a kit that is either rollable or foldable, aligning with the goal of minimizing the number of tools and storage space required. Further investigation into kit sealing mechanisms.

5.3.3 Workshop

The outcomes of the workshop were varied but gave new inspiration and out of the box solutions. Some of the outcomes were expected, such as the written suggestions from the brainstorm and the drawings from the sketching session. Since the scenario had been changed and the participants didn't know the background to the research, many of the ideas from the brainstorming were completely new and brought fresh perspectives to the challenges. The results from the first brainstorm were ideas regarding how the kit could be carried, how the tools can be attached in the kit, ways to store loose screws, and what materials could be used.

The second brainstorm added to the first one and answered some additional questions such as how the operator can access the kit, how the tools should be organized, and what the optimal size is. The full list of all of the suggestions from the participants can be found in Appendix F, but the most discussed and liked suggestions were:

Table 5.7: List of workshop design suggestions

1.	Creating a kit in the shape of a vest instead of a tool-bag
2.	A kit which could be attached similar to a slingbag and be rotated to be on either the stomach or back of the operator
3.	Using magnets to attach tools, screws and the tool-bag
4.	Retractable strings that would make the tools fall back into their place in the toolkit when dropped

During the brainstorming there were multiple questions regarding the scenario asked by the participants which gave a lot of new insights for the actual scenario. For example, a question was asked if there were any spare parts needed for the repair and if so, how the maintainers carry them. This was concluded to be outside of the scope for the tool-kit but gave refreshing perspectives and suggestions that could be implemented in the future.

The next step was for the participants to use the newfound ideas to sketch final concepts. The outcomes were varied and each participant had a different idea of how the tool-kit could be designed. The sketching also allowed the participant to play around with functions and details, which were easier to explain and understand when they had drawn their ideas. The majority of the sketches can be found in Appendix F.

One of the final solutions brought up by the participants was a harness used to carry the tools, where the bag could be flipped from the maintainers back to stomach without detaching it. The solution fit the scenario given to the participants perfectly as it allowed the maintainer to have the tool-kit on his back while climbing and didn't have to worry about dropping it when doing repairs.

5.3.4 Phase Conclusion

The Ideate phase yielded a diverse range of creative solutions aimed at addressing the identified usability challenges. A key area of focus emerged around the redesign and enhancement of the toolkit, with the goal of improving tool identification and accessibility.

The focus moving forward will be on prototyping and refining the most impactful ideas, ultimately leading to the development of a tangible solution that addresses the core usability needs of the Giraffe 1X radar system.

5.4 RESULTS OF PROTOTYPE

This chapter presents the results of creating and analyzing prototypes based on the iterative ideation of the toolkit. The resulting concept will serve as the foundation for further user testing. The prototypes were initially functional and made from paper, and later refined using fabric. Testing focused on aspects such as ergonomics, compartment design, visual communication, fold functionality, and more.

5.4.1 Mock-up

The results of this method can be found in Figure 5.15 below. The quickly made mock-up resulted in a rollable design, a paper lock to hold it in place, three compartments and straps to be able to attach it. A small test of different positions with the design attached to a user confirmed it was not obstructing accessibility. This means the concept is interesting to explore further.



Figure 5.14: Mock-up folded and unfolded. Photo: A. Sämfors

5.4.2 Tool Evaluation

An initial analysis of the maintenance manual for the Giraffe 1X system revealed that nearly all screws used in operator-level tasks were hex socket screws, but with varying dimensions and torque requirements. For each bit size, procedures typically specified both a low and a high torque setting, which increased the complexity of identifying a minimal yet sufficient toolset.

To organize this information, each documented maintenance task was entered into an Excel sheet, with rows representing individual procedures (not disclosed due to confidentiality), and columns representing the bit size and required torque. The numeric values in the cells indicated the number of screws of that particular type found in each task. This dataset (see Appendix L) served as the foundation for tool selection.

The collected data revealed a large spread of torque and bit-size combinations. To meet the practical limitations of field use, it was necessary to delimit the toolset to only the most essential tools. The analysis concluded that four main bit sizes were used most frequently, across eight different torque values.

Table 5.8: Most frequently used bits and torques

Bit size	M3	M4	M5	M6
High Torque	1.1	1.1	5.2	8.8
Low Torque	0.5	1.1	2.6	

This selection process was used to create initial toolset concepts, balancing two goals: covering all required operations and minimizing the tool count and weight to optimize usability in the field.

Concept Development and Expert Interviews

In the conceptual phase, a series of expert interviews were conducted to validate the initial findings and guide tool prioritization. Feedback confirmed that the most critical and frequently performed operator tasks involved accessing the antenna back cover and the central magazine, procedures which consistently relied on specific torque values for M5 and M6 screws. This supported the proposal to include fixed torque tools for these operations, while covering less common tasks with an adjustable torque screwdriver.

Interviews also highlighted usability concerns, particularly for novice operators. Adjustable torque screwdrivers, while flexible, were reported to be a common source of user error. To maintain high usability and reduce cognitive load, fixed tools were therefore prioritized for common tasks.

Tool selection was guided by the Wera product line, already used as a standard by the client organization. Wera's color-coding of tool sizes provided ergonomic advantages and served as inspiration for future visual cues in the layout. Available Wera tools were cross-referenced against procedural requirements to identify appropriate matches.



Figure 5.15: Color-coded Hex keys. Illustration: S. Tollesson

Multiple concept configurations were tested:

Many torque wrenches: This approach was quickly deemed unviable due to size and weight constraints.

Compact "knuckle" screwdrivers + 1 wrench: These were found to lack sufficient reach and robustness for certain tasks.

2 fixed torque screwdrivers + 1 small adjustable screwdriver + 1 large adjustable wrench**:** Evaluated with product experts, this setup was refined. The large adjustable tool was only needed for a single, extremely rare screw type. Due to its bulk and low relevance, it was excluded. It was determined that in such rare scenarios, regular hex keys could be used instead.

Additional auxiliary tools were also discussed, including:

- ESD protection (bracelets, bags)
- Strap wrench (standard size confirmed via Saab documentation)
- Brush for cleaning**:** The existing brush was found to be impractical for this specific tool-kit. A market scan identified a more suitable brush, which was also ESD-safe. It met both usability and procurement requirements.

During interviews and previous idea workshops, a need to store removed screws during repair was noted. An initial idea involved soft fabric pouches integrated in the tool-kit. However, concerns about spillage during transport or improper orientation led to a better solution: nesting screw containers which can be stored in a fabric pocket within the kit. These containers, when being used, could be placed in the antenna or on the CRM, depending on where the repairs were being done. No specific product has been researched or chosen, and the containers are currently being privately produced for the prototype. Further research needs to be done in this field and what is being presented is only a concept.

Final Tool Selection and Categorization
The final concept was structured around three usage categories, based on how often tools were expected to be used:

- Frequently Used

Two fixed torque screwdrivers with predefined torque and bit sizes, as these cover the majority of operator-level procedures:
M5 bit, 5.2Nm – for the central magazine.
M6 bit, 8.8Nm – for the antenna back cover.

Table 5.9: Bit size and Torque used by fixed screwdrivers

Bit size	M3	M4	M5	M6
High Torque	1.1	1.1	5.2	8.8
Low Torque	0.5	1.1	2.6	

ESD-safe cleaning brush: to remove dust and debris before opening sensitive components.

Set of hex keys (M2.5–M10): widely used in practice to quickly drive most of the screw length, with torque screwdrivers used only for the final tightening.

- Occasionally Used

Adjustable torque screwdriver with interchangeable bits (M3, M4, M5): used for procedures not covered by the fixed tools.

Table 5.10: Bit sizes and Torque used by adjustable screwdriver

Bit size	M3	M4	M5	M6
High Torque	1.1	1.1	5.2	8.8
Low Torque	0.5	1.1	2.6	

Long-reach M5 bit (15 cm): required for difficult-to-access screws, as identified during user- observations.

- Rarely Used / Supplementary items

ESD armband
ESD storage bags
Cable whisk
Adjustable wrench
Screw storage containers (stackable)

These supplementary items addressed less frequent, yet critical, scenarios and ensured the kit remained robust without being overburdened by unnecessary tools. A full list of the final concept tools can be found in Appendix J.

This final selection of tools was considered sufficient, based on both the earlier data analysis and feedback from the thesis supervisors. This concept allowed the process to continue to the next stage of creating new appropriate layouts and creating functional prototypes that could be further evaluated.

5.4.3 Functional Prototypes

To evaluate different design concepts, five physical prototypes were developed and tested iteratively. Each prototype aimed to explore specific aspects of the toolkit's usability, layout, materials, and attachment methods.

Black Prototype – Pilot Concept
This initial prototype aimed to explore the roll-up functionality and overall feasibility of a fabric-based toolkit. The tools were arranged side-by-side in sleeves, and foam padding was inserted between two fabric layers to increase structural stiffness and improve wearer comfort. The material used was relatively stiff and rough, in line with earlier benchmark studies that suggested some rigidity was necessary to maintain form when rolled.

Findings:
While the sleeve compartments successfully secured the tools, rolling the kit proved problematic. The tools stacked vertically on top of one another, making the roll bulky and unwieldy. The foam padding worsened this by increasing thickness and reducing flexibility. Additionally, it was determined that padding was unnecessary, as operators would likely wear protective or military gear between their bodies and the kit, rendering extra comfort features redundant. This iteration highlighted the need for a layout that prevented tool overlap and reduced total volume.

First Gray Prototype – Improved Layout and Tool Categorization

This prototype introduced a revised layout, optimizing the internal space by spreading tools out more efficiently. Importantly, it implemented a functional division of the toolkit into two main sections: one for frequently used tools and one for less frequently used items. The separation was physically realized through a foldable flap attached with buttons. This division followed the classification developed in Section 5.4.2 "Tool Evaluation", where tools were grouped into:

- Frequently used
- Occasionally used
- Rarely used

With the occasionally and rarely used being classed as the less frequently used items in the tool-kit. The new layout also replaced sleeves with sewn pockets. Hex keys were stored under a foldable flap using elastic bands, allowing for a two-layer arrangement: common keys on top, and less-used ones beneath. This flap also included a thin cardboard insert to simulate padding, intended to balance stiffness and rollability.

Findings:
The revised layout significantly improved rollability, though spacing between tools still required refinement. The two-part division of tools based on usage frequency was positively received in preliminary user discussions and was kept in later designs. The cardboard padding provided a suitable balance between stiffness and thickness, and a more sewable material with similar properties was sought for future iterations. The hex-key flap performed well, though its movement prompted a suggestion to add a Velcro strap to prevent it from flailing, an improvement planned for the next prototype. However, the general tool storage solution using fabric pockets was not well-received, prompting the need for a dedicated test of attachment mechanisms.

Tool Attachment Prototype – Evaluating Fastening Methods

This prototype was developed exclusively to test different attachment methods for the tools, particularly the screwdrivers. Three configurations were evaluated:

- Elastic bands only
- Pockets with Velcro-closing flaps
- A hybrid model combining small pockets with elastic bands

Other tools such as the brush, ESD wristband, and ESD bags were better suited for fixed pockets due to their irregular shapes and occasional use.

Findings:
While elastic bands were initially favored for their simplicity, the irregular shapes of Wera's tools made it difficult to secure them reliably. The most effective solution was a combination of elastic bands and Velcro straps, which offered both security and quick accessibility. This hybrid model was selected for the next prototype iteration.

Full-Concept Prototype – Integrated Design Evaluation

This prototype integrated lessons from earlier iterations into a comprehensive concept. A new leather-like material replaced the cardboard padding and struck a balance between flexibility and structural support. The layout included bendable zones (for rolling) and stiffer tool-holding zones (to maintain form). Spacing adjustments were made to prevent tool overlap. A carry handle was added, along with a Velcro-sealed fabric flap as a primary closing mechanism.

Findings:
The new padding material worked exceptionally well and was selected for continued use. The division between flexible and stiff sections helped enable smooth rolling without compromising tool support. The attachment methods proved effective, tools could now be detached using only one hand, a significant usability improvement. Based on this success, it was decided to update the foldable tool section (previously using buttons) to use Velcro instead, further enhancing one-handed operation.

However, the Velcro-based closing mechanism was not secure enough for transport or storage. While the handle functioned well, user suggestions prompted the addition of D-rings to allow attachment to other gear via carabiners or straps.

Final Prototype – Refined and Test-Ready

The final prototype consolidated all previous learnings into a test-ready format for user evaluation. The Velcro fastener was added to the hex-key flap as planned. The closing mechanism was upgraded with two external snap buckles that could be tightened, while the Velcro flap remained to hold the form during buckling. Tool slots were marked using embroidery, with a single consistent thread color.

Findings:
The new closure system resolved the security concerns of earlier iterations. Tool placement was clearly indicated through embroidery, but color-coding was intentionally avoided. Since the tools already differed in shape, size, and in some cases color, additional color codes would not enhance usability. In fact, it could increase cognitive load by requiring users to learn and remember a redundant visual layer. The choice to use a single embroidery color aligned with cognitive ergonomics principles by simplifying visual processing and minimizing mental effort, ultimately enhancing overall usability.

5.4.4 Phase Conclusion

The Prototyping phase resulted in the creation and testing of multiple tangible prototypes, providing valuable insights into the functionality and usability of the redesigned toolkit.

Feedback gathered from user interactions with these prototypes has informed key design refinements, leading to a more robust and user-friendly solution.

The next step will involve further user testing to validate the usability and effectiveness of the refined design and gather additional feedback that can be used for future work.

5.5 RESULTS OF TEST

This chapter presents the results of the user testing conducted. The data gathered will be used to confirm the usability of different aspects of the toolkit in various contexts. This includes evaluating the symbol subsystem of communication, aiming to determine which symbols are easiest to learn or intuitively understand, and thus incorporate into the final concept.

5.5.1 Think Aloud

The concepts were designed with different usability trade-offs in mind. The symbol-based test was expected to be the most intuitive and user-friendly, particularly for new or unfamiliar users, thanks to clear visual cues and a supporting reference table. The text-based concept prioritized visual simplicity and minimal clutter, assuming that more experienced users would be able to interpret grouped instructions with ease. The number-based test aimed to strike a balance between the two, providing immediate screw-specific information to reduce uncertainty, while accepting a slightly higher level of visual complexity. Figure 5.17 represent the concepts expected Usability versus Production Costs trade-offs. The Think Aloud sessions were intended to evaluate how these trade-offs impacted usability in practice, particularly in terms of guessability, learnability, and error rates.

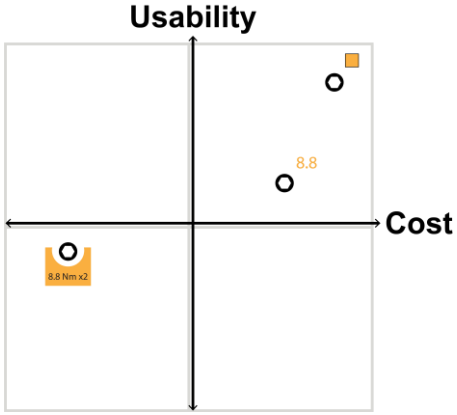


Figure 5.16: Usability-cost matrix of icons. Illustration: A. Sämfors

With these expectations in mind, the following section presents the findings from the think-aloud testing, highlighting how user interactions aligned with or diverged from the intended strengths and limitations of each concept.

The test was built around three distinct concepts, symbol-based, text-based, and number-based, each designed with different users and use cases in mind. These concepts, illustrated in Figure 5.17 and further detailed in Appendix G, were evaluated to explore usability trade-offs across guessability, learnability, and error rate.

The symbol-based concept displayed each screw with a unique symbol linked to a reference table showing torque and bit size. This approach aimed to be highly intuitive and accessible for all users through clear visual cues.

The text-based concept reduced visual clutter by tagging only one screw per group, with instructions such as "5.2 Nm × 4" indicating the required torque and quantity. While potentially less intuitive for first-time users, it was expected to suit more experienced operators.

The number-based concept placed the torque value directly beside each screw, eliminating the need for a reference table. This design aimed to balance directness and clarity, though it required more tags than the text-based alternative.

The test setup was intentionally kept simple to focus on core usability factors. Participants interacted with a layout of visualized screws and symbols (Appendix G) using standard tools and matching bits. The screw configuration was consistent throughout all test rounds to ensure comparability.



Figure 5.17: Symbol concept. Illustration: A. Sämfors

The **Symbol-Based Concept** was frequently praised for its intuitive structure and visual clarity, especially when paired with a reference table outlining torque and bit size. Many participants appreciated the recognizability of the symbols, particularly the distinction between reduced and full torque through visual cues like half and full squares. This aligns well with cognitive ergonomics principles around visual aids, which help reduce the need for internal memorization. The Think Aloud method revealed that users often relied on the table not only to confirm assumptions but also to build confidence in their interpretation, supporting the idea of learnability through self-directed exploration. However, some participants required time to understand the meaning of the symbols, indicating a slight initial learning curve. In one case, the symbols were not immediately understood until later in the task, demonstrating how gradual exposure to information during unstructured tasks can support user learning over time. Despite these strengths, the major drawback noted was the visual clutter created by marking each individual screw, raising concerns about feasibility in manufacturing. Notably, no errors occurred during the symbol test, suggesting that once understood, the system supported accurate performance and reduced the risk of misinterpretation.



Figure 5.18: Text concept. Illustration: A. Sämfors

The **Text-Based Concept** was generally considered clean and efficient, particularly for users with prior experience or familiarity with the system. The approach of marking a single screw to represent a group, along with an accompanying instruction (e.g., "5.2 Nm × 4"), was perceived as advantageous in reducing visual noise. This minimalism reflects ergonomic principles of cognitive load reduction, making it easier for users to focus on the task without being overwhelmed. The notes revealed users interpreted this method as more subtle and organized than the alternatives. However, challenges emerged in translating information into the correct action. Several users expressed uncertainty about which screws were included in the torque instruction, or whether to multiply the torque value itself. This suggests a risk of misinterpreting the information, particularly in high-pressure or distracted scenarios. Multiple participants suggested that the text-based concept could be improved by adding spatial grouping or visual aids, such as zones or color-coded backgrounds, to better link the instruction to the relevant screws. A total of three errors were observed during testing. One user picked the wrong torque but self-corrected before finishing the task. Another user failed to understand the bit-size color coding at first and selected the wrong bit for the first screw. These events point to

challenges in how users mapped the written instructions to physical action, especially without clear visual reinforcement.

The **Number-Based Concept** received mixed feedback. On one hand, it was appreciated for its directness as the torque requirement was presented next to each screw, eliminating the need to look in an external table. Several participants noted that this removed a step in the process, potentially improving efficiency and time-on-task, a key usability metric in this study. The Think Aloud method exposed how this advantage could sometimes be undermined by information overload. Having torque values and visual tags on every screw led to perceptions of clutter, especially among experienced users. From a cognitive ergonomics perspective, this reflects the concept of attention tunneling, where too much competing information can cause users to miss key cues or disregard the system entirely. Additionally, color contrast issues were mentioned, which impaired readability. The errors that occurred during these tests regarded a misunderstanding of bit size for the screw, and in another, the wrong torque was applied. These incidents suggest that even with direct labeling, visual overload or unclear associations between components may still hinder performance.



Figure 5.19: Number concept. Illustration: A. Sämfors

These insights from the Think Aloud sessions provided valuable qualitative input on how users experienced each concept in practice. To complement this and further guide decision-making, a more structured comparison was carried out using a Pugh matrix. The following chapter presents the results of this evaluation, highlighting how each concept performed against defined criteria to support the selection of a preferred direction.

5.5.2 Pugh Matrix

The resulting Pugh matrix is shown in Table 5.11. Based on the defined weighting and criteria, the text concept emerged as the winning design. Matrix with elaborated criterion is found in Appendix I.

Table 5.11: Pugh matrix of icon concepts

PUGH-MATRIX				
Criteria	Symbols	Number	Text	Weight
Speed	0	0	+	2
Cost	-	0	+	1
Time	0	0	-	2
Error rate	+	0	+	3
Guessability	-	0	+	3
Learnability	0	0	0	3
Area	-	0	+	2
Eastetics	+	0	+	1
Cognitive load	+	0	+	3
Clutter	0	0	+	2
Task success	0	0	+	2
Sum	1	0	17	
Rank	2	3	1	

The evaluation of each criterion was based on the performance observed during the think-aloud user tests, drawing upon the qualitative data collected and discussed in chapter 5.5.1.

Using this method provided a more objective and quantified comparison of the concept performances, moving beyond a purely qualitative evaluation based solely on the think-aloud sessions. The results suggest that the text concept, which used a single marking to represent the total number of screws, outperformed the numbers and symbols concepts, which used a marking for each individual screw. This latter approach resulted in a cluttered appearance, negatively impacting its aesthetic performance in the user tests.

It is crucial to highlight this fundamental difference between the concepts, as it may explain the text concept's superior performance. The ideal solution may lie in a middle ground. A single marking indicating the total number of screws, complemented by additional, low-visibility markings on individual screws.

5.5.3 Focus Group

The focus groups provided valuable and constructive feedback regarding the tool kit intended for use with the Giraffe 1X system. The overall response was highly positive, with participants highlighting the kit's strengths while also identifying areas for potential improvement and future development.

One of the most appreciated features was the compact and practical size of the kit, which was considered ideal for fieldwork. Participants also emphasized the intuitive organization of its contents, particularly the inclusion of a separate pocket for frequently used tools such as hex keys. This layout was seen as significantly reducing the risk of selecting the wrong tools during operation, thereby enhancing both efficiency and user confidence. The choice of materials was perceived as robust and durable, further contributing to the overall usability of the design.

When asked directly, all participants confirmed that the tool kit would be feasible to integrate into the current Giraffe 1X and Giraffe 1X CRM systems. More importantly, they unanimously agreed that such integration would make the Giraffe 1X easier to use for both operators and maintenance personnel. This clear endorsement highlights the kit's potential to improve usability in real operational contexts.

Several constructive suggestions were made to further enhance the design. In terms of indication and labeling, participants requested clearer compartment markings, such as the use of symbols or intuitive visuals. The ESD wrist strap was identified as needing greater visibility. The participants stated this should be implemented with consideration for users with color vision deficiencies, which needs to be researched in future work.

Users recommended functional improvements such as making the brush easier to remove and reinsert, and replacing the current straps with a more durable solution than Velcro, which can degrade over time. Adding a second handle was also suggested to improve balance and comfort when carrying the kit.

Content-wise, the addition of a separate compartment for consumables (e.g., screws, cable ties, bullets) was proposed. Furthermore, a multi-tool and a torque wrench with a broader torque range were recommended to better support tasks involving larger screws.

Regarding integration, participants expressed a desire for more seamless connection between the kit and the Giraffe 1X system, such as a designated storage location within the system itself. They also emphasized that future Giraffe 1X versions might benefit from simplified designs to further ease maintenance and troubleshooting, an area where this tool kit could provide strong support.

Looking forward, participants highlighted several directions for future development. These included studying ergonomic aspects such as maximum comfortable carrying weight, exploring modularity through a larger variant of the kit, and ensuring the use of high-quality materials to reflect Saab's standards. Real-world testing with end users was also strongly recommended to validate the kit's performance and usability under realistic conditions.

In conclusion, the focus groups clearly saw the tool kit as a valuable and practical enhancement to the Giraffe 1X system. Not only is the kit seen as feasible to integrate, but its thoughtful design and functionality also contribute directly to improved usability. By addressing the suggested refinements, Saab has the opportunity to deliver a user-centered solution that supports both operational efficiency and long-term innovation.

5.5.4 Phase Conclusion

The outcomes of this phase include the creation, testing, and evaluation of what will be refined into the final design. The results consist of a developed and functional fabric-based toolkit for operators, containing their most frequently used tools. The layout of pockets and tool placement was carefully designed and qualitatively evaluated by experts, who stated that the solution is both feasible to integrate and capable of enhancing the usability of the current radar system through an accompanying communication system.

The Pugh matrix indicated that a text-based icon system was the most effective indication system. However, the participants of the think aloud tests and experts favored a symbol-based system due to its workflow. This communication system was recommended for further development, as it had the potential to become even more efficient and user-friendly.



06 FINAL DESIGN

6.1 Integration & Storage
6.2 Layout & Tools
6.3 Pockets & Compartments
6.4 Icons & Communication

Photo Saab

6 Final Design

This chapter details the most developed and tested solution, which is the product that will be delivered to Saab. The final design is a foldable toolkit with a carefully curated set of tools and layout. It is delivered with a proposal for communicating which tool corresponds to each screw in the Giraffe 1X antenna and its designated location within the kit. The solution also includes a proposal for integrating the kit into the Giraffe 1X CRM for storage.

6.1 INTEGRATION & STORAGE

This product is designed for use in several scenarios: within a CRM, in a glove compartment, attached to a turntable unit, or carried by a person.

The kit includes a holder that can be attached to various points within a CRM. Currently, the only available placement is on the left wall of the CRM, as seen from the hatch. Another option is on top of a component rack, utilizing the same threads used to secure it.

The kit carrier is a stainless-steel bar or rail that is slightly bent on each side, enabling fixation of one degree of freedom. In combination with two additional rails, straps, and Velcro, which create a fork-like design, it is possible to fixate the remaining degrees of freedom to the holder. The holder itself is attached with two screws along the main rail, fixing the kit and holder to the rack.



Figure 6.1: Front of closed tool kit. Photo: S. Tollesson

The delivered prototype of the kit holder in the CRM is conceptual and a direct result of the brainstorming session. Although it is a proposal suggesting how it could be designed, it serves as a prototype to discuss, evaluate, and further explore.



Figure 6.2: Back view of toolkit. Photo: S. Tollesson

The kit (Figure 6.1) measures approximately 300mm in width and up to 150mm in diameter, with variations depending on the number of tools included. Its size and weight (under 4kg) allow for storage in compact spaces like glove compartments, backpacks, or boat cabins.



Figure 6.3: Toolkit attached to military equipment. Photo: S. Tollesson

The straps have multiple uses. In combination with the D-rings found in Figure 6.2, enable the kit to be secured to a turntable or attached to the loops. This allows the kit to be unfolded while attached to the turntable, even permitting tool usage. This is beneficial in situations where there's no space to lay out the kit, such as when working on systems not integrated with a CRM. Despite the straps being in use, the kit can still be folded using the additional strap with velcro, ensuring it remains closed even when hanging.

The straps also facilitate attachment to a person. By untying the buckle and straps creating an inverted loop, the kit can be fastened to a belt. The burdock on the backside allows for temporary attachment to a tactical vest worn by operators and maintainers. This setup permits folding and unfolding the kit while worn, ensuring constant tool accessibility.

6.2 LAYOUT & TOOLS

The included tools in this kit can be found in Figure 6.4, and are 3 torque keys, 7 hex keys (1), one brush (4), one wrench (8) and one strap wrench (10). In addition to that there are 5 bits (5) one ESD wrist strap (9) and an ESD bag (12).

These tools are distributed through the kit in a special and thought-out order. The first part of the kit holds the most frequently used tools, when doing light troubleshooting as an operator or maintainer. These tools include all the hex keys and two of the torque tools.

The second part of kit stores tools that can come in handy. These tools are used to perform task that the operator is not thought to do. However, due to the nature of the environment of the operator, it was deemed beneficial to also allow these additional tasks to be possible to perform if need would arise. These tools are the strap wrench, wrench, one torque key, the bits, the ESD wrist strap and the ESD bag.

Torque screwdriver 5.2 Nm – Number 2 in Figure 6.4, is one of the most frequently used torques and therefore, it gets its own tool, in order to save time and eliminating the time adjust to right torque. This torque is used on all outer screws of the TRM, but also the screw of the central housing. According to the analysis this is used nearly every time the operators interact with the antenna.

Torque screwdriver 8.8 Nm – number 3 in Figure 6.4, is the most frequently used torque tool because it is the one used on the back cover. Therefore, getting its own fixed torque tool.

Adjustable torque screwdriver – number 7 in Figure 6.4. To be able to meet the range of torque requirement of less interacted components.



Figure 6.4: Opened kit with tools. Photo: S. Tollesson

6.3 POCKETS & COMPARTMENTS

This kit features three types of pockets and compartments, designed to accommodate different items and purposes.

Elastic Band Compartments

Tools like torque wrenches are stored using two elastic bands. One band is fixed, while the other has an adjustable end secured with velcro. This design allows users to easily view tool markings, increasing the guessability of tool selection. More details on these compartments can be found in chapters 6.4 and 5.4.3.

Closable Pockets

These pockets offer protection against dust and are suitable for items that shouldn't be stored with elastic bands. For example, the brush's bristles need protection from bending, and the ESD wristband and bag have dedicated pockets due to their structure. Buckles are not used because they require more space and are harder to operate, especially with gloves.

All closable pockets are secured with velcro. Velcro are chosen over buttons because they require less fine motor skill and are easier to use with gloves.

Elastic Band Only Compartments

These compartments utilize only elastic bands and are designed to store cylindrical tools like bits, strap wrench, and L-keys. The cylindrical shape allows for easy insertion and removal while ensuring the tools remain securely in place. Avoiding velcro in this compartment simplifies assembly and reduces the number of components.

6.4 ICONS & COMMUNICATION

As part of the toolkit, an integrated conceptual icon system is also included. The kit is delivered with a main icon system, representing the current stage of product development in this project. It is intended to serve as a foundation for further development.

The icon system is based on two variables: the size of the screw, which requires a matching bit, and the torque requirement of the screw. Due to the possibility of combining different screw sizes and torque levels, screw size is represented by color, and torque is represented by an icon. The sizes mainly used in the kit are M6 (orange), M5 (yellow), M4 (purple), and M3 (light green). These colors are used consistently on the radar antenna, bits, hex keys, and torque tools.

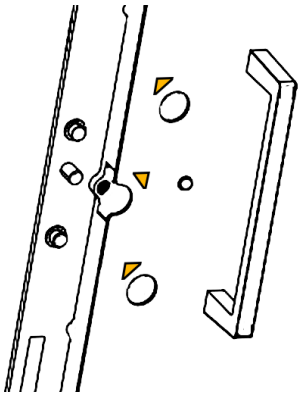


Figure 6.5: Icons implemented in component. Illustration: S. Tollesson

Each screw size can only be combined with two different torque settings: full or reduced. Therefore, full torque is represented by no symbol, and reduced torque is represented by a triangle. The torques for each screw size can be found in the solidified table. The table features holes shaped like the respective hex sizes, allowing the user to take a bit, match it with the screw, then the hole in the table, and finally read the torque. This systematic approach enables quick and accurate identification of the correct size and torque. The holes surrounding the table allow it to be sewn into place beside the bits.

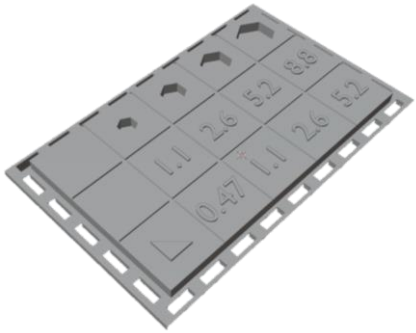


Figure 6.6: Solid Torque-Screw table. Illustration. A. Sämfors

The proposed scenario, or customer journey, when using the toolkit, this process is described in the steps outlined in Table 6.1 and illustrated in Figure 6.7. The previous process can be found in Figure 5.9. The user is in a situation where they need to reattach a component using screws with a specific torque requirement:

Table 6.1: Toolkit concept of choosing tool

1	**Observe**: They observe the screw for any markings, such as an orange triangle.
2	**Bit Selection:** The toolkit contains four bits. The third bit is estimated to be the one that matches the screw. The user removes this bit from its designated place in the toolkit.
3	**Bit Fit**: The user moves the bit to the screw, checking for a size match. If it doesn't fit, they would try another bit until they find one that matches.
4	**Torque Table**: The user then maneuvers the bit to the solid torque table and attempts to fit it into a hex-sized hole. This fit determines which column of torque values to use. Next, they choose the row based on the symbol (or lack thereof) beside the screw. In this case, the triangle indicates they should select the cell with 2.6 Nm torque.
5	**Torque Adjustment**: The user chooses an adjustable torque screwdriver and sets it to 2.6 Nm by moving the adjustment mechanism on the screwdriver.
6	**Bit Attachment**: The user attaches the correct bit to the adjustable screwdriver.
7	**Torque Application**: The combined tool is used to tighten the screw to the required torque.

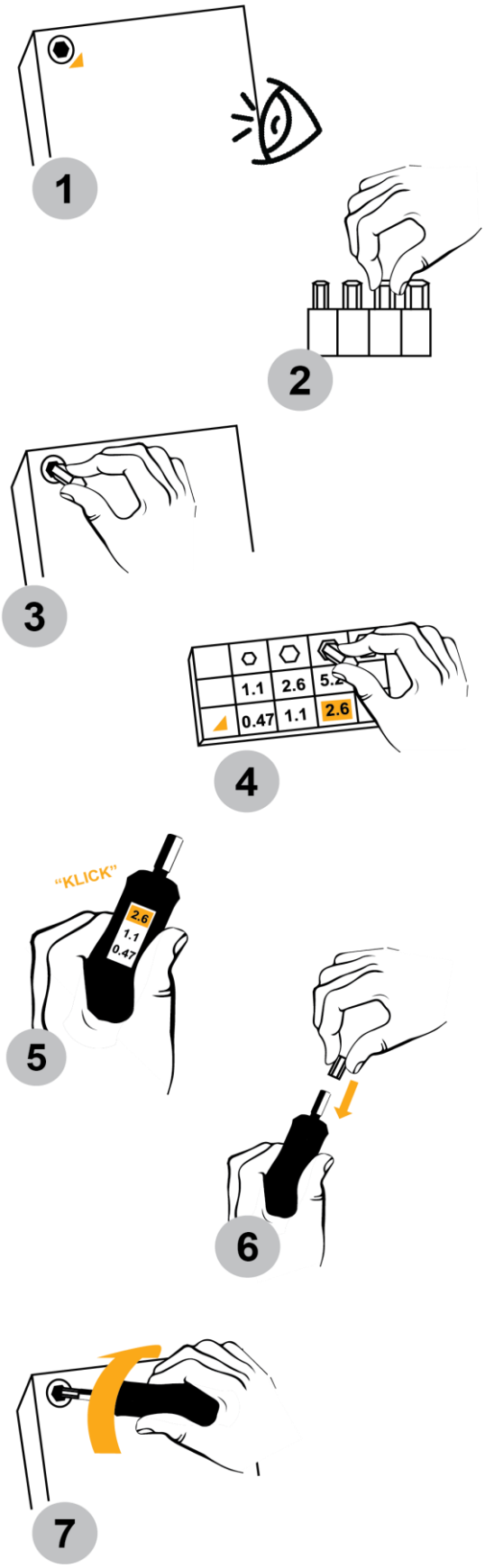


Figure 6.7: Toolkit process of choosing tool

07 Discussion

7.1 Positioning the Result
7.2 Relevance
7.3 Reflection
7.4 Conclusions
7.4.1 Project Objective & Aims
7.4.2 Research Question 1
7.4.3 Research Question 2
7.4.4 Research Question 3
7.5 Recommendations



Photo: Saab

7 Discussion

This chapter discusses the key findings and outcomes of the design process in this project. The results are interpreted and critically reflected upon, with a focus on usability, the design decisions and feedback from experts. Furthermore, opportunities for future improvements are identified and considered.

7.1 POSITIONING THE RESULT

This project is strongly rooted in our Industrial Design Engineering (IDE) education, especially through the use of Human-Centered Design. Wikberg-Nilsson (2015) describes this methodology as putting the user at the center and focusing on their needs. By involving users throughout the process, we ensured the results stayed relevant and addressed actual challenges. This reflects IDE's emphasis on empathy and working iteratively with users. Including input from participants outside the core development group also follows the co-design approach described by Wikberg-Nilsson (2015), which helped us gather more unbiased feedback and explore new ideas.

Our IDE background influenced many of our decisions, particularly in terms of usability. Concepts like guessability and learnability, discussed by Norman (1999), helped guide our design toward intuitive use. These ideas were especially important in creating solutions that work for operators with different levels of experience and technical knowledge.

Design Thinking was used as our main working method, and its structured yet flexible nature suited the project well. While our original timeline was a bit too optimistic, the process encouraged us to adapt and reflect continuously. Being able to respond to changes and feedback was key to moving the project forward in a meaningful way.

The theoretical tools we've learned during our education, like cognitive ergonomics, DFA, and general usability principles were helpful in analyzing and evaluating our work. They supported our decision-making and helped us improve the design with a clearer understanding of both technical and human factors.

Our collaboration with Saab gave us a valuable opportunity to apply these theories in practice. For us, it was a chance to test and strengthen what we've learned in a real-world setting. For Saab, the project resulted in new design concepts that could bring added value to their operations. This exchange of knowledge and experience benefitted both sides.

User input played an important role throughout the project. Even though confidentiality limited how many users we could include, we maintained close contact with Saab's operators and experts. Their feedback was essential in shaping the final concept and making sure it reflected real operational needs. It also helped build trust and confidence in the outcome among stakeholders.

Looking back, this project has helped us grow, especially when it comes to working across different disciplines and dealing with complex systems. Applying IDE methods gave us a clear structure while still allowing space for creative problem-solving. We feel that we've made a meaningful contribution, both by highlighting previously undocumented user needs and by developing a concept grounded in both theory and real-world insights. This work not only supports further improvements at Saab but also adds to the broader field of Industrial Design Engineering.

7.2 RELEVANCE

The relevance of this project is heightened by current global circumstances. Evaluating usability, focusing on ease of use and repair, is crucial in the radar's product development process. The product is mature enough to benefit from usability enhancements beyond mere functionality. Evaluating repairability aligns with sustainable design principles, as creating durable products extends their lifespan.

The project initially aimed to enhance the intuitiveness of the components, practically meaning reducing user errors and time needed to complete tasks. Observations during customer training revealed that users spent considerable time searching for the right tools. Even when they located them, they often forgot

where they stored them, leading to further delays. Interviews highlighted the need for readily accessible torque specifications, as searching for them during stressful situations was often neglected.

Therefore, the opportunity to reduce time and errors, enhance quality, and improve user satisfaction made a toolkit an excellent solution for enhancing the overall usability of the antenna.

7.3 REFLECTION

This master's thesis has been a dynamic and instructive journey, shaped by both academic theory and real-world collaboration with Saab. The process resulted in the development of a functional toolkit, a direction that evolved organically from the original aim of mapping usability. This shift, though unanticipated at the outset, was guided by the project's goals and justified by the insights gained through iterative user involvement. The result reflects a thoughtful response to real operational needs, and although it is not a finalized product, it establishes a strong foundation for future development and refinement.

Areas for Improvement and Lessons Learned

While the final concept successfully addresses key aspects of usability, the project also revealed areas for improvement in process planning and resource allocation. For example, hands-on development phases such as prototyping and working with textile-based materials required more time and iteration than initially expected. Additionally, academic milestones such as the mid-way presentation demanded more effort and preparation than anticipated. Combined with some underestimation of course-related administrative deadlines, this created a bottleneck in the later stages of the project.

In future work, more realistic scheduling and a broader consideration of time-consuming tasks and academic deliverables would improve planning accuracy and allow for a more iterative and reflective design cycle.

Motivation for Chosen Methods and Processes

The decision to apply Design Thinking as the primary methodological framework was a central strength of this project. This approach provided a clear yet adaptable structure, enabling the team to navigate the complexity of user needs while maintaining flexibility in addressing new insights. Design Thinking's user-centered nature aligned perfectly with the project's core goal, enhancing usability in a complex system—and offered valuable guidance for prioritization and evaluation.

Closely connected to this, the use of Human-Centered Design principles ensured that user needs remained at the forefront throughout the process. Methods such as user interviews, observations, and co-design workshops enabled inclusive decision-making and deepened the relevance of the design outcome. These practices, which are central to the Industrial Design Engineering (IDE) discipline, allowed us to draw on key theoretical concepts such as guessability, learnability (Norman, 1999), and cognitive ergonomics to inform both analysis and prototyping.

Constraints and Scope Considerations

As with any real-world design collaboration, the project encountered a number of contextual constraints. Confidentiality and information security protocols limited access to a broader user base and restricted some of the information that could be shared externally. However, this challenge was balanced by strong collaboration with Saab's internal operators and engineers, whose expertise was crucial to maintaining a valid and contextually grounded design.

Furthermore, the scope and timeframe of the thesis naturally required a degree of prioritization. Certain ideas, such as a fully modular toolkit or more advanced tool indication mechanisms were identified as having future potential but could not be fully realized within the current project. These remain valuable opportunities for continued development within Saab's internal processes.

Because the development of the kit took place after the radar was completed, it limited its potential impact. If the kit had been developed in parallel with the radar, taking into account torques and screw sizes, it would have allowed

the construction to be tailored to the kit's tools. This would have made the radar more unified and accessible, and provided helpful guidance for engineers to only use the screw sizes and torques existing in the toolkit.

Additionally, milling symbols or other indications into the components from the start, instead of using stickers, would make them more durable and resistant to weathering and abrasion.

Personal and Professional Growth

One of the most important outcomes of this thesis has been the professional development it offered. Collaborating with a leading industrial partner like Saab provided insight into interdisciplinary teamwork, technical communication, and design decision-making in high-stakes environments. It reinforced the importance of balancing creativity with practicality and theory with real-world feasibility.

7.4 CONCLUSIONS

The following headings explain and summarize the project's conclusions, ultimately answering the research questions established at the outset.

7.4.1 Project Objective & Aims

This thesis initially had a broad scope, aiming to analyze the Giraffe 1X's design from a user perspective and develop proposals to increase usability. As the project progressed, the focus narrowed to the Giraffe 1X's antenna.

The goal was to reduce errors and downtime during repairs, thereby reducing training costs and increasing availability. This would be achieved by providing concepts and prototypes for testing and comparison with the current system. Due to the complexity of the components, the initial prototype was envisioned as simple and functional, prioritizing testing over aesthetics and detail design.

As the project shifted towards creating a toolkit, which is less complex than prototyping an electrical component, the ambition for the prototype increased. Delivering a near-finished prototype would provide Saab with valuable input for the kit's product development process, simplifying manufacturing and future development.

7.4.2 Research Question 1:

What are the current usability issues, regarding repairability and assembly of the radar components?

This project began with the task of mapping current usability challenges in the repair and maintenance process of the Giraffe 1X radar system. The initial intention was to focus the analysis on the antenna. The process began with interviews and observations. The interview method followed a saturation approach, meaning interviews were conducted until no new insight emerged. At the point of saturation, the identified challenges were considered thoroughly explored, and the research question sufficiently addressed.

To gain a more detailed understanding of the factors affecting usability, a criticality analysis was conducted to categorize and map the types of human interactions that contribute to these challenges. This provided a structured description of each event, its underlying causes, and the resulting consequences. The analysis directly addresses and answers the first research question.

Regarding the mapping of assembly and repairability, the interviews involved people working on the current production line. These individuals have experience assembling the radar's components multiple times. Other sources of information included engineers working with requirements and directly with customers. Interviews were also conducted with field technicians sent to workshops to repair or replace components and with trainers. Trainers have experience with the challenges users face when first using the system and what they find difficult to learn.

However, the inability to conduct interviews with actual military operators and maintainers poses a risk that the research may not fully reflect the reality of their experiences. Conducting such interviews would require the ability and opportunity to travel or connect with these users, which is a complex undertaking given current global circumstances.

Although the individuals interviewed in this project have firsthand contact with these users, direct interaction with operators and maintainers would have strengthened the report's findings and made them even more robust.

7.4.3 Research Question 2:
How can we, with relatively small means, effectively increase the usability and make the product more intuitive to interact with?

Increasing the usability and intuitiveness of an existing product is a significant challenge. To address this, a theoretical exploration of usability is necessary, examining its definition, impact, and measurement methods. Understanding how to measure usability is crucial for identifying areas for improvement and achieving better results. This theoretical framework partially answers the research question of how usability can be increased but doesn't fully address how to make the product more intuitive.

Redesigning components with limited resources present a unique set of challenges. Balancing feasible implementation with creative problem-solving requires thoughtful consideration. The evaluation of the radar system, along with user testing of icon systems, revealed several areas for improvement. The design process generated concepts for redesigned components with the potential to further enhance usability.

Additionally, the project resulted in a toolkit that improved user satisfaction by simplifying the selection of tools and torque levels, thereby increasing overall usability. This outcome is supported by feedback from the expert focus group, adding legitimacy to the findings.

The redesigned components, together with the toolkit, address the research question of how usability can be improved through small, resource-efficient interventions.

If the concept had been implemented into the system earlier, it could have made a bigger impact. For example, integrating symbols directly into the chassis or structural components, rather than relying on stickers. It would allow the markings to be milled next to the screws, resulting in a more durable and long-lasting solution.

Another benefit of earlier implementation would be the opportunity to develop and standardize the toolkit from the beginning. Designers and engineers could then adapt components to align with the standardized torque settings, screw sizes, and communication features included in the kit. This approach would not only support the work of design engineers but also enhance the overall experience for end users.

7.4.4 Research Question 3:
How can design solutions be effectively integrated into the existing product?

At the outset of this project, the potential scope of the final result was broad. To ensure effective integration of the product into the existing system and maintain practicality for Saab, it was crucial to adopt a realistic approach to the thesis. As the project progressed, the challenge of integration became clearer, revealing a predominantly mechanical focus.

This project explored several ideas for enhancing usability, each varying in complexity depending on the concept and components involved. The investigation into integrating these solutions reached a high level of development in terms of feasibility. Given the time and resources of the project, the focus shifted toward developing a proposal with the greatest potential impact. As a result, the decision was made to narrow the project's scope to the design and proposal of a well-developed toolkit intended for integration into the existing system.

The result is a solution that can be integrated into the current system. Its straightforward design allows for flexibility, enabling it to be easily repositioned within the storage area. However, the nature of the product is subject to change, and the area designated for integration may be eliminated in the future. Nevertheless, after discussions with engineers responsible for developing this area, they assured us that integration would remain possible even in future systems, although minor product adaptations might be necessary.

7.5 RECOMMENDATIONS

This thesis lays the groundwork for what could become an intuitive aid for operators and maintainers in their service and repair tasks. While the project delivers a functional prototype, further development is needed to improve the solution and make it more robust and reliable.

One area for future research concerns the material selection for the toolkit. This thesis does not include testing or analysis of which materials would be most efficient for its intended use. Further investigation is recommended into how various materials respond to environmental conditions such as heat, cold, and rain, to ensure durability and functionality in different climates.

Another important direction for continued work is the quantitative evaluation of usability. While current expert feedback supports the claim that the toolkit would improve usability, no quantitative data has yet been collected to demonstrate how much improvement it provides. Additional user testing measuring factors such as task completion time, errors, and failure rates, would help validate the effectiveness of the toolkit and reinforce its legitimacy.

The holder for the toolkit originated directly from early brainstorming sessions. During focus groups, the idea of a dedicated holder emerged as highly promising. Experts discussed the potential value of a holder not only for storing the toolkit but also for including small spare parts such as screws, bullets and cable ties. Future development should involve designing a durable container based on research into suitable materials and structural properties. A thoroughly developed holder could eventually be implemented into the CRM, offering great value to both Saab and the stakeholders who rely on the product.

A crucial area for improvement in the toolkit involves consideration of color blindness and low-visibility conditions during maintenance. This aspect was outside the scope of the current thesis but should be addressed in future work. Specifically, research should investigate whether the color schemes used in the toolkit are distinguishable under typical working conditions for operators, including poor lighting and adverse weather. These findings would directly inform the design of visual indicators. Depending on the results, a rethinking of current color choices, or even a shift to alternative visual cues, may be necessary to ensure accessibility and clarity.

The communication system used within the toolkit also has potential for further refinement. This thesis evaluated three conceptual systems for icons: text, symbols, and numbers. Based on the Pugh matrix analysis, the text-based system was deemed the most effective in terms of communication. However, expert feedback emphasized the benefits of symbolic systems in supporting a structured, intuitive workflow for operators. Future development should focus on designing a hybrid system that combines the most appreciated aspects of all three approaches. Such a system could reduce the number of indicators required on the antenna, thereby minimizing cognitive overload while preserving clarity and usability.

This hybrid communication system should also be evaluated further and compared against the other indication methods, ideally in collaboration with actual operators working in field conditions. Comparative user testing should focus on key usability metrics such as guessability and learnability. For example, studies could measure how quickly users select the correct tool and how consistently they recall icon functions, then compare these results to current workflows to assess overall performance gains.

In summary, while this thesis presents a solid foundation and a functioning prototype, several key areas require further research and refinement. These include material selection, quantitative usability validation, kit-holder design, accessibility considerations, and indication systems. With continued development, this concept has strong potential to become a valuable asset for operators and maintainers alike.



08 REFERENCES

Photo: Saab

8 References

Besant, H. (2016). The Journey of Brainstorming. *Journal of Transformative Innovation, 2*(1).

Bevan, N., Kirakowski, J., & Maissel, J. (1991). *What is Usability.* Ireland: University College Cork.

Bhutta, S. K., & Huq, F. (1999). Benhcmarking - best practices: an integrated approach. *Benchmarking An International Journal*, 254-268. doi:10.1108/14635779910289261

Cirulis, G., & Chang, C. (2022, May 16). *14 tips to help you run a successful product design workshop*. Retrieved from Intercom: https://www.intercom.com/blog/running-design-workshops/

Coursera. (2024, November 14). *What Is Project Planning? Benefits, Tools, and More*. Retrieved from Coursera: https://www.coursera.org/articles/project-planning

Döringer, S. (2021). 'The problem-centred expert interview'. Combining qualitative interviewing approaches for investigating implicit expert knowledge. *International Journal of Social Research Methodology, 24*(3), 265-278. doi:10.1080/13645579.2020.1766777

Emaint. (2024, July 29). *Criticality Analysis in Maintenance*. Retrieved from Emaint: https://www.emaint.com/rank-assets-by-criticalness-for-a-more-effective-aca/

Gaver, W. W. (1991). Technology affordances. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, (pp. 79-84). doi:https://doi.org/10.1145/108844.108856

Gibbons, S. (2016, July 31). *Design Thinking 101*. Retrieved from Nielsen Norman Group: https://www.nngroup.com/articles/design-thinking/

Guler, K., & Petrisor, D. M. (2021). A Pugh Matrix based product development model. *Cogent Engineering, 8*. doi:10.1080/23311916.2021.1923383

Harte, R., Glynn, L., Rodríguez-Molinero, A., Baker, M. P., Scharf, T., Quinlan, L. R., & ÓLaighin, G. (2017). A Human-Centered Design Methodology to Enhance the Usability, Human Factors, and User Experience of Connected Health Systems: A Three-Phase Methodology. *JMIR Human Factors*. Retrieved from https://humanfactors.jmir.org/2017/1/e8/

IDSA. (n.d.). *What Is Industrail Design?* Retrieved from Industrail Designers Society of America: https://www.idsa.org/about-idsa/advocacy/what-industrial-design/

Interaction Design Foundation. (n.d.). Retrieved from Codesign: https://www.interaction-design.org/literature/topics/codesign?srsltid=AfmBOoqY_GoweySj__ZKT8iObgadmLbT532flovQ6m9nw3ssFTIKjpbz

Interaction Design Foundation. (2016, June 5). *What is Hawthorne Effect?* Retrieved from Industrial Design Foundation - IxDF: https://www.interaction-design.org/literature/topics/hawthorne-effect

Interaction Design Foundation. (2021, March 24). *How to Conduct User Observations*. Retrieved from Interaction Design Foundation - IxDF: https://www.interaction-design.org/literature/article/how-to-conduct-user-observations

Interaction Design Foundation. (n.d.). *User Centered Design (UCD)*. Retrieved from Interaction Design Foundation: https://www.interaction-design.org/literature/topics/user-centered-design?srsltid=AfmBOooNBEsWHiEkiVHTvfw6ULd8QKunynYWOq3gJ7yQEhdGcKptbUIZ

International Ergonomics Association . (n.d.). *What is Ergonomics (HFE)?* Retrieved from IEA: https://iea.cc/about/what-is-ergonomics/

Jordan, P. W. (1998). *An Introduction To Usability.* London: Taylor & Francis.

Kahneman, D. (2011). Thinking, Fast and slow. Farrar, Straus and Giroux.

Karolinska Institutet. (2023, 09 08). *Evaluating information*. Retrieved from Karolinska Institutet: https://kib.ki.se/en/search-evaluate/evaluating-information

Karolinska Institutet. (2025, 01 20). *Structured literature reviews – A guide for students*. Retrieved from Karolinska Institutet: https://kib.ki.se/en/search-evaluate/systematic-reviews-researchers/structured-literature-reviews-guide-students

Kennon, N., Howden, P., & Hartley, M. (2009). Who really matters? A stakeholder analysis tool. *Extension Farming Systems, 5*(2). doi:https://search.informit.org/doi/epdf/10.3316/informit.733413362842369

Korhan, O. (2023). *Ergonomics - New Insights.* London: IntechOpen. Retrieved from https://mts.intechopen.com/storage/books/11877/authors_book/authors_book.pdf

Martins, J. (2024, April 23). *How to conduct a stakeholder analysis*. Retrieved from Asana: https://asana.com/resources/project-stakeholder

Morgan, D. L. (2011). *Focus Groups as Qualitative Research.* Thousand Oaks: SAGE Publications, Inc. doi:https://doi.org/10.4135/9781412984287

Nielsen, J. (2012, January 15). *Thinking Aloud: The #1 Usability Tool*. Retrieved from Nielsen Norman Group: https://www.nngroup.com/articles/thinking-aloud-the-1-usability-tool/

Norman, D. A. (1999). Affordance, conventions, and design. *Interactions, 6*(3), 38–43. doi:https://doi.org/10.1145/301153.301168

Oguz, E., Kubicek, M., & Clelland, D. (2018). Failure modes and criticality analysis of the preliminary design phase of the Mars Desert Research Station considering human factors. In *Reliability Engineering & System Safety* (Vol. 178, pp. 247-254). Glasgow: Elsevier. doi:https://doi.org/10.1016/j.ress.2018.06.023

Onwuegbuzie, A. J., Dickinson, W. B., Leech, N. L., & Zoran, A. G. (2009). A Qualitative Framework for Collecting and Analyzing Data in Focus Group Research. *International Journal of Qualitativ Methods, 8*(3), 1-21. doi:https://doi.org/10.1177/160940690900800301

Osborn, F. A. (1954). Creative Thinking. *SageJournals, 6*(9), 23-25. doi:10.1177/216507995800600906

Phillips-Wren, G., & Adya, M. (2020). Decision making under stress: the role of information overload, time pressure, complexity, and uncertainty. *Journal of Decision Systems*. doi:10.1080/12460125.2020.1768680

Rojas, J. (2024). *Learn How to Use Sketching as an Ideation Method*. Retrieved from Interaction Design Foundation: https://www.interaction-design.org/literature/article/etch-a-sketch-how-to-use-sketching-in-user-experience-design#:~:text=Sketching%20is%20a%20time%2Dtested,with%20a%20new%20design%20challenge

Rosala, M. (2024, November 1). *Research Plans: Organize, Document, Inform*. Retrieved from Nielsen Norman Group: https://www.nngroup.com/articles/pm-research-plan/

Rosenbaum, M. S., Otalora, M. L., & Ramírez, G. C. (2017). How to create a realistic customer journey map. In *Business Horizons* (pp. 143-150). Retrieved from https://doi.org/10.1016/j.bushor.2016.09.010.

Saab. (n.d.). *Giraffe 1X*. Retrieved from Saab: https://www.saab.com/products/giraffe-1x

Shaukat, M. M. (2022). Investigating the influence of co-design on design process for developing new products. *Journal on Innovation and Sustainability RISUS. 13:84-91.* doi:10.23925/2179-3565.2022v13i2p84-91

Snyder, H. (2019). *Literature review as a research methodology: An overview and guidelines.* Oslo: Elsevier Inc. doi:https://doi.org/10.1016/j.jbusres.2019.07.039

Soest, C. v. (2023, March). Why do we Speak to Experts? Reviing the Strength of the Expert Interview Method. *Perspectives on Polictics*, pp. 277-287.

Tableau. (n.d.). *Root Cause Analysis Explained: Definition, Examples, and Methods*. Retrieved from Tableau: https://www.tableau.com/analytics/what-is-root-cause-analysis

Teamgantt. (n.d.). *What Is a Gantt Chart? A Practical Guide for Project Managers*. Retrieved from Teamgantt: https://www.teamgantt.com/what-is-a-gantt-chart

Toneva, M. (n.d.). *5-Whys analysis tool*. Retrieved from Businessmap: https://businessmap.io/lean-management/improvement/5-whys-analysis-tool

Ulrich, K. T., & Eppinger, S. D. (2019). *Product Design and Development.* McGraw Hill.

Westerlund, B. (2007). A Workshop Method that Involves Users Talking, Doing and Making. *International Conference on Human-Machine Interaction* (pp. 98-102). Timimoun, Algeria: Human07. Retrieved from https://bowesterlund.se/publications/westerlund-human07.pdf

Whitenton, K. (2013, December 22). *Minimize Cognitive Load to Maximize Usability*. Retrieved from NNgroup.com: https://www.nngroup.com/articles/minimize-cognitive-load/

Wikberg-Nilsson, Å., Ericson, Å., & Törlind, P. (2015). *Design: Process och metod* (1:5 ed.). Lund: Studentlitteratur.

Witzel, A. (2000). *The problem-centered interview.* Retrieved from Forum: Qualitative social research:

https://www.qualitative-research.net/index.php/fqs/article/view/1132/2522
World Design Organization. (n.d.). *Definition of Industrial Design*. Retrieved from World Design Organization: https://wdo.org/about/definition/



09 APPENDIXES

Photo: Saab

APPENDIX A: SERVICE STAKEHOLDER ANALYSIS

Appendix A contains information about each of the identified operator stakeholders. This goes into further detail than what described in the report.

The analysis focuses on the stakeholders involved in process of service and repair. The product has been delivered to the costumer and its stationed there. The process is generalized in order to grasp the scenario and process, although the stakeholders are quite the same, what stakeholders are involved, in reality, depends on the contract between Saab AB and their customer.

Classes of Maintaining Personnel

The number of levels of maintaining personnel can vary, but tend to be the same number of stakeholders. In this case, the stakeholders are sorted into 4 deferent levels, categorized by the tasks allowed to be done by the maintainer. The level of allowed task is sorted by complexity which directly correlates with time spent on service/repair task.

LEVEL 1

The Operator - The operating personal are the ones that are using the actual functionality of radar, such as spotting objects. When the radar stops functioning, their level of troubleshooting is to see if the cables are correctly mounted to the computer, and trying to reset the radar. If failures still exist, they call on the level 2 maintainer. Troubleshooting and services that requires opening the radar is too complex for this level. The operator is part of the On-board crew.

LEVEL 2

2-level maintainer – are mainly tasked with the responsibility to read error codes and analyze the what tier of problem that has occurred. If the error codes indicate a broken component that require about 1 hour repair time, they are tasked with doing the repair in general. This level of maintainer can be part of the On-board crew, if not, they travel to the radar site. If the error code requires a repair process of longer than 1 hour, they tend to send it to the 3-level maintainer.

LEVEL 3

3-level maintainer – are tasked with doing larger repairs and services that requires indoor environment/weather and heavy lifts. These maintainers are located in shops and are doing tasks that required more or about 1 day.

LEVEL 4

In-country service maintainers – This are supporting offices and workshops that is part of Saab. These stations have maintainers that is tasked with deep knowledge of the systems and are training future maintainers. These level-4 maintainers are able to change and service components that the customer maintainers are not. These components are usually not available "Spare parts" for the customer.

Saab AB – is more or less the back to the production site. These sites have employees that can repair the internal electronics of and re-use spare parts. They also have maintainers that is tasked with deep knowledge of the systems and are training future maintainers. These maintainers are able to change and service components that the customer maintainers are not. These components that are not available "Spare parts" for the customer.

APPENDIX B: SEMI-STRUCUTURED USER INTERVIEW TEMPLATE

Appendix B contains the structure and questions asked during the interviews.

Välkommen! Mitt namn är Albin Sämfors och med mig är Simon Tollesson, du är på väg att ta del av en semi-strukturerad intervju som huvudsakligen handlar om Giraffe 1X radarn. Vårat mål med intervjun är att få en bättre förståelse om hur radarn är uppbyggd, hur processen för montering samt demontering ser ut och vilka problem som finns med produkten. Med hjälp av denna informationen ska vi skriva ett examensarbete som har som mål att förbättra användbarheten samt användarupplevelsen vid bland annat reparation. Svaren du ger kommer skapa en grund för detta arbete och vi hade gärna kunnat återkoppla om vi har ytterligare frågor efter intervjun. Är det okej att vi tar dina kontaktuppgifter eller vill du förbli anonym i arbetet? (Vänta på svar)
Tack så mycket/ vi förstår.

Namn:
Mail:
Vi kommer nu att börja intervjun, du får gärna utveckla dina svar hur du önskar, ju fler detaljer vi får desto bättre kan vi förstå utmaningarna:

<u>Vem är du?</u>
Vad är din koppling till Giraffe 1X? (Arbetsuppgift/ intresse/ utbildare etc.)
-

Hur länge har du jobbat med Giraffe 1X? (Erfarenhet med systemet)
-

<u>Radarsystemet</u>
Vilken del och vilka komponenter har du jobbat mest med?
-

Har du märkt några tydliga problem när du arbetar med radarn? (Inte mjukvara, när man tar isär / sätter ihop)
-

(Mer specifikt när det kommer till montering och demontering av systemet…)
Vilka steg av svårast i processen, behöver du någonsin använda dig av manualen eller leta upp information på annat sätt?
-

Vad beror detta på? (Användarfel/ Otydligheter/ Fysiskt komplicerat att montera)
-

<u>Går saker sönder</u>
Går komponenter någonsin sönder?
-

Hur ofta relativt varandra är det på ett ungefär? (Behöver inte veta exakt tid, mer intressant att veta relativt till varandra)
-

Vilka delar är det och vad är anledningen?
-

Vilka delar är mest kritiska när de går sönder?
-

Om du försöker göra en avvägning till vilka delar som går sönder oftast och vilka som är mest kritiska, vilken del hade du sett var viktigast att förbättra? (Vilken vi del ska vi undersöka)

<u>Generella frågor</u>
Vilken förändring hade du helst sett i antennen? (Topp 3 om det är många)
-

Hur många olika sorters verktyg används vid service? (momentverktyg etc., rimlig mängd?)
-

Har de olika användningsområden eller gör de samma uppgift?
-

Är det något annat gällande användbarheten eller generellt du skulle vilja tillägga? (bra för oss att veta)
-

APPENDIX C: CUSTOMER JOURNEY DETAILS

Appendix C contains the aspects of experience for each component interacted with.

Table: Service – TRM swap

	Cover	TRM (old)	TRM (new)	Cover
Interaction	Screws, hex key	Screws, hex key	Screws, torque keys	Screws, Hex key
Do	Screwing, lifting, fitting	Screwing, pulling	Fitting, screwing	Lifting, fitting, screwing
Say	"How do I lift the cover?"	"Why isn't it possible to pull?"	"What screws need torque"	"Do I really need to use torque?"
See/hear				"Where did I put the tools?"
Experiencing	Excited, nervous	Nervous, anxious, annoyance	Annoyance	Repetition
Opportunities		Developing tools, the screws	Communication on torque	Indications for torque

Table: Service – Rotary Joint removal

	Cover	CM	Cassette	Turntable cover	TCU	Rotaryjoint
Interaction	Screws, hex key	Screws, hex key,	Screw, hex key, ESD-bracelet	Screws, hex key	Screws	
Do	Screwing, lifting, fitting	Screwing, opening	Screwing, pulling	Screwing, lifting	Lifting, disconnect	Lift
Say	"How do I lift the cover?"					
See/hear						
Experiencing	Excited, nervous	Nervous	Focused	Annoyance	Focus	Discomfort
Opportunities		Develop screws		Develop screws		Making it a 1-person removal

Concluding analysis:
The visualized maps are based on the answers in the interviews and observations from the customer training. When it comes to analyzing the journey while considering usability, the most frequent task is screwing. Naturally, if the goal is the reduce repair time, regardless of task, would be to either evaluate if there is any possibility to reduce number of screws or just streamline screwing to the user.

APPENDIX D: BENCHMARK

Appendix D contains the benchmarked solutions for the toolkit.







APPENDIX E: WORKSHOP

Appendix E contains the aim of the workshop and how it was planned to be executed.

Workshop aimed to be conducted with Industrial Design Engineers from Luleå University of Technology.

Information

Workshopen kommer att försöka framkalla idéer för reparation och verktyg. Kontexten till det är:
En tekniker ska reparera komponenter i ett elskåp på hög höjd i ett radiotorn. Förhållederna kan variera samt arbetsställning är ofta inte optimal. För att utföra reparationerna behöver han ta med sig ett verktygskit för att kunna öppna elskåpet samt skruva loss komponenterna som ska bytas ut. Uppgiften ska i förstahand utföras utan några fel eller misstag, och man vill även spendera så lite onödig tid som möjligt uppe i tornet.

Målet med denna workshop är då alltså att komma på idéer för hur man kan designa / skapa ett verktygskit som är så användarvänligt som möjligt.

Visa agenda, som kommer bestå av;

- Inspiration (Benchmark / Market-research)
- Brainstorming, ca 20min, för att generera skriftliga idéer.
- Paus, ca 5 min
- Braindrawing / Sketching, ca 20min, visuallisera idéer

Verktyg som kan komma att användas är Insexnycklar, Momentverktyg (ca 4st typ vanliga skruvmejslar) och multimeter.

1. What is the goal of the workshop?
    a. Gather ideas by implementing co-design
    b. Find new ideas for tool bag
        i. How can tools be attached?
        ii. How can they be arranged so you easily find them?
        iii. In what ways can you bring the toolbag with you? Backpack/ Handheld/ Bodyattached..?

2. Who needs to come?

The workshop is focused on generating quantitative ideas which requires the participants to not overthink their ideas. This can easily be solved by only involving participants who are not familiar with the system the ideas are being implemented in. Therefore, it was chosen to pick **peers and classmates** who had the design thinking but were not aware of the circumstances of the radar.

3. What are the tangible outcomes?
    a. New ideas on attachments and bags. (Details)
    b. Inspirational sketches which can contribute to the final design
4. Drafts to present to supervisors who can give further feedback and idea

5. When will the workshop be conducted?

Friday 21/3, aimed to take about an hour and will be held as an online meeting via Zoom. Tools such as Miro will be needed to brainstorm keywords for details and functions, as well as pen and paper to sketch ideas after the brainstorm.

6. Workshop time plan (Total: 57 min)
    a. Intro (5 min):
        i. Show Agenda for participants
        ii. Goals of the workshop
    b. Background of the workshop
        i. Present the background/ environment for the workshop
            1. Engineer climbs antenna to repair something in an electrical box at the top, needs certain tools to complete the repair. The tools needed will be about 4 screwdrivers, Torx and allen keys, and a multimeter.
    c. Warm-up exercise (5 min)
    d. Give inspiration by showing benchmarked solutions
    e. Brainstorm (warm-up for sketching ideation) (20-30min)
        i. Generate written ideas for: (5 min)
            1. What materials can be used for the box/ bag?
            2. How should it be carried?
            3. How can we prevent tools from falling out of the bag?
            4. Screws will get loose while repairing, how can we store them?
        ii. Quickly discuss the ideas (5 min)
        iii. Break (2 min)
        iv. Generate new ideas, still allowed to add to previous questions: (5 min)
            1. Should the tools be organized in the box/ bag? How?
            2. Does it need to be placed somewhere? How does it attach?
            3. Instead of one large bag, can a few smaller ones be better? Maybe modular?
            4. General ideas
        v. Quickly discuss the ideas (5 min)

f. Sketching (25 mins)

Using the words and ideas found in the brainstorm, now visualize them. It doesn't have to be what you wrote down yourself, it can be ideas from anyone.

i. Sketch-session 1: (5 minutes)
ii. Show ideas (5 min)
iii. Sketch session 2: (10 minutes)
iv. Show ideas (5 min)

7. Take into consideration:

a. Warm up the participants with some kind of small talk with each one. Basically, break the ice. Maybe even some warmup game?
b. Be mindful of about the brainstorm (IDEO's rules)
    i. Defer judgement
    ii. Encourage wild ideas
    iii. Build on the ideas of others
    iv. Stay focused on the topic
    v. One conversation at a time
    vi. Be visual
    vii. Go for quantity
c. Keep track of the time!!!

APPENDIX F: WORKSHOP BRAINSTORMING

Appendix F contains the ideas brainstormed from the participants of the workshop.

Brainstorming 1

General ideas

- Som en väst, men fack fram på bröstet
- Som en slingbag, den är på ryggen men man kan kasta den fram över axeln
- Själva väskan sitter fast i kroppen med magneter kan ta lös den och fästa väskan
- Förvara verktygen direkt i antennen
- Så kort avstånd som möjligt mellan där verktygen förvaras tills där man använder dem

Type of Material- or General properties for the kit

- Hård rygg med mjukare material för facken
- Hårt yttre och mjukare material innuti
- Mjuk
- Mjukt så det inte är obekvämt att bära när man klättrar

How can the kit be carried?

- På bröst
- På Armen
- Runt midjan
- På repet man sitter fast vid
- Around the leg
- Rygg / bröst
- Drönare
- Benfickor
- På huvudet
- Ett bälte som man kan snurra runt en själv och justera vart man vill ha den beroende på vad man gör

Prevent tools from falling out

- Vajer (som korthållare)
- Retractable keychain that is attached to the bag
- Verktygen sitter fast i väskan med ett snöre
- Magneter
- Dra in dra ut grej typ snöre

Storing loose screws from components

- Magnetdosa som sitter fast på bröstkorgen
- Skumaktigt material som man kan trycka in skruvarna i
- Bowl with magnet in the bottom
- Magneter
- Magnetisk handske
- Ett magasin som man kan trycka in skruvarna
- Magnetisk verktyg
- Typ en flugfångare, kan lägga i men inte få ut
- Skål med magneter

Brainstorming 2

General ideas

- Reservdelen bär man som Captain Americas sköld
- Öppning som sirapsflaska, man kan öppna åt ena hållet
- Om man har snöre borde det vara en sån som kan stanna vid ett visst läge

How can the tools be organized in the kit

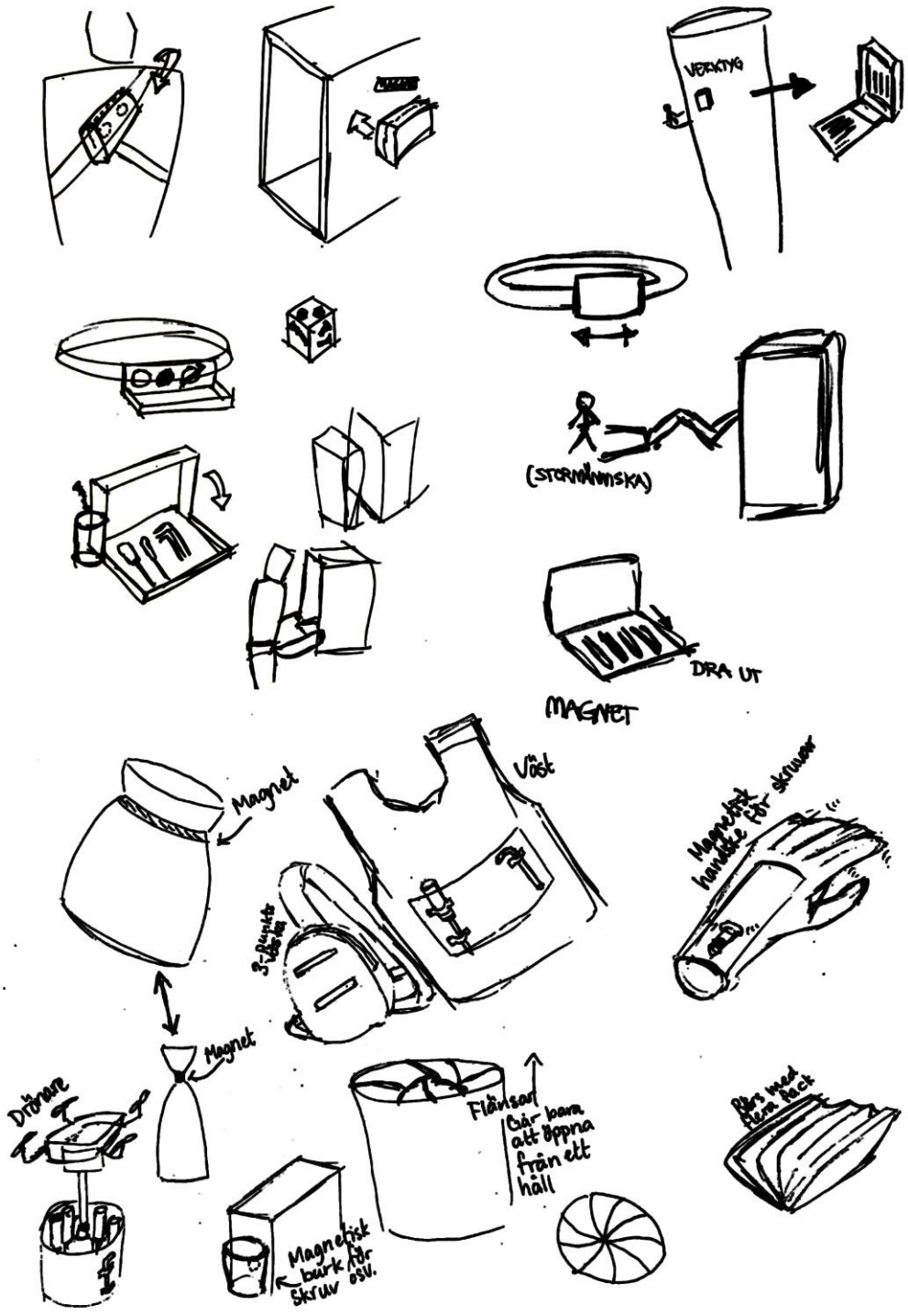
- Det går inte att lägga på fel plats som ett pussel
- Man kan på nått sätt se alla verktyg
- Som en stor börs med flera fack
- En ficka för varje verktyg med snöret som drar den till fickan
- Olika fack, verktygen passar bara i rätt fack

How does the operator access the kit?

- Det finns ett kit i antennen
- Det går att fälla ner så det funkar som ett bord
- Sitter fast på kroppen
- Fastsatt på skåpet
- Går att fälla ut med en arm

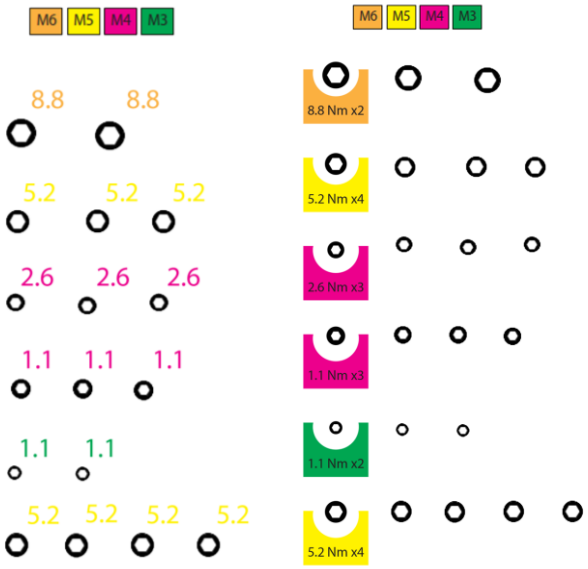
Maybe the solution is not one large kit?

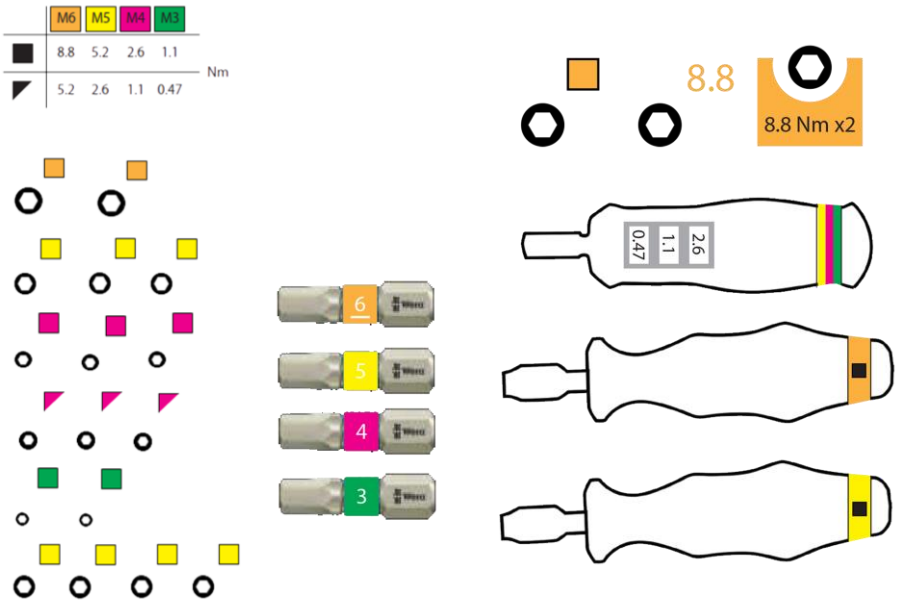
- Olika delar sitter på olika kroppsdelar
- Många små väskor, typ en på vardera ben
- Fördelad på benen


VERKTYG
(STORMÄNNISKA)
DRA UT
MAGNET
Magnet
Magnet
Vask
Magnetisk handtag för skruvar
Magnetisk burk för skruv osv.
Flänsar där bara att öppna från ett håll
Bärs med flera hack

APPENDIX G: THINK ALOUD – CONCEPT TESTS

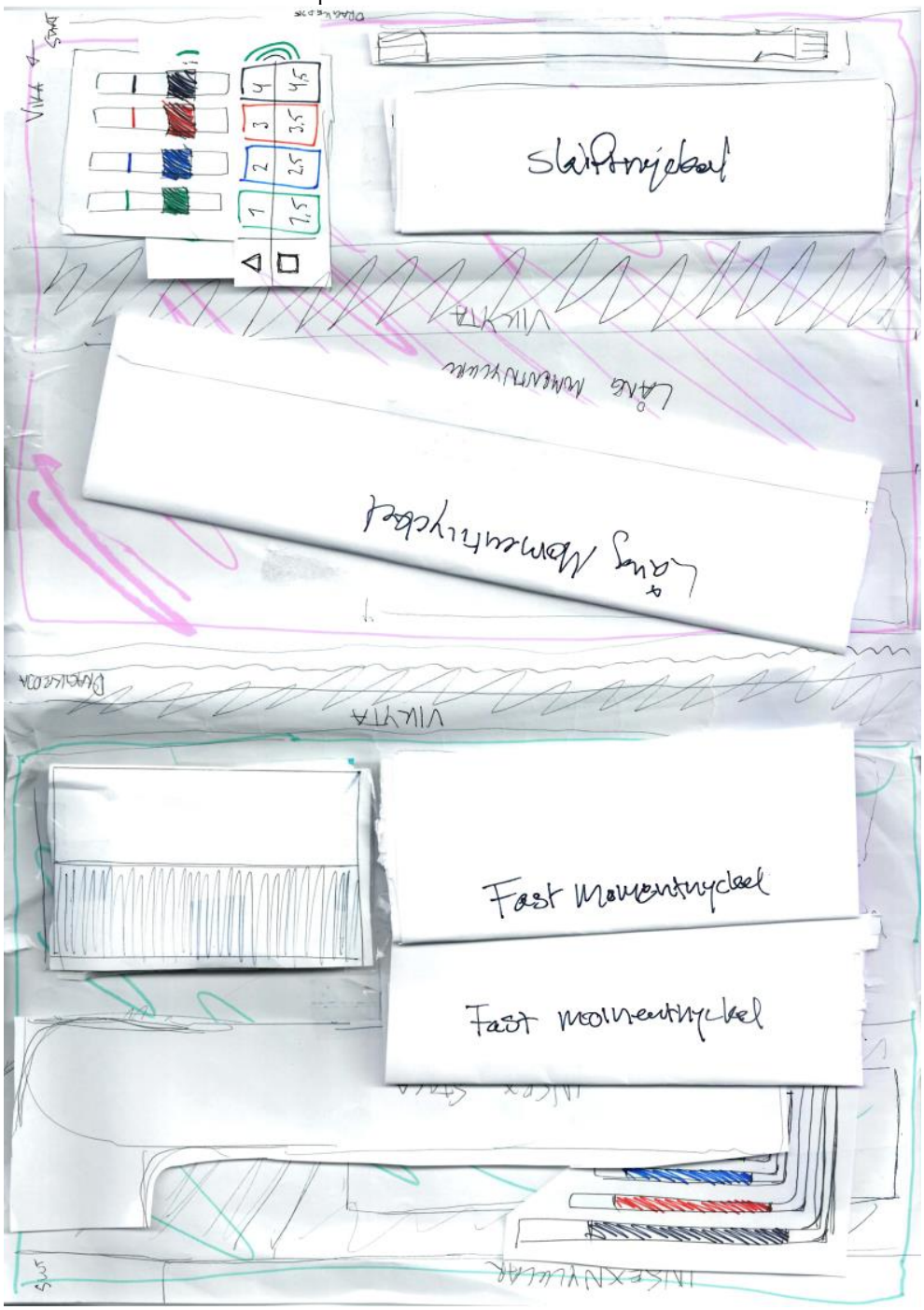
Appendix G contains how the Think aloud test was performed. One row of screws was displayed at a time and the participants had to figure out what screwdriver and torque was required to do the task.





APPENDIX H: PROTOTYPING

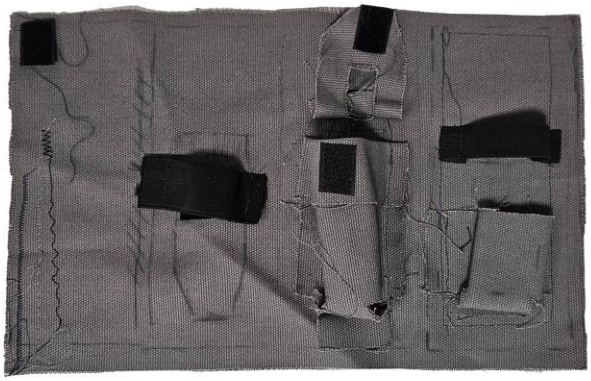
Appendix H contains images of the different prototypes, ranging from mock-up paper prototypes, to the real fabric ones. It also includes some prototypes of embroided tables to test if that could be possible to use instead of the hard one.

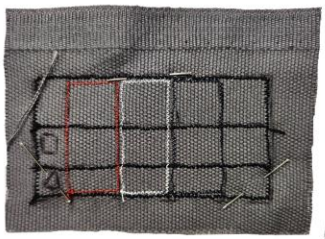


Paper mockup



Tests







Grey #1



Black Prototype



Grey #2



Final Prototype



APPENDIX I: PUGH MATRIX

Appendix I contains the PUGH matrix and how the aspects were valued. The Learnability row was chosen to not be considered as the data was not sufficient for a proper evaluation.

PUGH-MATRIX				
Criteria	Concept: Symbols	Concept: Number	Concept: Text	Weight
Concept implementation speed	0	0	+	2
Manufacturing and Implementing cost	-	0	+	1
Low time on choosing tools	0	0	-	2
Low error rate choosing tools	+	0	+	3
Guessability (Understand symbol at first glance)	-	0	+	3
Learnability (Understand symbolsystem fast)	0	0	0	3
Communication per area unit	-	0	+	2
Eastetical stasfaction	+	0	+	1
Cognitive load	+	0	+	3
Cluttered	0	0	+	2
Task done communication	0	0	+	2
Sum	1	0	17	
Rank	2	3	1	

APPENDIX J: TOOLS USED FOR FINAL CONCEPT

Appendix J contains the tools used for the toolkit

Image	Tool
	Series 7400 Imperial Kraftform torque screwdrivers with a customized factory pre-set measurement value, handle size 105mm
	7510 Kraftform Safe-Torque Speed Torque screwdriver, 1-3Nm
	3950/9 Hex-Plus Multicolour HF Stainless 1 L-key set, metric, stainless steel, with holding function, 9 pieces
	ESD 6" Drawing soft Flat Brush – Dissipative Bristles
	Siftnyckel IQ 150mm (4002IQ)
	Strap Wrench TG70
	Rosa Antistatiska Påsar (ESD) Med greppförsegling

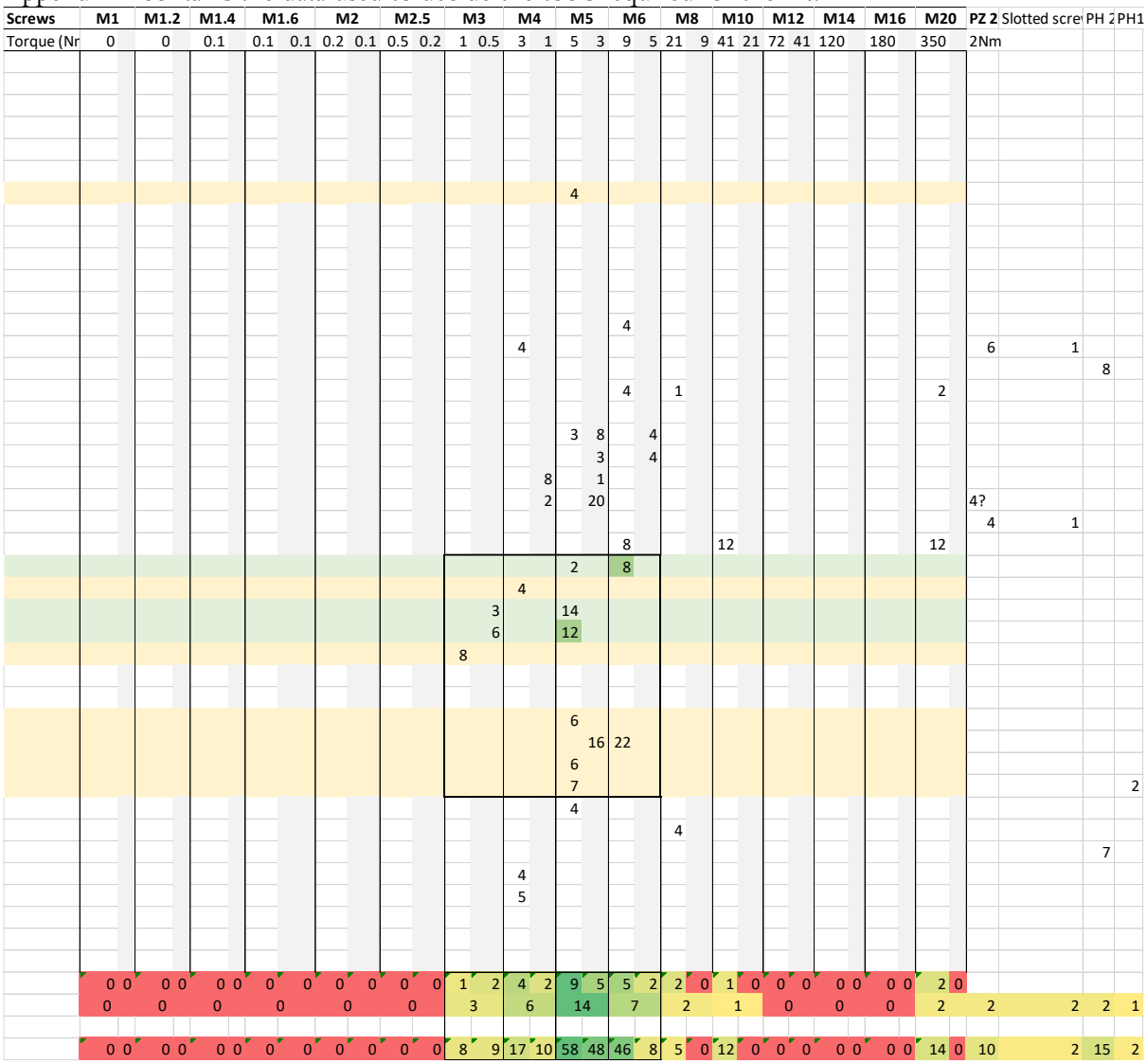
APPENDIX K: EXHIBITION POSTER

Appendix K contains the poster used for the exhibition presentation at LTU.



APPENDIX L: SCREW DATA FROM MANUAL

Appendix L contains the data used to decide the tools required for the kit.



Screws	M1	M1.2	M1.4	M1.6	M2	M2.5	M3	M4	M5	M6	M8	M10	M12	M14	M16	M20	PZ 2	Slotted scre	PH 2	PH1
Torque (Nr	0	0	0.1	0.1	0.1	0.2	0.1	0.5	0.2	1	0.5	3	1	5	3	9	5	21	9	41	21	72	41	120	180	350	2Nm			
									4																					
								4		4							6		1										8	
										4	1					2														
									3	8	4																			
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